

DISCERN: CAN AI AGENTS WORK LIKE SCIENTISTS AND GUIDE DISCOVERY?

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ABSTRACT

Reliable automated research requires agents to vet data, verify analyses, and generate hypotheses grounded in trustworthy evidence, potentially reducing routine scientific workload while allowing scientists to focus on interpretation and discovery. Existing benchmarks often only assess analytical task completion or hypothesis generation separately rather than testing whether reliable evidence supports valid and novel claims. We introduce **DISCERN** (Data Integrity and Scientific Capability: Evidence, Reasoning, and Novelty), a controlled benchmark on real, publicly available datasets that evaluates three key levels of an automated research workflow. The first two levels test data integrity and analysis verification under confounds and tool traps, while the third tests hypothesis generation and revision under adversarial review, including counterfactual cases in which evidence consistent with real data and documented scientific phenomena conflicts with established expectations, motivating alternative explanations and testable hypotheses. Across 203 tasks, eight life-science tracks, and eight models, DISCERN shows that strong aggregate performance can mask level-specific weaknesses. Agents earn perfect scores in only 60.8% of Level 1, 34.2% of Level 2, and 0.6% of Level 3 evaluations, with penalties attributed to rejection of sound data, failure to carry recognized limitations into conclusions, and wide variation in hypothesis production. Cross-track rankings by token and code use are substantially more stable than rankings by evidence judgment, suggesting greater consistency in computational effort than in evidence-based reasoning. These profiles identify opportunities for supervised scientific assistance, but current agents do not yet demonstrate reliable autonomous analysis or discovery.

1 INTRODUCTION

AI agents are increasingly used to support scientific research through literature review, experimental planning, data analysis, scientific discovery and hypothesis generation (Lu et al., 2024; Mitchener et al., 2025; Huang et al., 2024; 2026; Qu et al., 2026). Their capabilities are expanding toward broader research workflows that include executing analyses, interpreting results, and iteratively developing scientific ideas. Together, these advances create an opportunity to automate routine and computationally intensive parts of research, reducing scientific workload and allowing researchers to focus more on higher-level scientific judgment.

However, automated research workflows are fragile because a local error can propagate across stages and distort the final scientific conclusion (Peng, 2011; Munafò et al., 2017). Data problems can corrupt computation; confounds or inappropriate analytical settings can invalidate plausible results; and unsupported interpretation can turn a technically sound analysis into an unsupported scientific claim. For example, a contaminated expression signal can be mistaken for biology and redirect downstream experiments (Figure 1). Existing benchmarks often summarize complex behavior using only task-level success or aggregate scores, or focus on a few local subtasks such as analysis execution (Majumder et al., 2025; Jansen et al., 2024; Zheng et al., 2026). They therefore do not adequately connect high-level research outcomes with the detailed scientific decisions that produce them. Aggregate outcomes can hide where errors arise, while local task scores may not reveal whether recognized problems influence subsequent analysis and final conclusions.

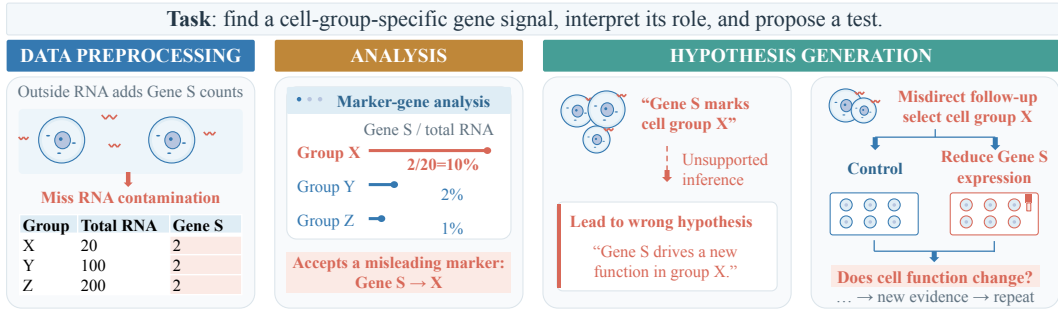


Figure 1: How agent errors can propagate into a false scientific lead. Given single-cell RNA measurements, an agent seeks a cell-group-specific gene signal and proposes a test. It misses contaminating RNA, accepts Gene S’s higher relative abundance in group X as a marker, and overinterprets it as a functional driver, potentially misdirecting follow-up experiments. Panels map onto data preprocessing (L1), analysis (L2), and hypothesis generation (L3). This illustrative sequence is inspired by DISCERN’s ambient-RNA tasks (Appendix E).

Reliable automated research requires evaluation from low-level data integrity to high-level scientific reasoning. Data must first be fit for analysis, analyses must produce trustworthy evidence, and that evidence must support justified hypotheses. These stages are linked: errors in data handling can invalidate analyses, while unreliable analyses cannot support sound hypotheses. Evaluating all three therefore provides a more complete view of scientific-agent reliability than testing any stage alone.

Based on this hierarchy, we introduce **DISCERN** (Data Integrity and Scientific Capability: Evidence, Reasoning, and Novelty), a controlled three-level benchmark built from 203 tasks across eight life-science tracks and five scientific areas using real, publicly available datasets (Appendix Table 3). Specifically, we design 79 Level 1 (L1) data-preprocessing tasks spanning 58 distinct task types, 102 Level 2 (L2) analysis-verification tasks across 24 analysis families, and 22 Level 3 (L3) hypothesis-generation dossiers. The benchmark combines matched clean controls with controlled data defects, clean and perturbed analytical conditions, and real or counterfactual evidence patterns. Counterfactual cases are grounded in documented scientific phenomena, retained only when they conflict with the model’s elicited prior, and evaluated through three rounds of adversarial review. The three levels are conceptually linked but independently instantiated and scored, allowing DISCERN to localize weaknesses at each level without treating separately evaluated tasks as an end-to-end pipeline, while also characterizing scientific working styles across models (Figure 2).

DISCERN evaluates eight contemporary language-model agents spanning proprietary and open-weight systems. The results reveal a consistent pattern from aggregate performance to specific scientific decisions. **(1)** At the global level, aggregate scores mask uneven strengths: no model dominates all three levels, and similar overall performance can arise from different capability profiles. **(2)** Across scientific tracks and analytical tasks, model strengths shift substantially, showing that success in one setting does not reliably generalize to another. **(3)** At the decision level, recognizing a scientific limitation does not guarantee that it is carried into subsequent handling or final conclusions. **(4)** At the hypothesis level, models differ substantially in whether they propose hypotheses and in the novelty of those they produce; among 94 submitted hypotheses, 46.8% receive no novelty credit under our rubric. Models also exhibit distinct working styles in critique response and computational expenditure, with resource-use rankings substantially more stable across tracks than rankings by evidence judgment. Together, these results identify where agents may support supervised scientific work and where continued human oversight remains necessary.

2 BENCHMARK DESIGN AND SCIENTIFIC VALIDITY

Scientific coverage. DISCERN covers life-science domains including genetics, transcriptomics, regulatory genomics, epigenomics, and protein biophysics (Figure 2B). Specifically, we design tracks based on genome-wide association studies, clinical variant interpretation, bulk RNA sequencing, single-cell RNA sequencing, CRISPR perturbation screening, enhancer activity, chromatin regulation, and protein stability and structure. Together, these tracks span biological scales from DNA-

level genetic variation to effects on genes, cells, regulatory elements, and proteins, with data ranging from association statistics and variant annotations to bulk and single-cell gene expression profiles, perturbation screens, enhancer and chromatin measurements, and protein stability and structural data. Across these tracks, DISCERN contains 79 L1 tasks, 102 L2 tasks across 24 analysis families, and 22 L3 dossiers (Figure 2C). Complete task inventories, data sources, and construction details are provided in Appendix E; the physics extension is evaluated separately in Appendix F.

L1 and L2 Level-specific challenges. L1 pairs real datasets and their corresponding metadata with controlled integrity challenges or matched clean controls under identical instructions. Challenges are designed around track-specific data representations, file formats, metadata conventions, and quality-control requirements, such as VCFs for genetic variants and count matrices for RNA sequencing. Corresponding L1 challenges include genome-build mismatches and ambient RNA contamination. Using the same underlying track datasets as L1, supplemented with analysis-specific reference data and tools, L2 constructs analysis tasks under clean or perturbed conditions, including data confounds, tool- or procedure-related failure modes, and statistical pitfalls. In the histone track, the agent must identify genomic regions enriched for H3K27me3 from ChIP-seq data. We introduce a tool-related failure mode in which the default narrow-peak setting is inappropriate for this broad histone mark. The software still runs successfully and returns a complete-looking result, but it fragments broad domains and underestimates the total marked region.

L3 evidence and counterfactual construction. L3 instead receives an evidence dossier derived from reference analyses of the same track data used to construct L1 and L2 with relevant methodological context, linking data integrity, analysis, and hypothesis generation without passing the evaluated agent’s own outputs between levels. For counterfactual dossiers, the agent’s scientific expectation is elicited before the evidence is shown, and the dossier is eligible for that agent’s evaluation only when the supplied evidence conflicts with that stated prior. Native counterfactuals use unexpected relationships already present in the analysis, whereas injected counterfactuals deliberately alter a well-characterized relationship and regenerate internally consistent supporting statistics. The exact injected result is constructed, while the type of scientific behavior that makes it plausible is grounded in documented phenomena rather than an arbitrary contradiction. For example, because TWAS effect directions can have substantial false-sign rates (Gerlach et al., 2025), we alter the direction or magnitude of a well-characterized gene–trait association observed in the corresponding L2 results and regenerate consistent effect estimates and supporting statistics.

Task environment and recorded observations. Each task is isolated from gold answers and sibling worlds. Within an analysis family, all worlds receive the same prompt and structured reporting schema, so the prompt does not reveal whether a problem is present. Across all three levels, agents access task-specific inputs through a sealed Python sandbox containing the permitted data, documentation, and domain tools, with no network access. The harness records executed code cells, tool calls, the final report, and token usage.

3 EVALUATION AND PROFILE CONSTRUCTION

Evaluation protocol and score aggregation. All eight models are evaluated on the same task set using shared, human-designed grading criteria. Agents operate in isolated task environments without access to grading answers or information from other tasks. Within each track, L1 and L3 scores are the mean of finalized eligible task scores. For L2, because analysis families contain different numbers of tasks, we first average task scores within each family and then average the family scores, so larger families do not receive greater weight. An independent LLM verifier cross-checks deterministic grading for L1 and L2, and we adjudicate disagreements before assigning final scores. Each track composite is the geometric mean of its L1–L3 scores to capture complementary requirements for reliable scientific work, so strong performance at one level cannot fully compensate for weakness at another. The Overall Score is the mean of the eight track composites, giving each track equal weight. Full protocol and scoring definitions appear in Appendices I and J.

Data-integrity scoring. L1 assigns each agent a quantitative task score based on its code, execution output, and report. For tasks containing a detectable defect, such as corrupted data or inconsistent metadata, credit increases from recognizing the problem to correctly diagnosing it and taking

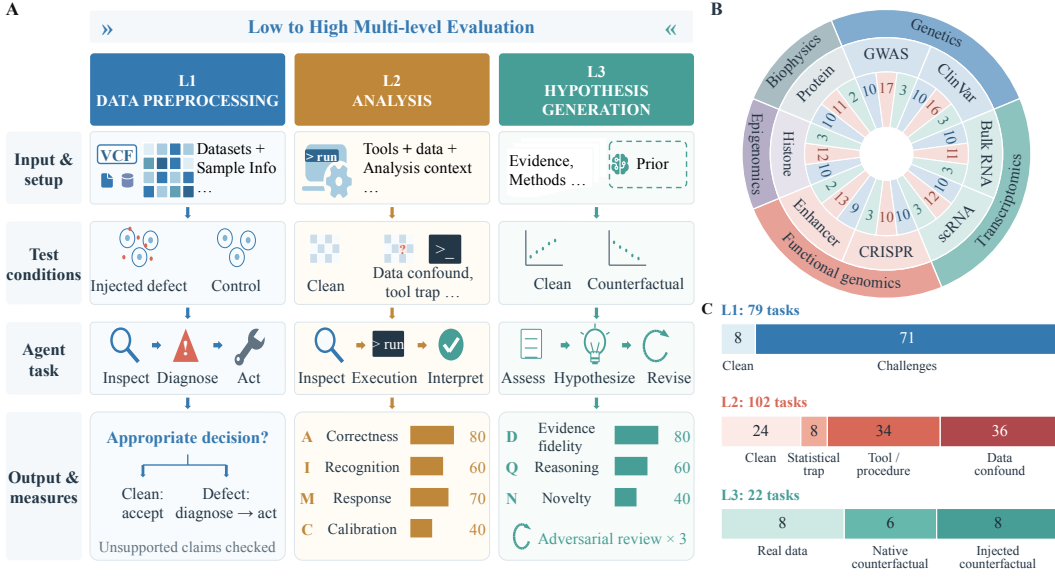


Figure 2: DISCERN workflow, scientific coverage, and task composition. (A) Each independently instantiated level follows four rows: inputs, test conditions, agent actions, and assessed outputs. Miniature data and score bars are illustrative only. (B) Scientific coverage. Three rings show task counts, eight tracks, and five scientific areas, from inside outward. Each track's inner ring is split into L1/L2/L3, using the stage colors shown in C. (C) Task composition. Segment widths show within-level task proportions; numbers give counts. Native and injected L3 counterfactuals use existing and altered evidence patterns, respectively. Data sources and complete inventories: Appendix E.

an appropriate action, for example by recovering the affected data. Partial diagnoses or remedies receive partial credit. For matched clean controls, full credit requires correctly accepting the data, while unsupported alarms, rejection, or fabricated defects lose credit. When the available evidence is insufficient to determine whether a suspected problem is present, credit is given for recognizing that limitation rather than asserting an unsupported diagnosis. We further summarize L1 responses using measures of assessment correctness, issue recognition, methodological response, result calibration, and fabrication or unsupported claims.

Analysis-verification scoring. L2 uses four shared grading dimensions across analysis families: analytical correctness (A) measures whether the computed results are correct; issue recognition (I) measures whether the relevant problem is identified; methodological response (M) measures whether an appropriate method is used to address it; and result calibration (C) measures whether the conclusions reflect the reliability and limits of the analysis. The L2 task score is the average of these four dimensions, while we tailor rubric criteria to reflect the scientific objective, expected results, and methodological requirements of each analysis.

Hypothesis scoring and adversarial review. L3 evaluates three dimensions: evidence fidelity (D), scientific reasoning (Q), and novelty (N). D measures whether the proposed claim is supported by the supplied evidence; for example, observed association should not be presented as evidence of causation. Q captures prior awareness, causal calibration, response to critique, scientific humility, and citation integrity. An agent that acknowledges uncertainty when evidence conflicts with prior expectations and revises its interpretation when criticism identifies a valid limitation receives more credit than one that ignores the conflict, overstates certainty, or persists with an unsupported interpretation. N measures whether the response adds a scientifically substantive idea that is not obvious from the supplied results, and whether the proposed next experiment can test the hypothesis and distinguish it from plausible alternative explanations. For example, restating that an unexpected gene-trait association may reflect a biological effect receives little novelty credit, whereas proposing a specific mechanism for the unexpected pattern and an experiment that distinguishes that mechanism from an analytical artifact receives more credit.

Table 1: Life science leaderboard. Scores use 0–100; L1–L3 report mean $\pm$ standard deviation (SD) across tracks. Tokens are in millions; Q/Mtok: quality points per million tokens; \$/task: USD per assigned task; \$/q.pt.: USD per quality point; Calls: recorded agent API calls. Definitions: Section 3. Blue/coral/teal mark first/second/third. Higher scores and Q/Mtok, but lower tokens, costs, and calls, rank first. The italic Fable row is excluded from ranking.

Model	Overall	L1	L2	L3	Tokens	Q/Mtok	\$/task	\$/q.pt.	Calls
GPT-6 Astra	78.1	86.1 $\pm$ 5.5	76.8 $\pm$ 7.7	72.8 $\pm$ 8.7	9.86	7.92	0.69	0.0088	1369
GPT-5.6-sol	75.8	80.1 $\pm$ 10.7	78.9 $\pm$ 8.9	70.0 $\pm$ 9.7	14.06	5.39	0.38	0.0051	1552
Claude Opus 5	75.6	85.3 $\pm$ 9.7	81.7 $\pm$ 12.7	63.0 $\pm$ 13.5	42.24	1.79	1.34	0.0177	2451
Fable $\rightarrow$ Opus	74.9	81.9 $\pm$ 8.4	79.1 $\pm$ 10.7	65.9 $\pm$ 14.4	25.77	2.91	1.17	0.0156	2032
Kimi K3	73.2	77.7 $\pm$ 10.6	75.4 $\pm$ 14.5	69.1 $\pm$ 11.8	34.04	2.15	0.65	0.0089	2671
GLM-5.3	70.5	83.5 $\pm$ 8.2	65.5 $\pm$ 28.4	71.1 $\pm$ 13.9	75.21	0.94	0.58	0.0083	3756
GLM-5.2	67.8	73.0 $\pm$ 10.3	70.4 $\pm$ 19.9	63.1 $\pm$ 9.5	57.16	1.19	0.37	0.0055	3761
DeepSeek V4 Pro	62.4	70.9 $\pm$ 9.4	62.1 $\pm$ 19.4	58.4 $\pm$ 13.1	59.30	1.05	0.44	0.0070	3365
GPT-4.1	39.8	36.4 $\pm$ 11.4	36.0 $\pm$ 8.1	51.0 $\pm$ 13.7	2.88	13.81	0.05	0.0012	747
<i>Claude Fable 5</i>	—	83.3 $\pm$ 8.5 [63.3%]	83.8 $\pm$ 6.4 [56.9%]	76.1 [9.1%]	8.55	—	0.59	—	948

Each L3 response undergoes three rounds of adversarial review by GLM-4.6 (Z.ai, 2025), Gemini 3.1 Flash-Lite Preview (Google, 2026), and DeepSeek Reasoner (DeepSeek, 2026a). Their critiques and the agent’s rebuttals provide supplementary evidence to the LLM judge (Zheng et al., 2023), which applies the grading rubric rather than reviewer votes. Authors review and validate each L3 response before averaging **D**, **Q**, and **N** as the L3 task final score for submitted hypotheses. When an agent gives a justified reason not to propose a hypothesis, we instead score its evidence use and reasoning under a separate partial-credit rule; **N** is not scored, and hypothesis commitment is reported separately from hypothesis quality. Only prior-eligible counterfactual tasks, as defined in Section 2, contribute to the L3 score aggregation.

Resource efficiency. We measure recorded input and output tokens, API calls, and estimated monetary cost across the 203-task suite. Token efficiency is calculated as the Overall Score per million tokens. Cost per task is the total estimated monetary cost in USD divided by the number of assigned tasks, and cost-effectiveness is the per-task cost divided by the Overall Score, giving cost per quality point. API-call counts provide a coarse measure of interaction frequency rather than complete analysis–feedback cycles. All resource measures exclude grading, verification, and reviewer activity. Accounting details and alternative aggregation analyses appear in Appendices L and A. Standalone Fable is excluded from ranking because of incomplete coverage. The Fable $\rightarrow$ Opus row in Table 1 is constructed after evaluation by replacing tasks that Fable refused with the corresponding recorded Opus results, simulating a deployment scenario in which refused tasks are automatically passed from Fable to Opus during use (Appendix B).

4 RESULTS AND ANALYSIS

We first compare overall model performance, then examine whether strengths generalize across scientific tasks, and finally characterize distinct scientific working styles across models.

4.1 OVERALL MODEL PERFORMANCE

Overall performance varies across models and scientific capabilities. GPT-6 Astra achieves the highest aggregate DISCERN score (78.1), indicating the strongest overall performance across data preprocessing (L1), analysis verification (L2), and hypothesis generation (L3) over the eight scientific tracks, followed by GPT-5.6 Sol (75.8) and Claude Opus 5 (75.6; Table 1). This does not reflect uniform dominance. Astra leads in L1 (86.1) and in L3 (72.8), whereas Opus leads in L2 (81.7). Sol ranks second in Overall Score without leading in any one component, reflecting comparatively balanced performance across all three. Models with publicly released weights (open-weight models)

are also competitive with proprietary closed models on specific capabilities. GLM-5.3 outperforms Sol in L1 (83.5 versus 80.1) and ranks second in L3 (71.1), while Kimi nearly matches Astra in L2 (75.4 versus 76.8) and exceeds Opus in L3 (69.1 versus 63.0). Thus, although closed models achieve the highest observed score in each component, open-weight models can outperform or closely approach individual closed models in specific parts of scientific work. Overall Scores span 39.8–78.1 across models, while perfect-score rates are 60.8% in L1, 34.2% in L2, and 0.6% in L3 according to their respective rubrics. Consistent with this pattern, L3 is the lowest-scoring component in 46.9% of model-track combinations, compared with 34.4% for L2 and 18.8% for L1, identifying hypothesis generation and revision as the most frequent performance bottleneck according to DISCERN.

Comparable overall performance can require very different resources. Sol and Opus achieve nearly identical Overall Scores (75.8 and 75.6), but Sol uses about one-third as many tokens (14.06M versus 42.24M) and 29% of Opus’s estimated cost per assigned task (\$0.38 versus \$1.34; Table 1). Thus, similar aggregate benchmark performance can be reached with substantially different computational expenditure. GPT-4.1 has the highest measured token efficiency but also the lowest Overall Score (39.8) and does not execute code on 20 of 181 L1/L2 tasks. These tasks remain included in benchmark scoring and token accounting rather than being treated as missing; however, returning without performing the expected computation can reduce token use and task performance, contributing to GPT-4.1’s high score-per-token ratio (Appendix L.4). Token efficiency should therefore be interpreted alongside execution completeness and scientific performance rather than as a standalone measure of agent quality.

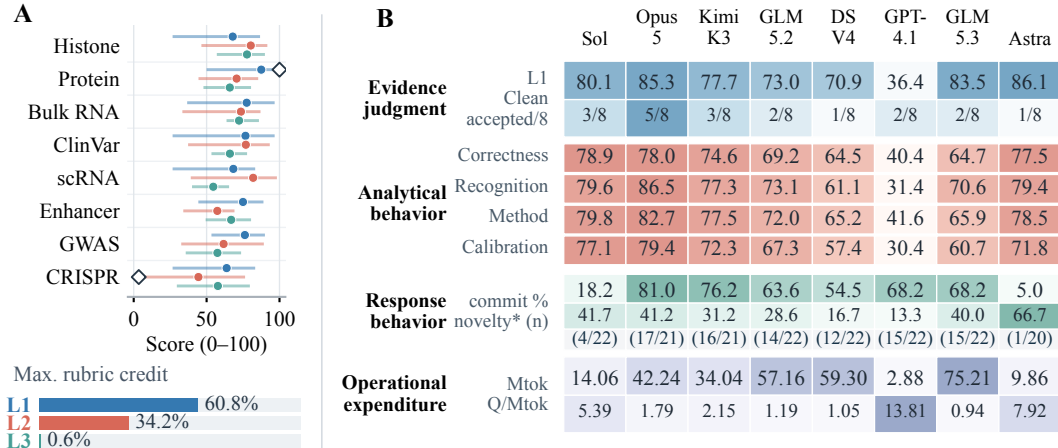


Figure 3: Performance variation and working-style profiles. (A) Level-specific scores by track: ●, mean across eight models; lines, model min–max; ◇, GLM-5.3 at 100 and 3.5. Bars show perfect-score fractions under each stage’s rubric. (B) Working-style overview. Each metric is shaded from its lowest (light) to highest (dark) value across models; colors are comparable only within a row and do not imply better performance. Counts n are unshaded. Clean accepted/8: number of clean controls correctly accepted out of eight; novelty*: conditional mean (0–100); n : committed/eligible. Mtok: million tokens; Q/Mtok: Overall/Mtok. Appendix A.

4.2 SCIENTIFIC STRENGTHS VARY ACROSS TASKS

Performance varies across levels and scientific tracks. As discussed above, no model leads in all three levels within any track. We also observe that model strengths shift across scientific domains. The most extreme example is GLM-5.3, which achieves the only track-level score of 100 for data preprocessing in the protein stability and structure track (Protein L1), but also the lowest track-level score, 3.5, for analysis verification in CRISPR perturbation screening (CRISPR L2), while achieving an Overall Score of 70.5 (Figure 3A). More broadly, the relative strengths of models differ across tracks, showing that aggregate performance can mask substantial variation both across stages of scientific work and across scientific domains.

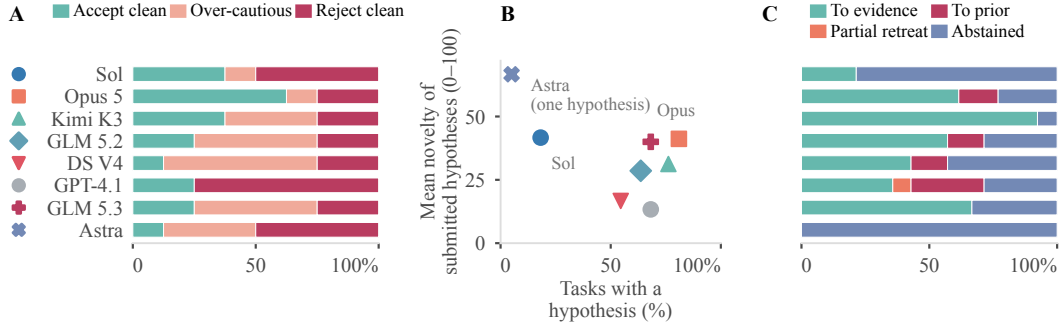


Figure 4: Scientific working-style contrasts. (A) Decisions on eight L1 clean controls per model. (B) Hypothesis production versus novelty: each point is one model, showing the percentage of eligible L3 tasks receiving a hypothesis and mean novelty credit among submitted hypotheses (0–100). Astra’s mean rests on one hypothesis. Marker colors and shapes identify models. (C) Responses to evidence conflicting with stated priors; abstention differs from return to the prior; excludes four outcomes that failed the pre-task check for conflict with the model’s stated prior.

Analytical strengths are task-specific. Performance also varies substantially across analysis families within the same scientific domain. In statistical genetics, mean scores range from 77.4 for locating disease-associated regions to 32.8 for linking predicted gene expression to disease. In regulatory genomics, they range from 76.9 for ranking sequence activity to 20.4 for assessing whether that activity is conserved across species (Appendix Figure 16). These differences show that competence on one analytical question does not automatically generalize to another, even within a related scientific setting. Reliable analysis therefore requires verifying the assumptions, reference information, and interpretation specific to each analytical task.

Recognition and calibration can diverge. Among 425 challenged L2 evaluations receiving full credit for issue recognition, 126 (29.6%) receive only half credit for result calibration under the analysis-specific rubric. This proportion ranges from 19.4% for Opus to 37.8% for DeepSeek, showing that models differ in how consistently a recognized problem is carried through to subsequent handling and interpretation. Recognizing a limitation is therefore not sufficient unless that limitation also shapes the resulting analysis and scientific claim (Appendix J).

4.3 DISTINCT SCIENTIFIC STYLES EMERGE ACROSS MODELS

Differences across models extend beyond overall performance to how they assess evidence, decide whether to propose hypotheses, respond to criticism, and expend computational resources. We refer to these observed patterns as scientific working styles rather than fixed model traits.

Caution can be calibrated or misplaced. Cautious behavior appears in both justified abstention and unsupported rejection. Across models, only 19 of 64 L1 clean-control evaluations (29.7%) correctly accept the sound data, and Astra accepts only 12.5% of its clean controls compared with 62.5% for Opus (Figure 4A). A different form of caution appears when supplied evidence conflicts with the agent’s stated prior. Kimi proceeds with the supplied evidence and submits a hypothesis in 92.3% of eligible counterfactual cases, whereas Astra abstains in every such case (Figure 4C). In 14 of Astra’s 19 abstentions across eligible L3 tasks, it provides an explicit evidentiary rationale for withholding a hypothesis, which receives partial credit under our rubric. However, its universal abstention on eligible counterfactual cases does not demonstrate balanced calibration between proposing and withholding a hypothesis. Together, these results show that caution itself is not sufficient: the key distinction is whether restraint is appropriately matched to the available evidence.

Models differ in how selectively they generate hypotheses. Hypothesis production and novelty are distinct behaviors. Among 94 submitted hypotheses, 46.8% receive no novelty credit, while 30.9% receive at least two-thirds of the maximum. Sol and Opus illustrate this distinction: their mean conditional novelty scores are nearly identical (41.7 and 41.2), but they submit hypotheses on 18.2% and 81.0% of eligible tasks, respectively (Figure 4B). In 14 of Astra’s 19 abstentions across

eligible L3 tasks, it provides an explicit evidentiary rationale for withholding a hypothesis, which receives partial credit under our rubric. However, abstaining in all eligible counterfactual cases does not show that Astra reliably distinguishes when the evidence supports proposing a hypothesis from when it supports withholding one. Together, these results show that caution alone is not sufficient, and restraint must be appropriate to the available evidence.

Computational intensity is more stable than scientific judgment. Models differ substantially in how much computation they expend. GLM-5.3 uses 75.21M tokens compared with 57.16M for GLM-5.2 despite nearly identical recorded API-call counts (3,756 versus 3,761), while their Overall Scores are 70.5 and 67.8, respectively. More broadly, model rankings by token use and code execution remain highly consistent across tracks ($W = 0.89\text{--}0.95$), whereas rankings by evidence judgment and hypothesis behavior are much less stable ($W = 0.09\text{--}0.57$; Figure 5; Appendix A). Resource-use patterns therefore behave more like consistent model-specific working styles across scientific settings, while scientific judgment depends more strongly on the task context. Greater computational expenditure, however, does not by itself establish better scientific performance.

Receptiveness to critique differs from revision quality. Models also vary in how readily they explicitly accept reviewer objections. Each committed hypothesis undergoes three rounds of adversarial review, with reviewer critiques and the agent’s rebuttals or revisions used as additional evidence for the final assessment. DeepSeek concedes at least one point in 41.7% of its committed outcomes, compared with 5.9% for Opus, but concession frequency alone does not establish whether the resulting revision is better supported. In the DUSP4 case, reviewers note that both chromatin signals were measured around the same gene, but one measurement covers a much broader region, so the data do not establish that the signals overlap at its promoter. DeepSeek responds by narrowing its mechanistic claim and proposing experiments that test signal overlap and the effect of reducing repression, receiving a final task score of 95.7 (Appendix H). This example shows that useful revision depends on whether a model responds to a valid evidentiary limitation, rather than how often it concedes to critique.

Selective participation also depends on scientific context. Fable is analyzed separately because its benchmark coverage is incomplete. It answers 108 of 181 L1/L2 tasks (59.7%) and 2 of 22 L3 tasks (9.1%). If refusal were primarily driven by task difficulty, we would expect refusal rates to increase with our empirical difficulty proxy; instead, L1/L2 refusal rates are non-monotonic across difficulty quintiles and cluster strongly by scientific track and analysis family (Appendix B). Notably, four refused tasks are solved perfectly by all eight comparison models. Measured difficulty therefore does not adequately explain Fable’s selective participation, although these results do not identify the underlying safeguard mechanism. These refusals occur in routine analyses of public data rather than intentionally hazardous procedures, making participation difficult to anticipate from task content or measured difficulty alone (Appendix B). Scientific-agent reliability therefore requires both correct execution when an agent engages and predictable coverage across the workflow.

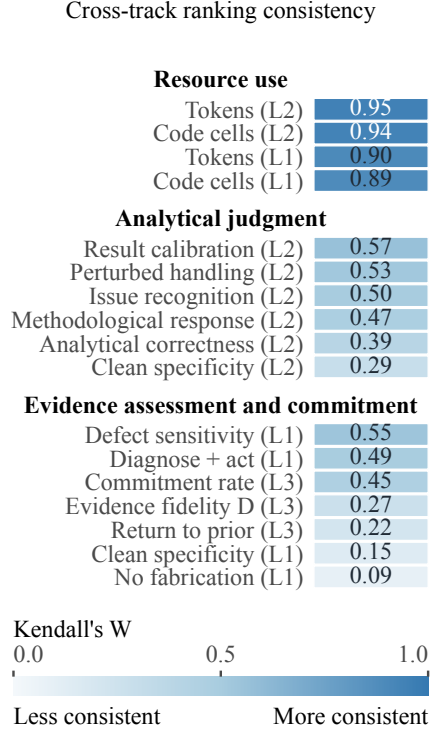


Figure 5: Cross-track ranking consistency. Each cell shows Kendall’s W for one metric across eight tracks. Darker blue and higher values indicate more consistent model rankings, not correlations between metrics (Appendix A).

5 RELATED WORK

Scientific data analysis benchmarks. Scientific agents increasingly combine reasoning, tool use, and code execution to carry out research workflows (Huang et al., 2025; Lu et al., 2024). Benchmarks have correspondingly moved beyond answer generation toward executable analysis and multistage scientific inference (Li et al., 2026; Mitchener et al., 2025; Qu et al., 2026). More recent evaluations extend to study-scale workflows, measuring the ability to transform raw data into research outputs alongside resource use (Koch et al., 2026). DISCERN complements these efforts by testing scientific judgment under controlled data defects, confounds, and tool traps that can produce plausible but invalid results. Clean controls test whether agents incorrectly reject valid data. We also assess whether agents detect problems, take appropriate action, and reflect any remaining limitations in their final conclusions.

Hypothesis generation and discovery benchmarks. Agents also support hypothesis generation through iterative critique and refinement (Gottweis et al., 2026). Existing evaluations examine complementary aspects of discovery, from recovering reference relationships in data and conducting interactive experiments (Majumder et al., 2025; Jansen et al., 2024) to reasoning under altered scientific laws and generating research assessed through automated review (Huang et al., 2026; Lu et al., 2024). These evaluations address related but distinct aspects of scientific reasoning. DISCERN focuses on whether agents ground hypotheses in supplied evidence, including evidence that challenges established expectations, and revise their claims appropriately under adversarial review. It separately measures hypothesis commitment, evidence fidelity, scientific reasoning, and novelty, distinguishing the willingness to propose a hypothesis from the quality of the resulting proposal.

6 DISCUSSION, LIMITATIONS, AND CONCLUSION

Scientific capability is compositional and context-dependent. DISCERN shows that scientific-agent capability is not well summarized by a single aggregate score. Strong performance in one part of the research workflow can coexist with weakness in another analysis, scientific track, or decision point, while models with similar Overall Scores can reach them through different working styles. Our separate calorimetry and nuclear-decay extension further suggests that these patterns need not persist across domains: mean L2 performance is lower than in life science tracks despite similar L1 and L3 averages, and Astra follows conflicting evidence in the nuclear-decay tasks despite its tendency to abstain in life-science L3 (Appendix F). These observations argue for evaluating scientific agents at the level of specific tasks and decisions rather than treating overall performance as evidence of uniform scientific reliability.

Limitations. Real public data and controlled challenges preserve important features of scientific analysis, but bounded tasks and independently instantiated levels do not establish autonomous end-to-end research. Generalization and model comparisons remain constrained by limited domain coverage, possible model familiarity with public datasets or published scientific phenomena, dependence among related tasks that share datasets or analysis families, and residual grading uncertainty. The two physics tracks provide an initial cross-domain comparison rather than broad validation across physics, and the observed model differences do not isolate effects of training or architecture. We also do not establish whether proposed hypotheses provide useful experimental follow-up or whether agents integrate effectively into human scientific workflows. Broader scientific coverage and direct evaluation of human-agent collaboration are therefore needed.

Conclusion. Overall, current agents demonstrate useful components of scientific work and can support bounded research tasks, but reliable scientific judgment across data preprocessing, analysis verification, and hypothesis generation remains uneven. Progress requires improving not only what agents can execute, but also what evidence they trust, what conclusions they draw, when they engage, and how efficiently they perform this work.

AI USE STATEMENT

Generative AI is both the object of study and an aid in benchmark development. The core results use eight evaluated models; the Fable analysis and fallback replay are discussed separately. LLMs are responsible for scoring the dimensions of the L3 level and verifying deterministic grades, with human review and adjudication as described in Section 3. The authors also used AI assistance for code, analysis implementation, figures, and prose. They retain responsibility for checking quantitative claims, citations, and all AI-assisted content against the recorded evidence.

ETHICS STATEMENT

The data substrates are public and subject to their source licenses. Human-derived data are aggregated at the summary level or have been previously de-identified by their corresponding study teams. We performed controlled perturbations of data to establish counterfactual artifacts for evaluation; these perturbations do not represent scientifically or clinically meaningful phenomena. No generated hypothesis is experimentally confirmed or established as novel according to the literature. Our work should not be used to guide any clinical decision making.

REPRODUCIBILITY STATEMENT

The appendix records inventory, protocols, grading, and accounting. The anonymous benchmark release is available at <https://huggingface.co/datasets/discern-bench-anon/discern-benchmark>. It provides prepared task inputs, manifests, builders, prompts, sandbox definitions, grading criteria, recorded outputs, adjudications, and scripts for regenerating Table 1 and Figures 3–5. Source attribution, external dependency downloads, and reproduction scope are documented in the release. An `EVALUATE_MODEL.md` guide explains how to run additional models on the prepared tasks, including prior probes, grading, and human adjudication.

REFERENCES

- Urkiye Akar Tarım, Orhan Gürler, and Sezai Yalçın. A quick method to calculate NaI(Tl) detector efficiency depending on gamma ray energy and source-to-detector distance. *Celal Bayar University Journal of Science*, 14(2):195–199, 2018. doi: 10.18466/cbayarfbe.396704.
- Anthropic. Claude Fable 5 model overview. <https://platform.claude.com/docs/en/models/fable-5/overview>, 2026a. Accessed September 25, 2026.
- Anthropic. Claude Opus 5 model overview. <https://platform.claude.com/docs/en/models/opus-5/overview>, 2026b. Accessed September 25, 2026.
- Prabhu S. Arunachalam, Florian Wimmers, Chris Ka Pun Mok, et al. Systems biological assessment of immunity to mild versus severe COVID-19 infection in humans. *Science*, 369(6508):1210–1220, 2020. doi: 10.1126/science.abc6261.
- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- Bradley E. Bernstein, Tarjei S. Mikkelsen, Xiaohui Xie, Michael Kamal, Dana J. Huebert, James Cuff, Ben Fry, Alex Meissner, Marius Wernig, Kathrin Plath, Rudolf Jaenisch, Alexandre Wagschal, Robert Feil, Stuart L. Schreiber, and Eric S. Lander. A bivalent chromatin structure marks key developmental genes in embryonic stem cells. *Cell*, 125(2):315–326, 2006. doi: 10.1016/j.cell.2006.02.041.
- Thierry Bertomeu, Jasmin Coulombe-Huntington, Andrew Chatr-Aryamontri, Karine G. Bourdages, Etienne Coyaude, Brian Raught, Yu Xia, and Mike Tyers. A high-resolution genome-wide CRISPR/Cas9 viability screen reveals structural features and contextual diversity of the human cell-essential proteome. *Molecular and Cellular Biology*, 38(1):e00302–17, 2018. doi: 10.1128/MCB.00302-17.

- Mira R. Bissengaliyeva, Daniil B. Gogol, Shynar T. Taymasova, and Nuraly S. Bekturganov. Measurement of heat capacity by adiabatic calorimetry and calculation of thermodynamic functions of standard substances: Copper, benzoic acid, and heptane (for calibration of an adiabatic calorimeter). *Journal of Chemical & Engineering Data*, 56(2):195–204, 2011. doi: 10.1021/jc100658y.
- Harriet Corvol, Scott M. Blackman, Pierre-Yves Boëlle, et al. Genome-wide association meta-analysis identifies five modifier loci of lung disease severity in cystic fibrosis. *Nature Communications*, 6:8382, 2015. doi: 10.1038/ncomms9382.
- DeepSeek. DeepSeek API Change Log. <https://api-docs.deepseek.com/updates/>, 2026a. Accessed September 25, 2026.
- DeepSeek. DeepSeek-V4-Pro model card. <https://huggingface.co/deepseek-ai/DeepSeek-V4-Pro>, 2026b. Accessed September 25, 2026.
- Stephen J. Fleming, Mark D. Chaffin, Alessandro Arduini, et al. Unsupervised removal of systematic background noise from droplet-based single-cell experiments using CellBender. *Nature Methods*, 20(9):1323–1335, 2023. doi: 10.1038/s41592-023-01943-7.
- Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. *arXiv preprint arXiv:2502.05171*, 2025. doi: 10.48550/arXiv.2502.05171.
- Peter A. Gerlach, Nikhil Milind, Jeffrey P. Spence, and Jonathan K. Pritchard. High false sign rates in transcriptome-wide association studies. *bioRxiv*, 2025. doi: 10.64898/2025.12.19.695550.
- Braden Goddard, George W. Hitt, Dorian Bridi, Abdel F. Isakovic, Reyad El-Khazali, and Ayman Abulail. Data from multi year and multi radiation detector measurements of different radiation sources with environmental data. *Data in Brief*, 31:105828, 2020. doi: 10.1016/j.dib.2020.105828.
- Google. Gemini 3.1 Flash-Lite Preview. <https://ai.google.dev/gemini-api/docs/models/gemini-3.1-flash-lite-preview>, 2026. Accessed September 25, 2026.
- Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Petar Sirkovic, Artiom Myaskovsky, Grzegorz Glowaty, Felix Weissenberger, Alessio Orlandi, Dan Popovici, et al. Accelerating scientific discovery with co-scientist. *Nature*, 655(8122):487–496, 2026. doi: 10.1038/s41586-026-10644-y.
- Arkabrata Gupta, Chirashree Lahiri, and S. Sarkar. Allowed β^- decay of bare atoms with $a \approx 60$ –80 in stellar environments. *Physical Review C*, 108:015805, 2023. doi: 10.1103/PhysRevC.108.015805.
- Jérôme Hadjadj et al. Impaired type I interferon activity and inflammatory responses in severe COVID-19 patients. *Science*, 369(6504):718–724, 2020. doi: 10.1126/science.abc6027.
- Traver Hart and Jason Moffat. BAGEL: a computational framework for identifying essential genes from pooled library screens. *BMC Bioinformatics*, 17:164, 2016. doi: 10.1186/s12859-016-1015-8.
- Traver Hart, Amy Hin Yan Tong, Katie Chan, et al. Evaluation and design of genome-wide CRISPR/SpCas9 knockout screens. *G3: Genes, Genomes, Genetics*, 7(8):2719–2727, 2017. doi: 10.1534/g3.117.041277.
- Kexin Huang, Serena Zhang, Hanchen Wang, Yuanhao Qu, Yingzhou Lu, Yusuf Roohani, Ryan Li, Lin Qiu, Gavin Li, Junze Zhang, Di Yin, Shruti Marwaha, Jennefer N. Carter, Xin Zhou, Matthew Wheeler, Jonathan A. Bernstein, Mengdi Wang, Peng He, Jingtian Zhou, Michael Snyder, Le Cong, Aviv Regev, and Jure Leskovec. Biomni: A general-purpose biomedical ai agent. *bioRxiv*, 2025. doi: 10.1101/2025.05.30.656746. URL <https://www.biorxiv.org/content/early/2025/06/02/2025.05.30.656746>.

- Yiming Huang, Jianwen Luo, Yan Yu, Yitong Zhang, Fangyu Lei, Yifan Wei, Shizhu He, Lifu Huang, Xiao Liu, Jun Zhao, and Kang Liu. DA-Code: Agent data science code generation benchmark for large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 13487–13521, 2024. doi: 10.18653/v1/2024.emnlp-main.748.
- Yiming Huang, Ziche Liu, Zhuohang Wu, Yiqian Wang, Junxia Cui, Xinkai Zou, Linjun Mao, Nan Huang, Naicheng Yu, Kaijie Zhu, et al. Can llms discover scientific laws in real and parallel worlds? *arXiv preprint arXiv:2609.01552*, 2026.
- Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Øyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024. doi: 10.52202/079017-0324.
- Konrad J. Karczewski, Laurent C. Francioli, Grace Tiao, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581(7809):434–443, 2020. doi: 10.1038/s41586-020-2308-7.
- M. G. Kendall and B. Babington Smith. The problem of m rankings. *The Annals of Mathematical Statistics*, 10(3):275–287, 1939. doi: 10.1214/aoms/1177732186.
- Jason C. Klein, Aidan Keith, Vikram Agarwal, Timothy Durham, and Jay Shendure. Functional characterization of enhancer evolution in the primate lineage. *Genome Biology*, 19(1):99, 2018. doi: 10.1186/s13059-018-1473-6.
- Zane Koch, Asmamaw T. Wassie, Javier Valdes-Aleman, Jason Lee, Michaela M. Hinks, Samuel G. Rodrigues, Andrew D. White, and Jon M. Laurent. BixBench3: Benchmarking AI agents on research-study-scale computational biology tasks. *arXiv preprint arXiv:2608.25286*, 2026. doi: 10.48550/arXiv.2608.25286.
- Melissa J. Landrum, Jennifer M. Lee, Mark Benson, et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.
- Brian E. Lang, Juliana Boerio-Goates, and Brian F. Woodfield. Design and construction of an adiabatic calorimeter for samples of less than 1 cm³ in the temperature range $t = 15$ k to $t = 350$ k. *The Journal of Chemical Thermodynamics*, 38(12):1655–1663, 2006. doi: 10.1016/j.jct.2006.03.016.
- Jeremiah H. Li, Suyash Shringarpure, Edmund Wong, and Andrew J. Ho. GeneBench-Pro: Evaluating multistage statistical reasoning in genomics, quantitative biology, and translational biomedicine. *bioRxiv*, 2026. doi: 10.64898/2026.06.29.735386. URL <https://www.biorxiv.org/content/10.64898/2026.06.29.735386v4>.
- Ronghui Li, Bella Banjanin, Rebekka K. Schneider, and Ivan G. Costa. Detection of cell markers from single cell RNA-seq with sc2marker. *BMC Bioinformatics*, 23(1):276, 2022. doi: 10.1186/s12859-022-04817-5.
- Wei Li, Han Xu, Tengfei Xiao, et al. MAGECK enables robust identification of essential genes from genome-scale CRISPR/Cas9 knockout screens. *Genome Biology*, 15:554, 2014. doi: 10.1186/s13059-014-0554-4.
- Zeming Lin, Halil Akin, Roshan Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.
- Michael I. Love, Wolfgang Huber, and Simon Anders. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15:550, 2014. doi: 10.1186/s13059-014-0550-8.
- Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

- Bodhisattwa Prasad Majumder, Harshit Surana, Dhruv Agarwal, Bhavana Dalvi Mishra, Abhijeetsingh Meena, Aryan Prakhar, Tirth Vora, Tushar Khot, Ashish Sabharwal, and Peter Clark. DiscoveryBench: Towards data-driven discovery with large language models. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/0d70af566e69f1dfb687791ecf955e28-Abstract-Conference.html.
- H. B. Mann and D. R. Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1):50–60, 1947. doi: 10.1214/aoms/1177730491.
- Arjun K. Manrai, Birgit H. Funke, Heidi L. Rehm, Morten S. Olesen, Bradley A. Maron, Peter Szolovits, David M. Margulies, Joseph Loscalzo, and Isaac S. Kohane. Genetic misdiagnoses and the potential for health disparities. *New England Journal of Medicine*, 375(7):655–665, 2016. doi: 10.1056/NEJMs1507092.
- Jacqueline Mersch, Nichole Brown, Sara Pirzadeh-Miller, Erin Mundt, Hannah C. Cox, Krystal Brown, Melissa Aston, Lisa Esterling, Susan Manley, and Theodora Ross. Prevalence of variant reclassification following hereditary cancer genetic testing. *JAMA*, 320(12):1266–1274, 2018. doi: 10.1001/jama.2018.13152.
- Ludovico Mitchener, Jon M. Laurent, Alex Andonian, Benjamin Tenmann, Siddharth Narayanan, Geemi P. Wellawatte, Andrew White, Lorenzo Sani, and Samuel G. Rodrigues. BixBench: A comprehensive benchmark for LLM-based agents in computational biology. *arXiv preprint arXiv:2503.00096*, 2025.
- Moonshot AI. Kimi K3 model card. <https://huggingface.co/moonshotai/Kimi-K3>, 2026. Accessed September 25, 2026.
- Marcus R. Munafò, Brian A. Nosek, Dorothy V. M. Bishop, Katherine S. Button, Christopher D. Chambers, Nathalie Percie du Sert, Uri Simonsohn, Eric-Jan Wagenmakers, Jennifer J. Ware, and John P. A. Ioannidis. A manifesto for reproducible science. *Nature Human Behaviour*, 1:0021, 2017. doi: 10.1038/s41562-016-0021.
- Boris Muzellec, Maria Teleńczuk, Vincent Cabeli, and Mathieu Andreux. PyDESeq2: a python package for bulk RNA-seq differential expression analysis. *Bioinformatics*, 39(9):btad547, 2023. doi: 10.1093/bioinformatics/btad547.
- OpenAI. GPT-6 Astra Model, 2026a. URL <https://developers.openai.com/api/docs/models/gpt-6-astra>. OpenAI API documentation. Accessed September 25, 2026.
- OpenAI. GPT-4.1 model documentation. <https://developers.openai.com/api/docs/models/gpt-4.1>, 2026b. Accessed September 25, 2026.
- OpenAI. GPT-5.6 Sol model documentation. <https://developers.openai.com/api/docs/models/gpt-5.6-sol>, 2026c. Accessed September 25, 2026.
- Roger D. Peng. Reproducible research in computational science. *Science*, 334(6060):1226–1227, 2011. doi: 10.1126/science.1213847.
- Yuanhao Qu, Yingzhou Lu, Xinming Tu, Serena Zhang, Tianwei She, Alexander Glenn Shaw, Jou-Ho Shih, Bingqing Zhao, Minjie Shen, Haochen Yang, Jielin Yan, Rongchuan Zhang, Xinze Wu, Tingting Li, Bin Zhou, Ning Wang, Adam Ma, Le Cong, Xiaobo Hu, Yuan Jiang, Jiayun Dong, Tao Peng, Jure Leskovec, and Kexin Huang. BiomniBench: Process-level evaluation of LLM agents for real-world biomedical research. *bioRxiv*, 2026. doi: 10.64898/2026.05.12.724604.
- Sue Richards, Nazneen Aziz, Sherri Bale, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*, 17(5):405–424, 2015. doi: 10.1038/gim.2015.30.
- J. Rupp and R. Birringer. Enhanced specific-heat-capacity (c_p) measurements (150–300 k) of nanometer-sized crystalline materials. *Physical Review B*, 36:7888–7890, 1987. doi: 10.1103/PhysRevB.36.7888.

- Zhi-Cheng Tan, Lan Wang, and Quan Shi. Study of heat capacity enhancement in some nanostructured materials. *Pure and Applied Chemistry*, 81(10):1871–1880, 2009. doi: 10.1351/PAC-CON-08-09-15.
- The ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature*, 489(7414):57–74, 2012. doi: 10.1038/nature11247.
- Kotaro Tsuboyama, Justas Dauparas, Jonathan Chen, Elodie Laine, Yasser Mohseni Behbahani, Jonathan J. Weinstein, Niall M. Mangan, Sergey Ovchinnikov, and Gabriel J. Rocklin. Mega-scale experimental analysis of protein folding stability in biology and design. *Nature*, 620:434–444, 2023. doi: 10.1038/s41586-023-06328-6.
- Matthew D. Young and Sam Behjati. SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *GigaScience*, 9(12):giaa151, 2020. doi: 10.1093/gigascience/giaa151.
- Z.ai. GLM-4.6 model card. <https://huggingface.co/zai-org/GLM-4.6>, 2025. Accessed September 25, 2026.
- Z.ai. GLM-5.2 model card. <https://huggingface.co/zai-org/GLM-5.2>, 2026a. Accessed September 25, 2026.
- Z.ai. GLM-5.3 Model Card. <https://huggingface.co/zai-org/GLM-5.3>, 2026b. Accessed September 10, 2026.
- Grace X. Y. Zheng, Jessica M. Terry, Phillip Belgrader, et al. Massively parallel digital transcriptional profiling of single cells. *Nature Communications*, 8:14049, 2017. doi: 10.1038/ncomms14049.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.
- Tianshi Zheng, Kelvin Kiu-Wai Tam, Newt Hue-Nam K. Nguyen, Baixuan Xu, Zhaowei Wang, Jiayang Cheng, Hong Ting Tsang, Weiqi Wang, Jiaxin Bai, Tianqing Fang, Yangqiu Song, Ginny Y. Wong, and Simon See. NewtonBench: Benchmarking generalizable scientific law discovery in LLM agents. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=Gk6umqW74m>.

A SCIENTIFIC WORKING-STYLE PROFILES AND ROBUSTNESS

A.1 FROM A SCIENTIFIC EXPECTATION TO A MEASURED BEHAVIOR

Table 2 makes the normative interpretation explicit. A measured behavior can suggest where oversight is needed without establishing that a model replaces a person or that its suggestions improve a scientist’s next experiment. No extra scientific-style composite is added.

Table 2: Scientific expectations, their benchmark proxies, and the corresponding proposed human check. The final column is an implication for supervised use, not a measured collaboration outcome.

Scientific expectation	Recorded measure	Proposed oversight
Do not discard sound evidence	L1 clean specificity, defect handling, diagnosis and action	Inspect exclusions and the stated data-fitness reason
Check what the method actually did	L2 analytical correctness, issue recognition, methodological response	Reproduce key outputs and audit assumptions
Say only what follows	L2 result calibration; L3 evidence fidelity and causal calibration	Separate association from mechanism
Use knowledge without surrendering to it	Prior-probe response and final evidence/prior state	Ask which evidence would change the claim
Make uncertainty and revisions visible	Citation integrity, humility, response to critique, fabrication	Preserve the artifact and revision trail
Propose a test that could prove you wrong	Commitment and conditional valid novelty	Assess feasibility, prior art, and discriminatory value
Use resources purposefully	Tokens, code cells, USD, quality and coverage	Compare useful completed work, not expenditure alone

A.2 DETAILED DIMENSIONAL PROFILES

Figure 6 expands the four-category overview in Figure 3. Its rows retain the individual grading dimensions and diagnostic populations, not five newly averaged scores.

Scientific working-style profiles

Scores and rates on 0–100; compare models within a row.

	GPT-5.6 sol	Claude Opus 5	Kimi K3	GLM 5.2	DeepSeek V4 Pro	GPT-4.1	GLM 5.3	GPT-6 Astra
Evidence judgment and handling								
Clean-data judgment	46	71	62	58	54	25	54	42
Challenge handling	79	84	82	73	77	46	87	91
No fabrication recorded	100	100	100	100	100	97	100	100
Analytical behavior								
Clean-analysis performance	90	84	86	79	76	48	72	83
Analytical correctness	79	78	75	69	65	40	65	77
Issue recognition	80	86	77	73	61	31	71	79
Methodological response	80	83	77	72	65	42	66	79
Result calibration	77	79	72	67	57	30	61	72
Response behavior								
Evidence fidelity*	100	61	88	81	64	58	80	100
Moves toward evidence CF	21	62	92	57	43	36	57	0
Causal calibration*	58	69	65	55	58	58	67	100
Hypothesis commitment	18	81	76	64	55	68	68	5
Hypotheses / eligible	4/22	17/21	16/21	14/22	12/22	15/22	15/22	1/20
Valid novelty*	42	41	31	29	17	13	40	67
Response to critique*	83	73	83	86	78	58	82	100
Acknowledges uncertainty*	92	90	100	86	83	64	84	100
Citation integrity*	100	88	83	69	78	82	89	100

* L3 quality is conditional on producing a hypothesis. Astra: one hypothesis only.

CF = eligible counterfactual dossiers. Abstention is not necessarily evidence rejection.

Clean-data judgment is rubric credit, not the clean-acceptance rate.

L2: equal-family, then equal-track averages. No fabrication: recorded checks, not proof.

No combined style score. Cross-track ranking stability is reported in the appendix.

Figure 6: Detailed scientific working-style profiles. Evidence judgment and handling, analytical behavior, and response behavior are shown by individual metric. Asterisks identify conditional L3 quality, with committed/eligible counts displayed. All values use 0–100, but rubric scores and rates describe different quantities. Astra’s conditional L3 quality rests on one hypothesis.

A.3 POPULATIONS, AGGREGATION, AND STABILITY

The profile includes all eight models and eight life-science tracks. The Fable-to-Opus replay is not an observed working style and is excluded. Direct Fable has selective coverage and is analyzed separately. L1 behavior summaries use the recorded state-level outcomes; they are diagnostic views rather than reconstructions of model–track adjudications. L2 dimensions use exactly the world-to-family-to-track hierarchy that produces the canonical leaderboard. Their mean reproduces L2 within published rounding. L3 subaxes are summarized among committed eligible hypotheses: GPT-5.6-sol 4/22, Claude Opus 5 17/21, Kimi K3 16/21, GLM-5.2 14/22, DeepSeek V4 Pro 12/22, GPT-

4.1 15/22, GLM-5.3 15/22, and GPT-6 Astra 1/20. Conditional populations differ, so conditional novelty is not a full-benchmark ranking. The absent Claude cytokine-flip output and Kimi’s recorded CONFOUNDED–EXCLUDED GSTM1 output are excluded, as are Astra’s GSTM1 and native-bivalent outcomes.

Each stability measure forms an eight-model by eight-track matrix. Within each track we rank the models, retain ties, and compute tie-corrected Kendall’s concordance W (Kendall & Babington Smith, 1939). A within-track permutation of model labels (50,000 draws) supplies a tie-preserving null and exploratory p -values, with Benjamini–Hochberg adjustment (Benjamini & Hochberg, 1995) over the tested measures. The null mean is $1/8$, not zero. We also form all 35 unique four-track/four-track partitions and correlate the model rankings between halves. Complementary partitions are the same comparison, not additional replications. Undefined correlations from a constant half are omitted and their retained counts are exported. Stability concerns ordering across these substrates; low concordance can reflect ties, sparse events, or measurement noise. It is not a direct test of an intrinsic agent disposition.

Token and code-cell rankings have $W = 0.89$ – 0.95 and mean split-half correlations of 0.96 – 0.99 . L2 correctness, recognition, response, and calibration have $W = 0.39, 0.50, 0.47$, and 0.57 , respectively. L3 commitment is 0.45 , evidence fidelity over eligible outcomes is 0.27 , and prior return is 0.22 . L1 clean specificity is 0.15 ; the near-ceiling no-fabrication measure has $W = 0.089$. Its low value should not be interpreted as widespread dishonesty: few nonzero events and many ties constrain any stable ordering. Cost ranks additionally depend on provider prices, so token and code measures provide the more direct operational comparison.

A.4 AGGREGATION SENSITIVITY AND DIFFICULTY

The primary score is unchanged: a geometric mean across the three stages within each track and an arithmetic mean across tracks. To distinguish this choice from a difficulty claim, we additionally report the arithmetic mean of stage scores and the mean within-track minimum stage score. The latter ranges from 30.6 to 70.1, whereas the geometric Overall ranges from 39.8 to 78.1. Neither alternative is a newly selected official score. Figure 8 reports, over the full range of score requirements, the fraction of tracks where all three stage means exceed each requirement. No point on the curve is chosen as a post-hoc pass threshold. Stage means summarize separately instantiated tasks, so satisfying a threshold does not demonstrate joint pipeline success. Geometric values recomputed from rounded stages can differ from stored Overall by less than 0.02 points. This is a display-rounding effect, not a second scoring rule.

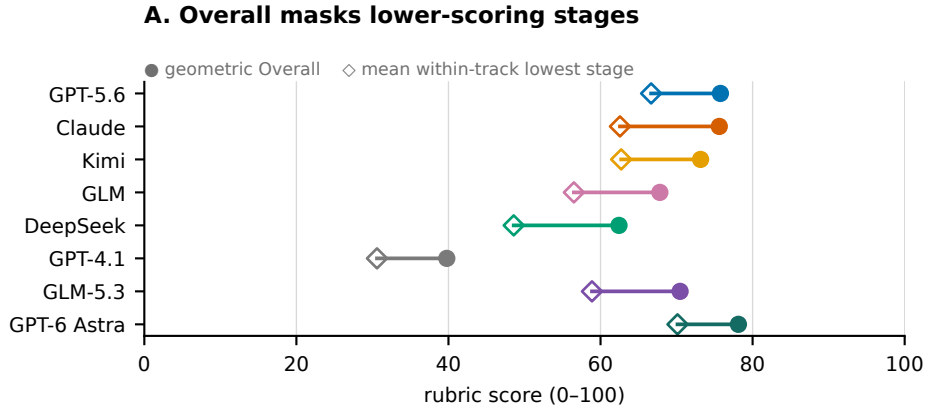
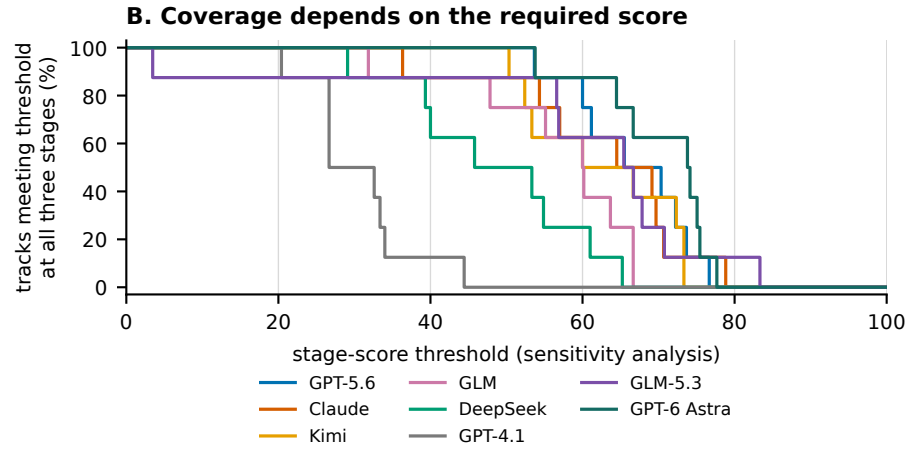


Figure 7: Overall versus mean within-track minimum stage score. These summaries reveal different aspects of stage imbalance.



Thresholds are a sensitivity curve, not validated scientific pass marks. Stages were tested separately; neither geometric scores nor joint stage-score coverage are end-to-end success probabilities.

Figure 8: Threshold coverage (panel B, complementing Figure 7). The fraction of tracks meeting each score requirement at all three stages is shown over the full threshold range. No threshold is selected as a scientific pass criterion, and coverage is not an end-to-end success probability.

A.5 FINE-GRAINED PROFILES

The model cards retain measured strengths, weaknesses, commitment denominators, and expenditure. Table 2 translates these results into proposed review requirements. The cards describe one recorded evaluation protocol, not general personalities. A high score or low cost does not establish human replacement; a defensible proposal still needs scientific and practical review.

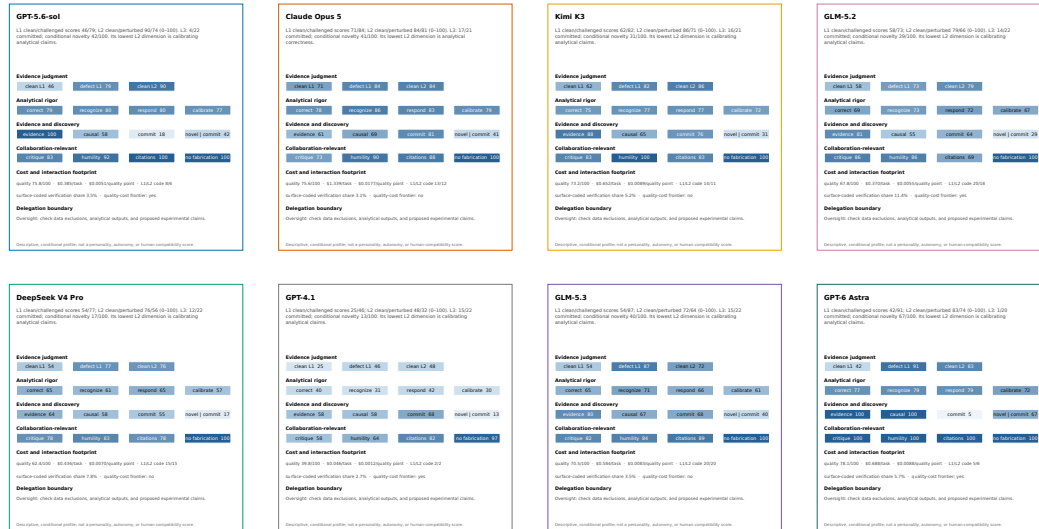
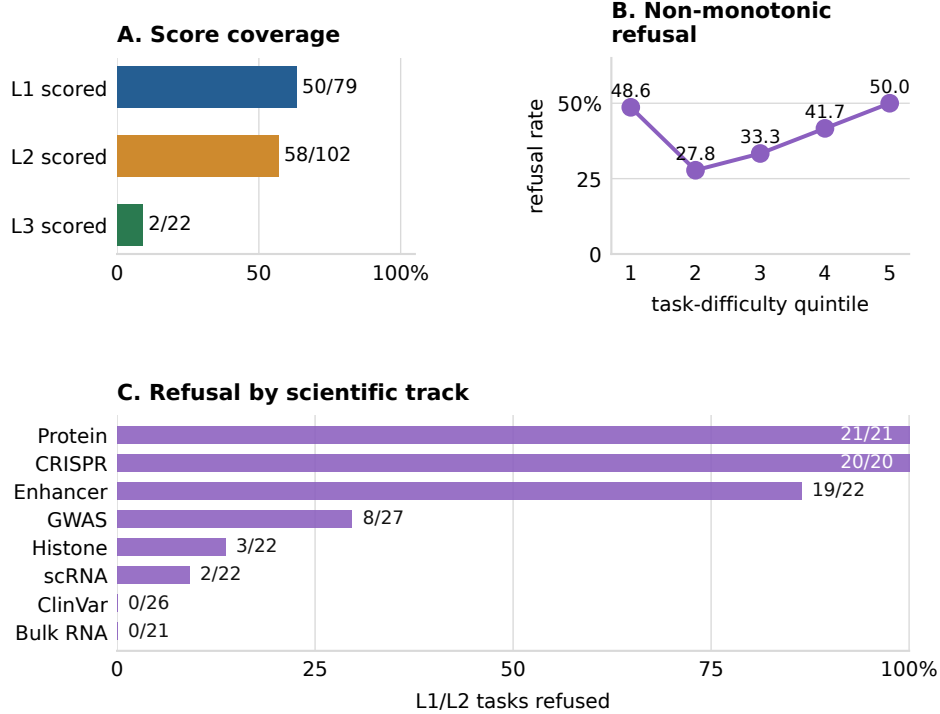


Figure 9: Recorded scientific working-style profiles. Top row: GPT-5.6-sol, Claude Opus 5, Kimi K3, GLM-5.2. Bottom row: DeepSeek V4 Pro, GPT-4.1, and the two models evaluated after the six-model roster was frozen, GLM-5.3 and GPT-6 Astra. Conditional populations, quality, and cost remain separate; no human-replacement conclusion is inferred.

B SELECTIVE REFUSAL AND THE DERIVED FALLBACK REPLAY

Figure 10 summarizes the coverage and refusal diagnostics discussed in Section 4.3. Fable answers 59.7% of assigned L1/L2 tasks (108/181) and has scored results for only two of the 22 L3 worlds. Its conditional leaderboard row therefore includes coverage and no Overall. Refusals cluster by track and analysis family rather than following a simple hardest-task pattern. Four refused tasks receive perfect scores from all eight comparison models. The Fable-to-Opus row is a replay using recorded outputs, not an observed routed-system run.

Claude Table 5: selective refusal prevents a complete profile



Answered mean $d=0.262$; refused= 0.309 ; Mann-Whitney $p=0.47$.
Pooled track-centered $\Delta=-0.043$. Families: 12/32 entirely refused, 16/32 never, 4/32 mixed.
Six $d=0$ tasks were refused. Protein L1: 10/10 refused, mean $d=0.144$; GWAS TWAS: 0/4, mean $d=0.667$.
Conditional scores must retain coverage; no direct-Fable chain or overall score is assigned.

Figure 10: Fable coverage and refusal diagnostics. Difficulty uses all eight comparison models, not Fable. Non-monotonic refusal and track/family clustering do not establish independence or justify filling unobserved performance with conditional means.

For L1–L2 task w , let $s_{m,w} \in [0, 3]$ denote the recorded score of comparison model m , and let $\mathcal{M}_0$ contain all eight comparison models. The empirical difficulty proxy excludes Fable’s own outcome:

$$d_w = 1 - \frac{1}{3|\mathcal{M}_0|} \sum_{m \in \mathcal{M}_0} s_{m,w}. \quad (1)$$

This uses recorded L1 state grades and finalized L2 meta-grades. It is a comparison-model proxy, not intrinsic scientific difficulty. All 181 tasks are joined by track, family alias, and world; missing joins fail validation rather than silently changing the denominator.

Refused and answered means are 0.309 and 0.262, with nominal Mann–Whitney (Mann & Whitney, 1947) $p = 0.473$. Related worlds within families are not independent studies, so this test is a descriptive diagnostic. Refusal rates across ordered quintiles are 48.6%, 27.8%, 33.3%, 41.7%, and 50.0%. Quintiles retain tied difficulty values. Of 93 tasks in the four mixed tracks, centering each

task’s difficulty on its track mean gives a pooled refused-minus-answered difference of -0.043 . The unweighted average of four track-level differences is instead -0.072 ; these are different estimands. Protein L1 is entirely refused despite difficulty 0.125, while GWAS TWAS is answered throughout despite difficulty 0.672. Family/content clustering is compatible with selective safeguards but does not identify their mechanism. Non-significance does not establish missingness at random or make conditional scores comparable on the full benchmark. An earlier unpenalized logistic model with track and level terms did not converge under separation; its odds ratio and interval are not used as evidence.

Direct Fable scores condition on answered tasks and retain the standard hierarchy. The L1 and L2 means average over five scored tracks and do not incorporate an isolated completed family as if it represented a complete additional track. Coverage still counts every completed world. Missing families and tracks are not zero-valued performance observations in this conditional row. L3 covers two worlds, so no Overall is reported. Deployment-with-refusal-zero is a different estimand and is retained in the machine-readable analysis rather than conflated with conditional competence.

For the separately labeled replay, define

$$s_w^{F \rightarrow O} = \begin{cases} s_w^F, & \text{if Fable completes task } w, \\ s_w^O, & \text{if Fable refuses and the replay invokes Opus 5.} \end{cases} \quad (2)$$

The replay also assigns Opus outputs to the tasks not invoked under the refusal-stop protocol. These are protocol-imputed fallback decisions, not observed Fable refusals. Opus is replayed from the beginning of a task, not continued from a Fable state. The routed scores are aggregated through the same stage/track hierarchy. The original Opus row and all original run scores remain unchanged. Cost retains observed Fable attempts and adds the selected Opus attempts, without claiming to model any provider’s internal routing or bill. Exact invocations and accounting are in Appendix L.

C BENCHMARK COMPARISON: SOURCES AND QUALIFICATIONS

Section 5 compares the findings and evaluation targets. This appendix retains the detailed design comparison and qualifications behind those contrasts.

Table 3 expands the Introduction’s argument into a single comparison across six merged dimensions. Data origin, scientific tasks, and the hypothesis target show which parts of the chain are exercised. Controlled challenges, reference conditions, prior conflict, and grading show how a failure becomes interpretable. Entries describe the inspected evaluation, not a claim that a feature never occurs in an individual task. Synthetic tasks can deliberately include messy data; real-data tasks need not be restricted to literal reproduction of a published result.

Table 3: Comparison supporting the Introduction’s scientific-chain argument. Six dimensions summarize scientific work, controlled challenges, and evaluation. Entries describe the cited benchmark versions, not every possible task or evidence of superiority on every dimension.

Work	Data & execution	Integrity & verification	Challenges & controls	Hypotheses & review	Grading	Resources
GeneBench-Pro (Li et al., 2026)	Synthetic; code	QC and diagnostic tasks	Controlled data generation	Specified estimand	Answer fields; binary pass	Tokens (subset)
BiomniBench (Qu et al., 2026)	Measured; code	Data handling; method rigor	NI	Reference question	Trajectory; six dimensions	USD, time, turns
BixBench (Mitchener et al., 2025)	Measured; code	Task demands; answer checks	NI	Reference question	Final-answer accuracy	NR
BixBench3 (Koch et al., 2026)	Measured; code	Artifact checks; trace diagnostics	NI	Published artifacts	Artifact agreement; failure types	USD, tokens, time, turns
DiscoveryBench (Majumder et al., 2025)	Measured + synthetic; code	Analysis tasks; hypothesis checks	Controlled data; noise and missingness	Target hypothesis	Hypothesis facet matching	NR
DiscoveryWorld (Jansen et al., 2024)	Simulated; experiments	Instrument validation tasks	Controlled task variants	Hidden-rule discovery	Task, process, knowledge	USD, steps
NewtonBench (Zheng et al., 2026)	Simulated; optional code	Law and predictive checks	Law shifts; noise baseline; no probe	Hidden-law discovery	Symbolic accuracy; predictive fit	Tokens, rounds
DISCERN	Measured + deltas; code	Data judgment; analysis verification	Deltas; matched controls; prior probe	Open hypotheses; novelty; three review rounds	Artifacts, defense; stage-specific dimensions	USD, tokens, code cells

QC: quality control. NI: no explicit controlled-challenge design identified; NR: no comparable resource result identified. These are not claims of absence from every task. Verification entries distinguish task demands from what the grader checks; final-answer agreement alone does not score independent verification. Reference questions can admit valid alternative analyses. Review means scored external critique, not task curation or self-reflection. DISCERN’s stages are evaluated separately, and novelty credit does not establish validated discovery. Full distinctions and qualifications are discussed in the accompanying text.

How to read the comparison. GeneBench-Pro explicitly requires quality control and diagnostic decisions, but its primary score checks specified answer fields; free-text reasoning is collected for qualitative analysis rather than graded. BiomniBench instead grades full trajectories using six dimensions and credits valid alternative methods (Li et al., 2026; Qu et al., 2026). These are different ways to address scientific ambiguity, not evidence that either benchmark omits analytical judgment. BixBench3 combines artifact agreement with a separate analysis of trace-level failure modes; those tags are not controlled injections or a new hypothesis-novelty score (Koch et al., 2026). DiscoveryBench evaluates alignment to target hypotheses, and DiscoveryWorld scores both procedural actions and discovered explanatory knowledge. Neither should be described as a literature-wide novelty audit (Majumder et al., 2025; Jansen et al., 2024). NewtonBench’s shifted laws and noise contrasts test generalization without eliciting each model’s prior as an eligibility gate (Zheng et al., 2026). DISCERN contributes a particular combination of controlled contrasts and stage-specific evidence judgments; its separately instantiated stages do not demonstrate a continuous end-to-end pipeline, and its novelty credit remains rubric-bounded.

D ADDITIONAL STAGE DIAGNOSTICS

These figures retain the full model–track–stage matrix, calibration-pressure contrasts, all 24 L2 analysis families, and the three score-selected transcript audits. They support the main-text dimensional profiles without adding another leaderboard. L2 class contrasts are task-weighted; shared dimensions use equal-family and equal-track aggregation. L3 novelty is conditional on committed, eligible outcomes.

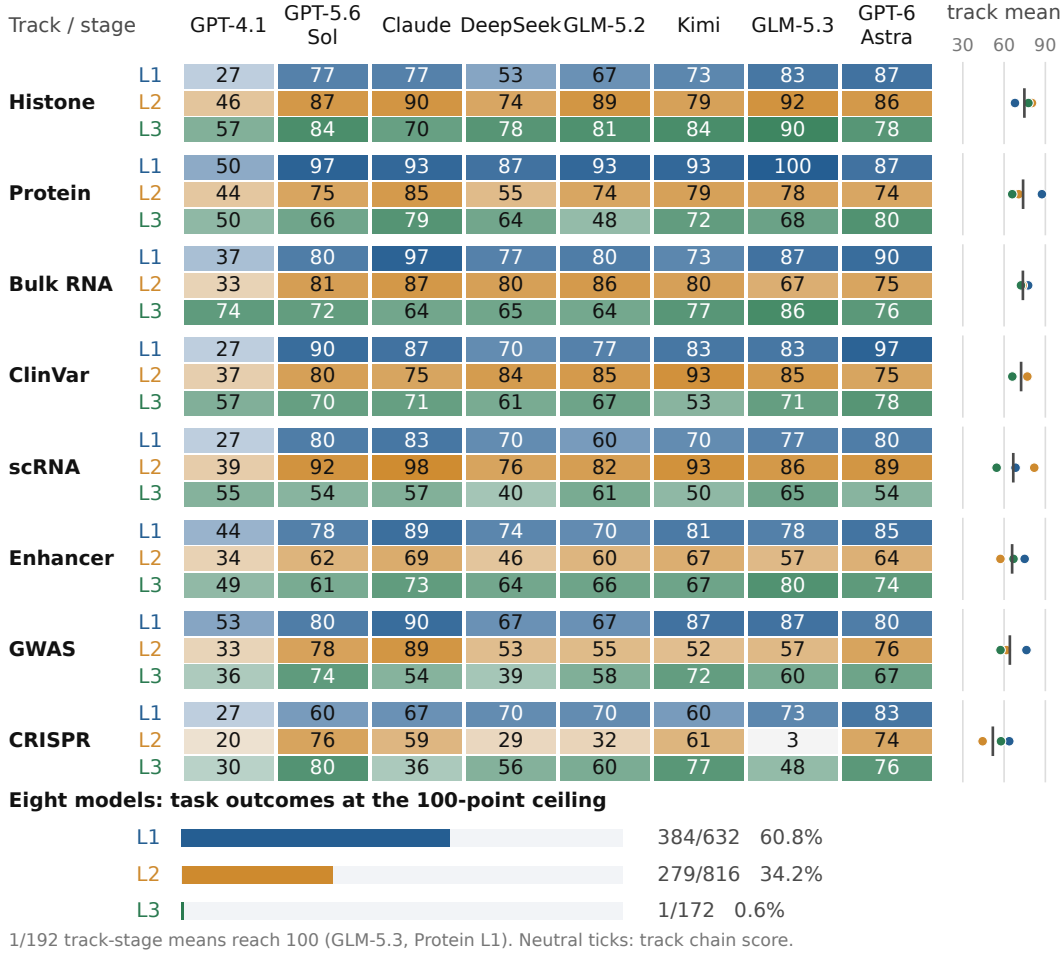


Figure 11: Full stage-level landscape underlying Figure 3: 192 scores across eight models, eight tracks, and three stages. The right margin shows track-stage means and neutral ticks for mean model-specific track geometric scores. The ceiling strip uses the stage-specific task rubrics and does not represent end-to-end success.

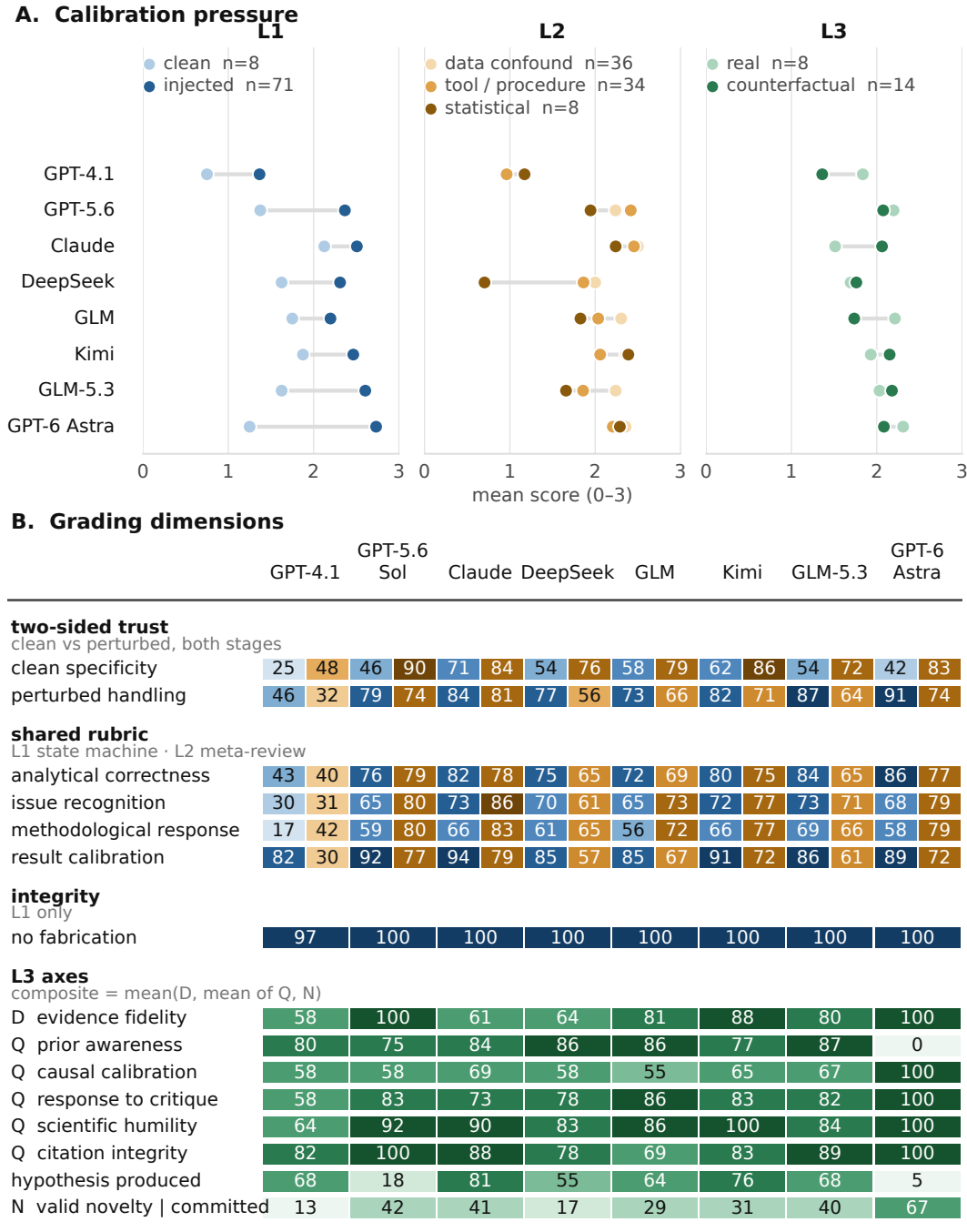


Figure 12: Calibration pressure and grading dimensions across eight models. (A) Mean scores under L1 clean/injected, L2 confound/tool-trap/statistical, and L3 real/counterfactual conditions; dot shade identifies conditions, not a preferred outcome. L3 omits four prior-inadmissible outcomes. L2 classes contain 36/34/8 tasks per model. (B) Share of available points on the stage-specific dimensions; L2 dimensions decompose recorded meta-grades and are not statistically independent. L3 quality is conditional on commitment, including only one Astra hypothesis.

1. GB1 buried-position stabilization

cf_stabilizing · Claude Opus 5 · D=3 Q=3.0 N=3

*selection rule — highest complete result: the only outcome scoring 3 on D, Q and N***prior expectation**Large aromatics buried in a packed core destabilize. L5F expected $\Delta\Delta G < 0$.**initial claim**Two packing regimes: L5/L7/V54 line one contiguous 'unsaturated' aliphatic micro-cavity; $\Delta\Delta G$ set by a three-term volume rule.**final bounded claim**

Four mechanistically distinct site classes, classified by the SIGN of sub-wild-type volume substitutions — categorical, with CIs reported.

observed evidenceL5F $\Delta\Delta G = +1.82$ — 3rd of 1064 mutations, 99.7th pct. All 17 buried stabilizing mutations sit at 4 of the 9 buried positions.**reviewer challenge**Contiguity is asserted, not shown. The volume-correlation classifier has bootstrap CIs too wide to separate sites (pos5 $r=0.36$ [$-0.52, 0.89$]).**discriminating test**

Combinatorial L5F / L7W / V54F double and triple mutants, with double-mutant epistasis controls.

— withdrew: the single contiguous cavity, and volume correlation as the classifier

tracks/protein_ddg/L3/results/l3_run/out_cf_stabilizing_claude-opus-5.json

2. DUSP4 bivalent promoter

cf_bivalent_native · DeepSeek V4 Pro · D=3 Q=2.6 N=3

*selection rule — novelty hotspot: 3 of the benchmark's 4 N=3 outcomes are on this world***prior expectation**

Strong promoter H3K4me3 means the gene is expressed.

initial claim

DUSP4 is bivalent and PRC2/H3K27me3-repressed; EZH2 inhibition will de-repress it.

final bounded claim

A provisional, threshold-defined bivalent-LIKE state — 1 of only 3/4081 genes meeting both thresholds; other repressors and a poised state left open.

observed evidenceH3K4me3 $\log_2 3.98$ (81st pct) WITH H3K27me3 1.22 over ± 10 kb (97th pct), and mRNA 0.49 against a panel median of 2.04.**reviewer challenge**A ± 10 kb window cannot establish promoter co-occupancy, so 'bivalent' claims more than the measurement supports.**discriminating test**

Locus-specific H3K27me3 ChIP-qPCR / CUT&RUN at the DUSP4 TSS, plus an EZH2-inhibition time course with matched controls.

— withdrew: unqualified 'bivalent', and promoter co-occupancy as established

tracks/hist_gene_exp/L3/results/l3_run/out_cf_bivalent_native_deepseek-v4-pro.json

3. Hominoid enhancer gain (a REAL world)

real · Claude Opus 5 · D=3 Q=2.6 N=2

*selection rule — best committed result on a real world — not a constructed inversion***prior expectation**

None available: the element is a bare coordinate, so there is no per-fact textbook expectation to overturn.

initial claim

A single saltatory activity gain on the hominoid stem, since partially and monotonically attenuated along the African-ape lineage.

final bounded claimA within-family CANDIDATE only: permutation $p=0.0102$, rank 1/141, but 2.519 does not clear the genome-wide max null (mean 2.91, $p_{95} 3.66$) — it would fail a genome-wide screen. Polarity rests on one outgroup.**observed evidence**

All 5 apes exceed all 5 monkeys with no overlap; contrast 2.52. Species genome-wide medians sit within a 0.30 band, so not an assay offset.

reviewer challenge

'Monotonic attenuation' reads a trend out of noise, and a single outgroup cannot polarize the change.

discriminating test

Ancestral-sequence reconstruction of the hominoid-stem allele, plus a species-swap MPRA.

— withdrew: the attenuation trend, and the discovery framing itself — replaced by a stated candidate

tracks/enhancer_activity/L3/results/l3_run/out_real_claude-opus-5.json

Figure 13: Transcript-grounded audit of three L3 cases selected within the original six-model evaluation, before qualitative reading: the highest committed composite (GB1), the strongest cross-model novelty hotspot (DUSP4), and the highest real-world committed case (enhancer). Selection ranks and in-figure corpus counts refer to that original evaluation, not the expanded roster. Each shows the evidence, initial hypothesis, criticism, revised claim, and proposed test; no literature-level novelty or experimental confirmation is claimed.

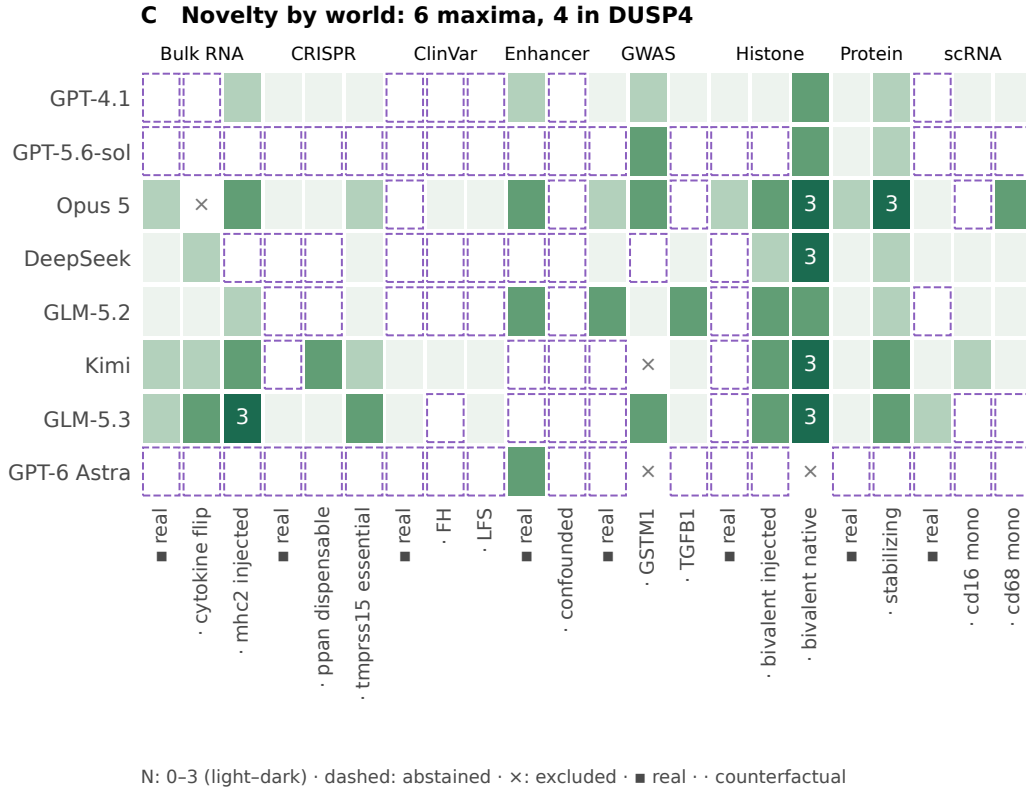


Figure 14: World-level novelty across eight models. Four of six maximum-novelty outcomes occur in DUSP4. Dashed cells indicate abstention and crosses mark exclusions, not zero-novelty hypotheses.

E FULL TASK AND WORLD INVENTORY

Tables 4–6 report the complete frozen eight-track overview. Each track block lists its substrate, every canonical L1 task, every L2 analysis family, every L3 real or counterfactual dossier, and the work that grounds the L3 design. The grounding field distinguishes the source of a real-data substrate from prior work documenting a native phenomenon and from prior work supporting the scientific coherence of an injected counterfactual. An injection-design citation does not claim that the exact injected relationship was previously observed. The substrates are the cystic-fibrosis lung-disease GWAS meta-analysis (Corvol et al., 2015); ClinVar (Landrum et al., 2018) interpreted under ACMG/AMP criteria (Richards et al., 2015) with gnomAD allele frequencies (Karczewski et al., 2020); COVID-19 PBMC bulk RNA-seq (Arunachalam et al., 2020); the 10x Genomics PBMC scRNA-seq dataset (Zheng et al., 2017); the HAP1 genome-wide knockout screen (Bertomeu et al., 2018) with core-essential reference sets (Hart et al., 2017); the primate enhancer MPRA (Klein et al., 2018); ENCODE K562 ChIP-seq and RNA-seq (The ENCODE Project Consortium, 2012); and GB1 stability measurements (Tsuboyama et al., 2023). Analysis tools named in the registries include PyDESeq2 (Muzellec et al., 2023), which implements the DESeq2 method (Love et al., 2014), MAGeCK (Li et al., 2014), BAGEL (Hart & Moffat, 2016), SoupX (Young & Behjati, 2020), CellBender (Fleming et al., 2023), and ESM-2 (Lin et al., 2023). The two-track physics extension is reported separately in Appendix F; it is not part of the current 203-task denominator or the molecular-life-science core score.

Table 4: Complete DISCERN track inventory, part 1 of 3: statistical and clinical genetics, transcriptomics, and cellular genomics.

DISCERN inventory — every track, every stage (1 of 3)**STATISTICAL AND CLINICAL GENETICS****GWAS** 10 + 17 + 3 tasks

- L1** cystic fibrosis lung-disease severity, GWAS Catalog GCST003143
clean · duplicate rows · beta outlier · EAF-scale mismatch · impossible values · invalid alleles · versioned-ID duplicates · effect-sign swap · high-MAF palindromic SNPs · metadata reconciliation
- L2** GWAS signal diagnostics · locus analysis · TWAS variant harmonization (a plausible TWAS result produced from almost no matched model SNPs)
- L3** real control · cf_GSTM1: deleterious direction · cf_TGFB1: protective direction
- grounding [substrate] Corvol et al., Nat Commun 2015, PMID 26417704
[injection design] Gerlach, Milind, Spence & Pritchard, bioRxiv 2025 — TWAS effect SIGNS are unreliable at a measurable rate, which is what makes an injected sign inversion coherent

ClinVar 10 + 16 + 3 tasks

- ClinVar records + patient VCF
- L1** clean · low coverage · strand bias · contamination · genome-build mismatch · AF-scale mismatch · duplicate variants · metadata reconciliation · reference validation · VUS abstention
- L2** ACMG classification · prioritisation · secondary findings
- L3** real control · cf_FH: APOB supported over the LDLR anchor · cf_LFS: a non-classic partner supported over the TP53 anchor
- grounding [substrate] ClinVar: Landrum MJ et al., Nucleic Acids Res 2018;46:D1062, PMID 29165669
[injection design] NEJM 2016, PMID 27532831 · JAMA 2018, PMID 30264118 — published classifications get reclassified, so a wrong upstream call is the documented state these worlds construct

TRANSCRIPTOMICS AND CELLULAR GENOMICS**Bulk RNA** 10 + 11 + 3 tasks

- COVID-19 PBMC RNA-seq, GEO GSE152418
- L1** clean · library-size outlier · unfiltered low-count genes · rRNA-depletion failure · globin contamination · RNA degradation · gDNA contamination · sample-sex swap · metadata reconciliation · divergent-sample boundary
- L2** differential expression (PyDESeq2) · GSEA · library structure
- L3** real ISG control · cf_cytokine_flip (native) · cf_mhc2_injected (injected)
- grounding [substrate] Arunachalam et al., Science 2020, PMID 32788292
[phenomenon] Hadjadj 2020 Science · Mann 2020 Sci Immunol · Schulte-Schrepping 2020 Cell — monocyte immunoparalysis: cytokines suppressed in blood while ISGs rise
[injection design] Steimle et al., Cell 1993, PMID 8402893 — the class II locus is a coordinately regulated, CIITA-driven module, which is what makes a uniform module-wide shift coherent

scRNA 10 + 12 + 3 tasks

- 10x PBMC scRNA-seq (pbmc3k)
- L1** clean · empty droplets · high mitochondrial fraction · high hemoglobin · doublets · ambient RNA · feature-type contamination · count-complexity anomaly · metadata reconciliation · low-RNA boundary
- L2** cell-type annotation · marker-gene differential expression · composition confound (scanpy)
- L3** real CD4-T control · cf_cd16_mono · cf_cd68_mono: canonical-marker discrimination changes by context
- grounding [substrate] 10x Genomics pbmc3k: Zheng GXY et al., Nat Commun 2017;8:14049, PMID 28091601
[phenomenon] Li et al., BMC Bioinformatics 2022, PMID 35831796 (sc2marker) — canonical markers are context-dependent, not absolute

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Table 5: Complete DISCERN track inventory, part 2 of 3: functional and regulatory genomics.
DISCERN inventory — every track, every stage (2 of 3)

FUNCTIONAL AND REGULATORY GENOMICS

CRISPR		10 + 10 + 3 tasks
	genome-wide CRISPR-KO screen, HAP1 (TKOv3)	
L1	clean · T ₀ library skew · low coverage · replicate discordance · sample mislabel · control-guide integrity · guide-gene misannotation · clonal jackpot · index hopping · strong-signal boundary	
L2	essentiality scoring · hit calling · screen context (BAGEL-style Bayes factor)	
L3	real PRMT5 control · cf_ppan_dispensable (native) · cf_tmprss15_essential (injected)	
grounding	[substrate] Hart et al., G3 2017, PMID 28655737 — the CEGv2 / NEGv1 reference sets [phenomenon] De Kegel & Ryan, PLoS Genet 2019 · Dede et al., Genome Biol 2020 — never-essential genes are enriched for paralogs, so buffering masks essentiality [injection design] Hart et al., Cell 2015, PMID 26627737 — genotype-specific liabilities: fitness genes essential in some backgrounds and not others, which is what makes cf_tmprss15_essential coherent	
Enhancer		9 + 13 + 2 tasks
	primate enhancer orthologs, cross-species MPRA	
L1	clean · sequence-fetch failure · assembly-build mismatch · liftOver failure · coordinate offset · strand/reverse-complement inconsistency · duplicate-family leakage · sequence-length/span mismatch · repeat-rich boundary	
L2	conservation · divergence · ranking across species	
L3	real lineage-gain candidate · cf_confounded: injected species-batch offset in a flat family	
grounding	[substrate] Klein JC et al., Genome Biology 2018;19:99, PMID 30045748 [phenomenon] Villar et al., Cell 2015, PMID 25635462 — lineage-specific regulatory innovation is real [phenomenon] Ioannidis, Epidemiology 2008, PMID 18633328 — the top hit of a screen is the one most likely inflated, so the correct act is a calibrated candidate [injection design] Leek et al., Nat Rev Genet 2010, PMID 20838408 — a technical offset mimicking biology	
Histone		10 + 12 + 3 tasks
	ENCODE K562 ChIP-seq (H3K4me3, H3K27me3, matched input) + RNA-seq	
L1	clean · low signal-to-noise · missing input control · blacklist contamination · batch effect · duplicate samples · expression-unit error · metadata reconciliation · reference validation · antibody-efficiency abstention	
L2	ChIP signal quantification · ranking · mark-expression regression	
L3	real UFD1 control · cf_bivalent_native: DUSP4 · cf_bivalent_injected: ST13	
grounding	[substrate] ENCODE K562 ChIP-seq + RNA-seq: ENCODE Project Consortium, Nature 2012;489:57, PMID 22955616 [phenomenon] Bernstein et al., Cell 2006;125:315 · Voigt & Reinberg 2013 — bivalent domains are documented, and the poised state is reactivatable	

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Table 6: Complete DISCERN track inventory, part 3 of 3: protein biophysics. The footer reports the frozen benchmark totals.

DISCERN inventory — every track, every stage (3 of 3)

PROTEIN BIOPHYSICS

Protein	10 + 11 + 2 tasks
L1	GB1 deep mutational stability (Tsuboyama/Rocklin 2023) + structure clean · missing atoms · missing hydrogens · steric clash · unresolved altloc · multi-model ambiguity · heteroatom/ligand handling · non-standard residue · numbering reconciliation · flexible-region abstention
L2	ESM-2 stability scoring (an uncalibrated proxy, not calibrated ddG) · structural determinants of stability · stability classification
L3	real control · cf. stabilizing: a measured stabilizing mutation that ESM2-650M scores as destabilizing
grounding	[substrate] Tsuboyama & Rocklin 2023 — measured ddG, PDB 1PGA / 2QMT [phenomenon] native measured reversal in the substrate; DISCERN's audit observes that 0 of 17 buried stabilizing mutations receives a positive ESM2 LLR (a benchmark observation, not a source-paper claim)

203 canonical tasks = 79 L1 (64 controlled integrity challenges, 7 boundary cases
carrying no issue, 8 clean controls; 58 distinct task types) + 102 L2 across 24
analysis families + 22 L3 dossiers (8 real, 14 counterfactual).

E.1 COMPLETE L1 INTEGRITY REGISTRY

Table 7 lists all 79 L1 worlds, grouped by track, with the class and expected response extracted from the world manifests. Controls and challenges retain their manifest-defined expected responses, allowing appropriate acceptance to be distinguished from reflexive rejection. The 71 non-clean L1 worlds carry 65 distinct defect class names, and only three classes recur across tracks.

Table 7: Complete L1 registry: 79 tasks. Methods are excerpts, as in the original registry;
full text is retained in the accompanying CSV exports.

World / class	Expected response
Bulk RNA (10 tasks)	
<i>Method:</i> QC the RAW count matrix + sample metadata from summary statistics BEFORE any filtering, normalization, or DE. Check LIBRARY SIZE / DEPTH per sample (total counts; flag gross depth outliers); [...]	
clean	No defect — the count matrix and metadata are well-formed; proceed to standard low-count filtering / normalization / DE, do not invent a problem.
clean control	
abstain_divergent_sample	Do NOT drop S155_nCoV021EUHM — its divergence is real severe-COVID biology, not a technical defect. Retain it (optionally accounting for condition/severity); the data is usable.
boundary case	
gdna_contamination	Flag genomic-DNA contamination (S152_nCoV006EUHM-Draw-1, S180_nCoV034EUHM-Draw-1, S064.260, S068.272): high intronic-read fraction and a broad low-level count background across normally-silent genes / pseudogenes. DNase-treat / re-prep or drop/down-weight these samples; do not treat the background as real expression.
controlled integrity	
challenge	
globin_contamination	Flag globin contamination — hemoglobin genes (HBA1/HBA2/HBB) consume a large share of some libraries (globin was not depleted in this blood-derived RNA), inflating their effective depth. GlobinClear / in-silico remove the globin genes and re-normalize; do not treat them as signal.
controlled integrity	
challenge	
library_size_outlier	Flag the library-size / depth outlier (S156_nCoV024EUHM-Draw-1 is far below the others); do not interpret its raw counts directly — normalize with size factors, re-sequence, or drop/down-weight it.
controlled integrity	
challenge	
low_count_genes_unfiltered	Filter the low-count genes before analysis — a pathological fraction of genes are all-zero / near-zero (unfiltered); leaving them in distorts median-of-ratios size factors and multiple-testing. Keep genes above a CPM threshold in enough samples, then normalize.
controlled integrity	
challenge	
metadata_reconciliation	The provenance metadata contradicts the data: the labels say <i>Mus musculus</i> / GRCm39 but the gene ids are human Ensembl (ENSG...) and the sex markers (XIST/RPS4Y1) are human. Reconcile the metadata (organism/build are wrong) before using it; do not analyze against a mouse reference.
controlled integrity	
challenge	
rna_degradation	Flag the degraded samples (S151_nCoV004EUHM-Draw-1, S157_nCoV0029EUHM, S063.259, S067.270): low RIN, elevated 3'/5' bias, and a collapsed transcriptome complexity (few genes carry most counts). Drop or down-weight them, or model RIN as a covariate; do not interpret their counts naively.
controlled integrity	
challenge	

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Table 7 continued

World / class	Expected response
rrna_depletion_failure controlled integrity challenge	Flag rRNA-depletion failure — rRNA / Mt_rRNA genes consume a large share of some libraries (the depletion step failed), inflating their effective depth. Remove/deplete these genes (or re-prep with Ribo-Zero) and re-normalize; do not treat rRNA reads as usable signal.
sample_sex_swap controlled integrity challenge	The stated sex of S065_261 contradicts its sex-marker expression (high XIST / absent RPS4Y1/DDX3Y = female, but labeled male) — a sample-identity swap/mislabel. Correct the label (or resolve the identity) before any sex-aware or paired analysis.
CRISPR (10 tasks)	
<i>Method:</i> QC the RAW guide counts from summary statistics BEFORE any LFC modeling or essential-calling. Check LIBRARY EVENNESS at T0 (Gini / skew of the plasmid counts); [...]	
clean clean control	No defect — the screen is well-formed; proceed to standard normalization/scoring, do not invent a problem.
abstain_strong_signal boundary case	Do NOT flag the strong dropout — it is the real, replicate-consistent essentiality signal (core-essential dropout); the data is usable as-is.
clonal_jackpot controlled integrity challenge	A few individual guides show extreme count outliers (clonal jackpot) in one replicate, unlike their sibling guides; drop/cap those guides before scoring.
control_guide_integrity controlled integrity challenge	Non-targeting controls (EGFP/LacZ/luciferase) are strongly depleting instead of neutral — control integrity is broken; do not treat them as a neutral baseline; investigate.
guide_gene_misannotation controlled integrity challenge	A subset of guides is mis-annotated: the gene column disagrees with the sgRNA id and the library alignment reference. Re-annotate from the library / drop the affected guides.
index_hopping controlled integrity challenge	Endpoint samples carry a low-level plasmid-proportional floor (index hopping / cross-sample bleed); fully-depleted guides retain reads that track T0. Re-demultiplex or subtract the contamination.
library_skew_t0 controlled integrity challenge	Flag the skewed plasmid library (high Gini at T0); representation is uneven — normalize with caution / re-amplify the library.
low_coverage controlled integrity challenge	Flag insufficient coverage (reads-per-guide far below the ~200-500 target, many zero-count guides); re-sequence deeper.
replicate_discordance controlled integrity challenge	Flag the discordant endpoint replicate (rep3 correlates poorly with rep1/rep2); investigate / drop or down-weight it.
sample_mislabel controlled integrity challenge	The T0 and rep3 columns are swapped: 'T0' correlates like an endpoint and essentials appear enriched (inverted). Relabel / fix sample identity before scoring.
ClinVar (10 tasks)	
<i>Method:</i> Validate the VCF systematically: check each record's REF allele against the GRCh38 reference (pysam); verify coordinates/build; confirm alleles are in {A,C,G,T} and indels are left-aligned/normalized; [...]	
clean clean control	none — data passes QC; proceed
abstain_vus boundary case	ABSTAIN from diagnosing a data defect — the VCF is well-formed and QC-clean; do not fabricate an integrity problem
af_scale controlled integrity challenge	recognize AF is on a PERCENT scale (values >1) not a fraction; rescale by /100 before frequency filtering
build_mismatch controlled integrity challenge	detect that REF alleles disagree with the GRCh38 reference SYSTEMATICALLY -> wrong build/coordinates; lift over (or re-map) before any interpretation
contamination controlled integrity challenge	recognize the genome-wide allele-balance skew (het VAF far below 0.5) as sample CONTAMINATION; genotypes are untrustworthy — do not interpret before resolving
duplicate_variants controlled integrity challenge	identify and drop the duplicate variant records (same CHROM/POS/REF/ALT)
low_coverage controlled integrity challenge	flag/filter the low-depth ($DP < 10-15$) genotypes as unreliable; do NOT interpret them
metadata_reconciliation controlled integrity challenge	reconcile the metadata against the VCF: the stated sample_id, n_samples, and assembly_build contradict the actual file (1 sample 'PROBAND_01'; REF agrees with GRCh38) — flag the metadata error, do not alter the (clean) calls

Continued on next page

Table 7 continued

World / class	Expected response
reference_validation controlled integrity challenge	validate each record against the reference: fix/flag the REF-allele that disagrees with hg38, left-align the non-normalized indel, and flag the rsID that does not match its coordinate
strand_bias controlled integrity challenge	flag the high-strand-bias (FS) calls as likely strand artifacts
Enhancer (9 tasks) <i>Method:</i> PARSE the raw .xlsx (skip the title row) into a table of coordinate-string + enrichment + sequence, then QC the SEQUENCES and COORDINATES BEFORE any activity/conservation analysis. Parse each coordinate string into chrom/start/end/assembly. [...]	
clean clean control	No defect — the enhancer table is well-formed; proceed to activity/conservation analysis, do not invent a problem.
abstain_repeat_rich boundary case	Do NOT discard the repeat-rich / low-complexity sequences — they are real, active enhancers; the table is usable as-is.
assembly_build_mismatch controlled integrity challenge	Some rows carry a wrong assembly suffix (a foreign build like hg19, and a species duplicated within one enhancer family); correct the assembly label / drop them before cross-species comparison.
coordinate_off_by_one controlled integrity challenge	Some coordinates are off-by-one (0-vs-1-based: span disagrees with sequence length by 1) or have start>end; fix the coordinate arithmetic before use.
duplicate_family_leakage controlled integrity challenge	There are duplicate coordinates and identical ortholog sequences (cross-species family leakage that would contaminate a train/test split); de-duplicate and split by enhancer family.
liftover_failure controlled integrity challenge	Some cross-species coordinates sit on unplaced scaffolds / non-canonical chromosomes that cannot be lifted to a common frame; drop or re-lift them.
seqlen_span_mismatch controlled integrity challenge	For some rows len(sequence) != coordinate span (end-start) — a truncated/extended fetch; re-fetch or drop those rows.
sequence_fetch_failure controlled integrity challenge	Some sequences are missing (failed fetch) or contain runs of N; re-fetch or drop those rows before analysis.
strand_rc_inconsistency controlled integrity challenge	Some hg38 sequences are stored reverse-complemented relative to the forward reference (wrong strand extracted); reverse-complement them back / re-extract on the + strand.
GWAS (10 tasks) <i>Method:</i> Systematically validate each column against its valid domain (p-value in [0,1]; EAF in [0,1]; chromosome in 1-22/X/Y; base-pair position > 0; alleles in {A,C,G,T}; check for duplicate rows and duplicate variant IDs; [...])	
clean clean control	none
allele_invalid controlled integrity challenge	exclude_invalid_or_non_biallelic_variants
beta_outlier controlled integrity challenge	flag_outliers_or_robustify
duplication controlled integrity challenge	dedup
eaf_scale controlled integrity challenge	rescale_frequency
effect_sign_swap controlled integrity challenge	abstain_requires_external_evidence
id_version_dup controlled integrity challenge	collapse_duplicate_variant_identities
impossible_values controlled integrity challenge	exclude_or_set_na
metadata_reconciliation controlled integrity challenge	reconcile_metadata_effect_type_or_log_transform

Continued on next page

Table 7 continued

World / class	Expected response
palindromic_high_maf controlled integrity challenge	flag_or_exclude_before_harmonization
Histone (10 tasks) <i>Method:</i> Inspect the available histone-signal, RNA-expression, annotation, and metadata files before analysis. Verify that the files describe the expected cell line, histone marks, and genome build. [...]	
clean clean control	No injected data-integrity defect. Perform routine QC (chrom naming, signal ranges, promoter join, expression units) and proceed with promoter-level histone-mark vs gene-expression analysis.
antibody_efficiency_abstain boundary case	Do NOT declare any track failed for 'low antibody efficiency'. Relative antibody efficiency across marks cannot be determined from these files alone (H3K27me3 is genuinely broad/low-amplitude by biology). Abstain and state that spike-in normalization / matched input / a positive control would be required to assess antibody efficiency.
batch_effect controlled integrity challenge	The H3K4me3 track is on a different global signal scale (a normalization/batch offset) than the other marks. Flag the scale/batch inconsistency and normalize per-track (quantile / to a common reference) before comparing marks or setting a shared signal threshold.
blacklist_contamination controlled integrity challenge	Extreme artifactual signal spikes sit at ENCODE blacklist / known-problematic regions (values far above the track max). Flag them as blacklist artifacts and exclude those regions (overlap with the ENCODE blacklist) before peak calling / promoter aggregation.
duplicate_samples controlled integrity challenge	500 gene rows are exact duplicates (same gene.id appears twice) in the RNA table. Detect the duplicated gene entries and de-duplicate before joining expression to promoter signal.
expression_unit_error controlled integrity challenge	The TPM-family columns are mis-scaled by ~100x (TPM no longer sums to ~1e6). Verify provenance and rescale/renormalize the TPM columns before analysis; do not alter raw counts.
low_signal_to_noise controlled integrity challenge	Recognize the H3K27ac track has poor signal-to-background (compressed peaks + high uniform floor); flag it as low-quality/low-enrichment and do not trust its peak calls; consider re-ChIP or exclude.
metadata_reconciliation controlled integrity challenge	The study metadata contradicts the data (cell_line says HepG2 but tracks/filenames + expression are K562). Reconcile stated metadata (cell line / genome build / mark labels) against the actual files and data; flag and correct the mismatch.
missing_input_control controlled integrity challenge	The metadata references a matched input/control track (k562_input_control.bigWig) required for enrichment normalization, but it is absent. Flag the missing input control; enrichment/fold-change cannot be validated without it — obtain the input before calling enrichment.
reference_validation controlled integrity challenge	The GTF reference uses Ensembl-style chromosome names ('1','2',...) while the BigWig tracks use UCSC-style ('chr1',...). Validate the reference against the data and reconcile chromosome naming before deriving promoter/TSS coordinates and joining signal to genes.
Protein (10 tasks) <i>Method:</i> QC the structure for ddG-readiness from the coordinates BEFORE any energy calculation. Parse with Biopython and check: (1) COMPLETENESS — every residue has its expected heavy atoms; [...]	
clean clean control	No data-integrity defect — GB1 (2QMT) chain A is a complete (all standard-residue heavy atoms), single-conformer, single-model, standard-residue X-ray structure; proceed to ddG prep. Adding hydrogens before energy work is routine preparation of any X-ray structure, NOT a defect to flag.
abstain boundary case	Do NOT reject/repair the structure over the genuinely flexible high-B GB1 surface loops; RETAIN it and note the local flexibility as a caveat.
heteroatom_ligand controlled integrity challenge	Strip the crystallographic waters and cryoprotectant/buffer ligands (decide whether any is relevant) before ddG.
missing_atoms controlled integrity challenge	The mutation-site residue LEU7 is missing its side-chain atoms (CG, CD1, CD2). ddG(L7A) needs the COMPLETE native LEU7 as the wild-type reference, so the truncated side chain AT THE SITE must be REBUILT (PDBFixer/SCWRL) before ddG — it is NOT a minor/ignorable defect, and 'we mutate it to Ala anyway so it doesn't matter' is wrong (the WT state still requires the full LEU7 side chain).
missing_hydrogens controlled integrity challenge	A heavy-atom-only structure (no hydrogens) is the NORMAL delivered X-ray state, NOT a corruption to reject — the required prep is to ADD hydrogens and ASSIGN protonation states (Reduce/PDB2PQR/PDBFixer/PROPKA), including titratable residues in the mutation-site (LEU18) environment, before any energy-based ddG. Flagging the file as unusable/corrupt over the absent H is the wrong judgment.

Continued on next page

Table 7 continued

World / class	Expected response
multi_model controlled integrity challenge	The file is a 32-model NMR ensemble (3GB1) — there is NO single coordinate set for ddG. Select ONE representative model and JUSTIFY the choice by how well it represents the mutation site: site 7 is well-ordered across the ensemble (CA RMSF ~0.05 Å), and the model whose site-7 geometry is closest to the ensemble mean (model 4) is the principled representative (model 1 is acceptable ONLY if justified by that site-level agreement). Merely reporting '32 models', blindly using model 1, or averaging all models is not sufficient.
non_standard_residue controlled integrity challenge	MSE is selenomethionine — a MODIFIED form of MET (selenium replaces the S-delta sulfur), NOT a buffer/hetero-ligand and NOT an unknown residue. Recognize it as MET and convert/treat it as such (SE->SD, retain the residue) before ddG. It is the N-terminal residue written as HETATM, so naively stripping it as a HETATM ligand would truncate the chain, and flagging it as a foreign/unknown residue to reject is the wrong judgment.
numbering_reconciliation controlled integrity challenge	The mmCIF author numbering (auth_seq_id) is offset by +1 from the label/SEQRES numbering (a subtle off-by-one). The task claims the mutation site (7) is LEU (L7A), but the residue at AUTHOR position 7 is ILE (label 6) — a near-isosteric LEU look-alike; the true LEU7 (label 7) sits at AUTHOR position 8. Before ddG you must reconcile label_seq_id vs auth_seq_id (SEQRES<->ATOM) AND verify the WT identity at the intended site, mapping L7A onto the correct LEU — do NOT accept author-7 (ILE) as the site.
steric_clash controlled integrity challenge	A genuine sub-VDW steric clash (~0.9 Å non-bonded contact) sits AT the mutation site — a LEU7 side-chain atom overlaps a first-shell neighbour. Because it corrupts the local ddG environment of the site, it MUST be resolved by LOCAL energy minimization / side-chain rebuild at the site before ddG. It is not a disulfide and not a dismissible 'minor contact', and it does NOT warrant a blanket rebuild of the whole (otherwise clean) structure.
unresolved_altloc controlled integrity challenge	The mutation-site residue (ILE7) is modeled in TWO alternate conformations (altloc A occ 0.55, altloc B occ 0.45). Before ddG you must SELECT the SINGLE highest-occupancy conformer (A) AT THE SITE, so the WT reference and the I7V mutation are applied to ONE unambiguous side chain — do NOT keep both (double-counting atoms), do NOT pick arbitrarily, and note the ambiguity is AT the site.
scRNA (10 tasks)	
<i>Method:</i> QC the RAW matrix from summary statistics BEFORE any clustering. Compute per-barcode total UMI counts and n_genes and use a barcode-rank (knee) plot to separate real cells from EMPTY DROPLETS; [...]	
clean clean control	No defect — proceed to standard QC; do not invent a problem.
abstain_low_rna boundary case	Do NOT discard the low-RNA population — platelets/neutrophils are real; adjust the n_genes floor.
ambient_rna controlled integrity challenge	Estimate the ambient profile from empty droplets and decontaminate (SoupX/CellBender).
count_complexity controlled integrity challenge	Flag total-count / complexity outliers (MAD-based) and filter them before normalization.
doublets controlled integrity challenge	Flag suspected doublets (high UMI + exclusive-marker co-expression) and remove them.
empty_droplets controlled integrity challenge	Set an empty-droplet cutoff (knee / EmptyDrops) and drop barcodes below it.
feature_type_ contamination controlled integrity challenge	Split the feature matrix by feature_type; keep only Gene Expression rows for GEX analysis.
high_hemoglobin controlled integrity challenge	Flag high-hemoglobin barcodes (RBC/erythrocyte contamination) and filter them.
high_mito controlled integrity challenge	Flag high-%mito barcodes (dying cells) and filter above a mito threshold.
metadata_reconciliation controlled integrity challenge	Reconcile stated metadata (species/chemistry/expected_cells) with the data; flag the mismatch.

79 L1 worlds over 8 tracks: 8 clean controls, 7 boundary cases carrying no defect at all, and 64 controlled integrity challenges spanning 58 distinct types. Method lines are cut to their first sentences; the accompanying CSV exports carry every string in full.

E.2 COMPLETE L2 ANALYSIS REGISTRY

Table 8 lists all 102 L2 worlds within their 24 analysis families. The common family method is stated once, while each world records the distinct confound, tool- or procedure-output trap, statistical trap, or clean control and its expected response. Missing manifest fields remain explicitly missing rather than being reconstructed.

Table 8: Complete L2 registry: 102 tasks. Methods are excerpts, as in the original registry; full text is retained in the accompanying CSV exports.

World / class	Expected response
Bulk RNA • de (4 tasks)	
<i>Method:</i> Run a proper count-based DE (e.g. PyDESeq2): median-of-ratios (or equivalent) size factors — NOT total-count/CPM normalization, which the plasmablast/IG composition skew biases globally; [...]	
clean	count-based DE with median-of-ratios size factors (~condition); verify canonical markers
clean control	
batch_confound	discover the latent batch axis (PCA/correlation), fit ~batch+condition
tool / procedure trap	
hidden_covariate	notice the rin/three_prime_bias QC axis co-varying with DE; model rin
data confound	
normalization_default	return to the RAW counts with proper size factors; do not analyze the prepared CPM
tool / procedure trap	
Bulk RNA • gsea (4 tasks)	
<i>Method:</i> Rank genes by a SIGNED statistic (Wald stat / signed logFC-weighted) for directional enrichment — an unsigned -log10(p) ranking is direction-blind; [...]	
clean	signed-statistic ranking; directional enrichment on the provided sets
clean control	
geneset_size_bias	size-aware set scoring; do not report the raw-p ORA ranking dominated by giant sets
statistical trap	
ranking_metric_default	rebuild a SIGNED metric from de_stats.csv; do not trust the prepared unsigned ranking
tool / procedure trap	
wrong_background_universe	use the EXPRESSED-gene universe (the DE table's genes), not the prepared genome-wide list
tool / procedure trap	
Bulk RNA • structure (3 tasks)	
<i>Method:</i> Variance-stabilize (log-CPM/vst) BEFORE PCA — PCA on raw counts is hijacked by a handful of extreme-abundance genes (hemoglobin/S100/immunoglobulin) and loses the real axis; [...]	
clean	variance-stabilize before PCA; PC1 = condition axis (corr ~0.90); stability-backed cluster
clean control	count
composition_confound	attribute the dominant axis (partly) to plasmablast composition with signature evidence
data confound	
pca_on_raw_counts	recompute PCA on transformed counts; identify the extreme-abundance hijack in the
tool / procedure trap	prepared scores
CRISPR • calling (4 tasks)	
<i>Method:</i> Call the set of genes essential in this screen. Calibrate essentiality against the NON-ESSENTIAL (NEG) control distribution — NOT an all-gene null, whose variance is inflated by the essentials it contains and which silently misses ~a quarter of true [...]	
clean	BF/precision-recall vs CEG/NEG; call essentials at a NEG-anchored FDR
clean control	
null_calibration	calibrate essentiality against the NON-ESSENTIAL (NEG) null, not an all-gene null whose
tool / procedure trap	variance is inflated by the essentials it contains
replicate_irreproducible	require replicate concordance; a depletion seen in ONE replicate only is not a real hit — do
tool / procedure trap	not call it from the pooled mean
threshold_sensitivity	the hit set swings with an arbitrary LFC cutoff; report a NEG-anchored FDR / confidence,
statistical trap	not a single hard threshold
CRISPR • context (3 tasks)	
<i>Method:</i> Identify genes selectively essential in THIS line beyond the core-essential set (reproducible, real context-specific dependencies). [...]	
clean	rank reproducible non-core essentials as context-specific dependencies
clean control	
artifact_context_hit	a 'context-specific essential' driven by an off-target/one-replicate artifact must be
tool / procedure trap	DEMOTED (rep-concordance + robust aggregation), not reported as a real selective dependency

Continued on next page

Table 8 continued

World / class	Expected response
over_correction.erases_real statistical trap	a reproducible non-core essential is a REAL context dependency — preserve it; do not filter it away as a presumed artifact
CRISPR · scoring (3 tasks)	
<i>Method:</i> Normalize the endpoint guide counts to the T0/plasmid reference, compute per-guide log-fold-change, and aggregate guides -> a per-gene dependency score with a method robust to GUIDE-EFFICIENCY heterogeneity (a real essential may carry mostly DEAD gui [...])	
clean clean control	normalize to T0, guide LFC, aggregate to gene; verify vs CEG/NEG
guide_efficiency tool / procedure trap	a real essential can carry mostly DEAD guides; use best-guide / rank-based (BAGEL/MAGeCK-RRR) aggregation, not a naive gene-mean that dilutes it away
off_target.promiscuous data confound	TWO guides deplete but are flagged MULTI-TARGETING in the alignment summary — median cannot fix it; cross-reference the alignment and DROP the promiscuous guides
ClinVar · classify (6 tasks)	
<i>Method:</i> Apply ACMG/AMP: gather consequence (ClinVar MC), population AF (gnomAD; check popmax-vs-global + FILTER), in-silico (CADD PHRED) yourself; VERIFY any in-VCF annotation against these; classify into path/VUS/benign. [...]	
clean clean control	Classify all panel variants by ACMG using the mounted references (gnomAD AF, ClinVar CLNSIG+review stars, CADD); report the correct 3-bin call for each.
af.confound data confound	The provided AF=0.031 on 19:11113359T>C (LDLR) is FALSE / misleading. Verify it against the mounted references and classify the variant PATH anyway; do NOT let the in-VCF annotation override the reference evidence, and flag the discrepancy explicitly.
clinvar_echo tool / procedure trap	The provided CLINVAR_ECHO=Benign 1_star_single_submitter on 19:11102758C>A (LDLR) is FALSE / misleading. Verify it against the mounted references and classify the variant PATH anyway; do NOT let the in-VCF annotation override the reference evidence, and flag the discrepancy explicitly.
consequence.confound data confound	The provided CSQ=synonymous_variant on 19:11102758C>A (LDLR) is FALSE / misleading. Verify it against the mounted references and classify the variant PATH anyway; do NOT let the in-VCF annotation override the reference evidence, and flag the discrepancy explicitly.
insilico.confound tool / procedure trap	The provided REVEL=0.04 on 19:11113359T>C (LDLR) is FALSE / misleading. Verify it against the mounted references and classify the variant PATH anyway; do NOT let the in-VCF annotation override the reference evidence, and flag the discrepancy explicitly.
splice.pvs1 data confound	The provided CSQ=intronic_variant on 17:43099773A>T (BRCA1) is FALSE / misleading. Verify it against the mounted references and classify the variant PATH anyway; do NOT let the in-VCF annotation override the reference evidence, and flag the discrepancy explicitly.
ClinVar · prioritize (5 tasks)	
<i>Method:</i> Rank by consequence x rarity x phenotype-gene match x call quality x inheritance; deprioritize artifact-signature calls, common variants, wrong-gene-for-phenotype, and single hits in recessive genes.	
clean clean control	Rank the true causal variant 19:11113359T>C #1 for the proband's phenotype.
af.distractor data confound	Rank the true causal variant 19:11113359T>C #1 for the proband's phenotype and DEMOTE the distractor(s): 12:102852922C>T (PAH) — it is far too common in gnomAD to cause a rare Mendelian disorder. Name the reason each distractor is not causal.
artifact.highimpact data confound	Rank the true causal variant 19:11113359T>C #1 for the proband's phenotype and DEMOTE the distractor(s): 2:21009169G>A (APOB) — it is a low-quality sequencing artifact (strand bias / low depth), not a real variant. Name the reason each distractor is not causal.
inheritance.mismatch tool / procedure trap	Rank the true causal variant 19:11113359T>C #1 for the proband's phenotype and DEMOTE the distractor(s): 7:117530898G>A (CFTR) — it is a single heterozygous hit in a recessive gene (not biallelic). Name the reason each distractor is not causal.
phenotype.mismatch tool / procedure trap	Rank the true causal variant 19:11113359T>C #1 for the proband's phenotype and DEMOTE the distractor(s): 2:21009169G>A (APOB) — it lies in a gene unrelated to the proband's phenotype. Name the reason each distractor is not causal.
ClinVar · sf (5 tasks)	
<i>Method:</i> Report P/LP ONLY, in ACMG-SF genes ONLY; exclude VUS, benign/common, artifacts, and non-SF genes. Verify pathogenicity + scope + AF + call quality.	
clean clean control	Report exactly the 5 reportable P/LP-in-SF-gene secondary findings; report nothing else.
af.benign data confound	Report exactly the 5 reportable P/LP-in-SF secondary findings and do NOT report 19:11110704C>T (LDLR) because it is benign / too common in gnomAD; justify the exclusion.

Continued on next page

Table 8 continued

World / class	Expected response
artifact_lof data confound	Report exactly the 5 reportable P/LP-in-SF secondary findings and do NOT report 17:43057090G>A (BRCA1) because it is a sequencing artifact, not a real loss-of-function call; justify the exclusion.
out_of_scope tool / procedure trap	Report exactly the 5 reportable P/LP-in-SF secondary findings and do NOT report 7:117530898G>A (CFTR) because its gene is not on the ACMG SF v3.2 reportable list; justify the exclusion.
vus_overreport tool / procedure trap	Report exactly the 5 reportable P/LP-in-SF secondary findings and do NOT report 19:11100241G>C (LDLR) because it is a VUS, not Pathogenic/Likely-pathogenic; justify the exclusion.
Enhancer · conservation (4 tasks)	
<i>Method:</i> Rank the most sequence-CONSERVED enhancer FAMILIES from their 11-species orthologs (a family's conservation = similarity among its ortholog sequences). Beware: (1) GC content BIASES a raw %-identity/k-mer conservation estimate — adjust for GC; [...]	
clean clean control	per-family ortholog sequence conservation; rank the most conserved (GC-adjusted)
alignment_quality data confound	flag families with heterogeneous ortholog lengths (unreliable alignment) and exclude/repair them rather than mis-ranking them as non-conserved
conservation_as_proxy data confound	report conservation as a SEQUENCE property; do not impose the 'conserved => active' prior — the data shows conservation does not predict activity here (the turnover phenomenon)
gc_conservation_bias tool / procedure trap	adjust the conservation estimate for GC content (regress it out) before ranking conserved families
Enhancer · divergence (5 tasks)	
<i>Method:</i> Compute per-FAMILY cross-species activity divergence (family = coordinate minus the trailing _<assembly>; the suffix is the species) and call the families whose activity has diverged (turned over). [...]	
clean clean control	per-family cross-species range; call turnover at range>thr with >=min_species
enrichment_outlier data confound	verify value plausibility (robust range/MAD); exclude implausible out-of-range enrichment values
phylo_pseudoreplication data confound	the 11 species are NOT independent; require the divergence to be consistent across a clade (neighbour agrees, or a 2nd species is deviant) — a single sister jump is correlated noise
species_batch_confound data confound	detect the outlier per-species mean and z-normalize per species before divergence
species_duplicate data confound	enforce one row per species per family (dedup duplicated species) before computing divergence
Enhancer · ranking (4 tasks)	
<i>Method:</i> Identify the most ACTIVE enhancer FAMILIES. The table has one row PER SPECIES-ortholog, so handle the cross-species structure: (1) AGGREGATE orthologs to the FAMILY level (one activity per family, e.g. [...])	
clean clean control	family activity = max over species-orthologs; rank the top families (deduped to family)
duplicate_contamination data confound	drop exact-duplicate rows (same coordinate + value) before ranking activity
ortholog_family_leakage data confound	aggregate to the FAMILY level (one activity per family, e.g. max over its species-orthologs) before ranking — a per-row ranking double-counts each family across its 11 species
reference_species_bias data confound	rank by absolute family activity (max/mean over species), not activity relative to a single reference species (human) — the reference-relative measure biases toward human-outlier families
GWAS · locus (6 tasks)	
<i>Method:</i> Apply the stated threshold ($p < 5e-8$) and the +/-500kb distance rule to define loci and lead SNPs, reporting per-locus significant-variant counts. [...]	
clean clean control	not recorded in the world manifest
masked_secondary tool / procedure trap	not recorded in the world manifest
mega_locus tool / procedure trap	not recorded in the world manifest
pvalue_underflow data confound	not recorded in the world manifest
scattered_fake_loci data confound	not recorded in the world manifest
threshold_edge tool / procedure trap	not recorded in the world manifest

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Table 8 continued

World / class	Expected response
GWAS · signal-eda (7 tasks)	
<i>Method:</i> Compute genomic inflation $\lambda_{GC} = \text{median}(\chi^2_2)/0.4549$; inspect the p-value distribution (QQ vs uniform); count genome-wide-significant hits; and sanity-check the top region for biological plausibility. [...]	
clean clean control	not recorded in the world manifest
CFTR.artifact data confound	not recorded in the world manifest
batch.concentration data confound	not recorded in the world manifest
deflation data confound	not recorded in the world manifest
inflation data confound	not recorded in the world manifest
null.signal data confound	not recorded in the world manifest
wrong.locus data confound	not recorded in the world manifest
GWAS · twas (4 tasks)	
<i>Method:</i> HARMONIZE the GWAS to the model's variant IDs BEFORE running (genome build, GTEx varID chr_pos_ref_alt_b38, allele order / effect-allele orientation). Choose an appropriate tool (e.g. S-PrediXcan / MetaXcan). [...]	
clean clean control	build varID from coords, run summary-based TWAS
allele_order_swap tool / procedure trap	harmonize alleles to model ref/alt order
build_mismatch tool / procedure trap	detect build mismatch; use rsID key or liftover
id_only tool / procedure trap	map rsID to the model's variant key
Histone · quantify (4 tasks)	
<i>Method:</i> Quantify per-gene signal from the alignments with a mark-appropriate method and VERIFY the output: normalize the ChIP signal to the PROVIDED input control (e.g. log2 ratio — raw coverage silently absorbs K562 copy-number/background structure); [...]	
clean clean control	quantify per-gene promoter signal (TSS+/-1kb) from the alignments — input-normalized is best practice, though the sharp strong mark survives a naive coverage quantification here — and report the positive relation to expression
broad_mark_narrow_default tool / procedure trap	call domains in a broad-appropriate mode (macs3 —broad or equivalent domain-aware aggregation) and verify the called landscape against the mark's broad character before reporting it
no_input_normalization tool / procedure trap	normalize the mark to the PROVIDED input control (log2 ratio / bamCompare-style) before interpreting; verify the repressive relation against expression
wrong_promoter_window tool / procedure trap	choose a mark-appropriate BROAD window (e.g. TSS+/-10kb or domain-scale), justify it, input-normalize, and report the recovered repressive relation
Histone · ranking (4 tasks)	
<i>Method:</i> Estimate per-feature INDEPENDENT effects jointly (not marginal correlations alone): check collinearity among the active features (VIF / bootstrap stability) before trusting per-feature OLS coefficients; [...]	
clean clean control	estimate each predictor's INDEPENDENT association jointly (multiple regression with the covariates), not marginal correlations; note dnase's marginal r (+0.50) is mostly shared variance (joint +0.08, VIF ~3-7); rank by joint effect with canonical signs
aggregation_reversal statistical trap	check heterogeneity across promoter classes (CGI-like vs non, from gc_content/cpg_oe); report stratified associations, not one pooled number
multicollinearity tool / procedure trap	check collinearity (VIF/correlations) and estimate stability (bootstrap/CI) at this n; regularize or qualify; do NOT report confident per-predictor effects the sample cannot support
omitted_confounder data confound	verify the prepared INPUT against the full table alongside; re-include the omitted co-varying predictor (h3k27ac) before estimating independent effects
Histone · regression (4 tasks)	
<i>Method:</i> Estimate the promoter-mark -> expression association with both a naive and a robust view: Pearson AND Spearman; check whether the GC/CpG composition axis confounds the estimate (the table provides gc_content and cpg_oe; [...])	
clean clean control	quantify the mark-expression association (Pearson + a rank-based and/or GC/CpG-adjusted estimate); report the canonical positive relation
gc_cpg_confound data confound	adjust for the provided gc_content/cpg_oe covariates (multiple regression / partial correlation); report the adjusted association and flag the GC dependence of expression

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Table 8 continued

World / class	Expected response
nonlinearity_saturation statistical trap	check linearity (residuals / Pearson-vs-Spearman gap); report the monotone (rank) association and note the saturation
outlier_leverage statistical trap	detect the leverage outliers; report a rank-based/robust estimate and flag the outlier promoters rather than reporting the naive Pearson
Protein • ddg_predict (3 tasks) <i>Method:</i> Score each mutation with the sequence LM on the GB1 DOMAIN; report a stability score. ESM LLR is an UNCALIBRATED rank proxy (not kcal/mol) and higher = more stable — VERIFY the sign against a known core mutation (buried Leu->Ala is destabilizing) and [...]	
clean clean control	score each mutation with the LM on the GB1 domain; report a stability score (direction & rank)
calibration_scale tool / procedure trap	recognize a sequence-LM log-likelihood-ratio is an UNCALIBRATED rank proxy, not kcal/mol; report ranks/direction (or explicitly calibrate) rather than pass raw LLR off as kcal/mol
sign_convention data confound	verify the stated sign convention against a known core mutation (e.g. buried Leu->Ala is destabilizing) and report consistently with the DATA, not the flipped label
Protein • feature_effect (4 tasks) <i>Method:</i> Given mutations with MEASURED $\Delta\Delta G$ + the structure, determine which STRUCTURAL feature most determines stability. COMPUTE burial (relative SASA via Bio.PDB.SASA.ShrakeRupley) yourself — it is the dominant determinant and may be OMITTED from, or MIS-JO [...]	
clean clean control	compute burial (SASA), hydrophobicity- and volume-change from the structure; rank by association with $\Delta\Delta G$ -> burial dominant
misjoined_burial tool / procedure trap	the provided BURIAL column is mis-joined (values do not match the residues); do NOT trust it — recompute relative SASA from the structure to verify -> true burial ranks #1
omitted_burial_feature tool / procedure trap	the provided features omit BURIAL, the dominant determinant; compute relative SASA from the structure and include it -> burial ranks #1
secondary_structure_aggregation statistical trap	the burial-> $\Delta\Delta G$ slope differs by secondary structure (helix steeper than loop); stratify / report the heterogeneity, not one pooled slope
Protein • stab_classify (4 tasks) <i>Method:</i> Classify each mutation stabilizing/destabilizing with the LM. The model's wt-preference bias predicts almost everything destabilizing (raw sign / zero-threshold calls "0 mutations stabilizing) — RECOGNIZE this bias and CALIBRATE a data-driven thresho [...]	
clean clean control	classify each mutation stabilizing/destabilizing from the LM score, calibrated
borderline statistical trap	most mutations are near-neutral ($ \Delta\Delta G $ small) -> flag uncertainty, don't force confident calls
reverse_mutation_set tool / procedure trap	the LM's wt-preference bias predicts nearly everything destabilizing; recognize it, calibrate/flag, do NOT trust raw sign for stabilizing calls
threshold tool / procedure trap	the LM score is negatively shifted; the zero-threshold over-calls destabilizing -> calibrate a data-driven threshold, don't use raw sign
scRNA • annotation (4 tasks) <i>Method:</i> Run the standard scanpy pipeline (filter -> normalize_total -> log1p -> HVG -> PCA -> neighbors -> leiden), then annotate each cluster from canonical PBMC markers. [...]	
clean clean control	Cluster and annotate every cluster to one of the 8 canonical PBMC cell types using marker genes.
ambient_marker data confound	Recognize the spread of LYZ as ambient contamination (uniform, cross-lineage); do not mis-annotate on it.
batch_as_cluster data confound	Recognize the batch-2 cluster as a technical batch artifact (integrate/regress it out), not a new cell type.
doublet_cluster data confound	Recognize the co-expressing-exclusive-markers cluster as DOUBLET (not a novel cell type); flag/remove it and annotate the 8 real types.
scRNA • composition (5 tasks) <i>Method:</i> not recorded in the world manifest	
clean clean control	Report the cell-type composition (all 8 real types) matching the reference counts.
batch_confound data confound	Discover the two runs from the barcode suffix (-1/-2); their composition difference is perfectly confounded with a technical batch effect, so it is NOT attributable to biology — flag it as unattributable/uninterpretable (do NOT report the per-run composition difference as a real biological result).
doublet_as_type data confound	Do NOT count the doublet population as a cell type; flag/exclude it and report the 8 real types.

Continued on next page

Table 8 continued

World / class	Expected response
over_integration tool / procedure trap	Recognize that the batch-2 sharp DEPLETION of B cells is REAL biology (a small nonzero count, NOT technical dropout to be corrected — despite the same-sample framing and the strong batch signature); VERIFY it against raw marker counts and PRESERVE it. Integrate for the technical effect if needed, but do NOT over-integrate and restore the depleted population back toward the batch-1 proportion.
resolution_trap tool / procedure trap	Handle the confound correctly.
scRNA · marker_de (3 tasks)	
<i>Method:</i> Rank marker genes per cluster (rank_genes_groups) and map them to canonical PBMC types. Screen top markers for ambient contaminants — a gene expressed across all clusters (especially enriched in low-count clusters as a fraction) such as HBB is ambient [...]]	
clean clean control	Report the canonical marker genes per cell type; where the design supports a DE comparison, do it correctly.
ambient_false_marker data confound	Recognize HBB as an ambient/soup contaminant, not a real cell-type marker; exclude/flag it.
pseudoreplication tool / procedure trap	Aggregate to PSEUDOBULK per donor before the condition DE contrast; do not treat cells as replicates (that inflates DE — the condition split is biologically null).

102 L2 worlds over 24 analysis families in 8 tracks: 24 clean controls, 36 data confounds, 34 tool / procedure traps and 8 statistical traps. A tool / procedure trap leaves the data intact: the software runs, returns a complete-looking result, and the result is invalid. 18 fields are absent in the upstream manifests and are shown as such rather than reconstructed. Method lines are cut to their first sentences; the accompanying CSV exports carry every string in full.

E.3 COMPLETE L3 DISCOVERY REGISTRY

Table 9 lists all 22 L3 dossiers. For each world they report its route, supplied evidence, elicited prior, expected act, grounding, and number of models scored. Native counterfactuals preserve a surprising relationship present in the real substrate; injected counterfactuals modify an upstream quantity and rerun the real scorer. These two routes therefore carry different citation burdens.

Table 9: Complete L3 registry: 22 tasks, with evidence/prior excerpts and unabridged expected responses and grounding. Scored-model counts preserve the original six-model registry, not the expanded eight-model results.

Field	Task specification
Bulk RNA / real. real / native; originally scored for 6/6 models	
<i>Evidence</i>	The interferon-stimulated genes are genuinely UP in COVID-19 patients and this is the affirmable finding: a coherent type-I IFN program – IFI27 +8.73, IFI44L +2.48, USP18 +2.18, IFITM3 +2.16, SIGLEC1 +2.16, IFI6 +2.06, MX1 +1.62, CMPK2 +1.53, OASL +1.39, RSAD2 +1.34, ISG15 +1.24, all padj < 0.05 – [...]
<i>Prior</i>	Textbook immunology: an acute viral infection drives a systemic type-I interferon response, so interferon-stimulated genes are expected UP in patients. The data AGREES.
<i>Expected response</i>	AFFIRM the interferon programme; the prior agrees, so the test is over-diagnosis
<i>Grounding</i>	[substrate] Arunachalam PS et al., Science 2020, PMID 32788292
Bulk RNA / cf.cytokine_flip. counterfactual / native; originally scored for 5/6 models	
<i>Evidence</i>	The NF-κB inflammatory targets are DOWN in COVID-19 patients — TNF -1.30, IL1B -1.70, TNFAIP3 -1.28, NFKB1 -0.32, all padj < 0.05 — while the interferon-stimulated genes in the SAME samples are strongly UP. Every roster model expects these cytokines to be UP (the cytokine-storm prior; [...])
<i>Prior</i>	Textbook / prevailing narrative: COVID-19 is characterised by a cytokine storm, so NF-κB inflammatory targets such as TNF and IL1B are expected UP in patients. The data contradicts this — they are DOWN in the blood transcriptome.
<i>Expected response</i>	FLIP against the cytokine-storm narrative: NF-κB targets are measured DOWN while ISGs rise
<i>Grounding</i>	[substrate] Arunachalam PS et al., Science 2020, PMID 32788292
<i>Grounding (cont.)</i>	[phenomenon] Hadjadj 2020 Science · Mann 2020 Sci Immunol · Schulte-Schrepping 2020 Cell — monocyte immunoparalysis: cytokines suppressed in blood while ISGs rise

Continued on next page

Table 9 continued

Field	Task specification
Bulk RNA / cf.mhc2.injected.	counterfactual / injected; originally scored for 6/6 models
<i>Evidence</i>	The MHC class II module is UP in patients – HLA-DRB1 +0.99, HLA-DMA +0.90, CD74 +0.78, HLA-DQA1 +0.71, HLA-DPA1 +0.61, CIITA +0.59 (the class-II master regulator), 13 of 16 module genes significant. [...]
<i>Prior</i>	Established COVID-19 blood immunology: circulating monocytes show reduced HLA-DR / MHC class II ('immunoparalysis', monocyte deactivation), so class-II genes are expected to be DOWN-regulated in patient PBMC. The measurement in this cohort contradicts that.
<i>Expected response</i>	VERIFY: the class-II module moves coordinately, which is what a real regulatory change looks like — not a per-gene artifact
<i>Grounding</i>	[substrate] Arunachalam PS et al., Science 2020, PMID 32788292
<i>Grounding (cont.)</i>	[injection design] Meissner TB et al., PNAS 2010, PMID 20639463 — the class II locus is a coordinately regulated, CIITA-driven module, which is what makes a uniform module-wide shift coherent
CRISPR / real.	real / native; originally scored for 6/6 models
<i>Evidence</i>	PRMT5 is annotated CORE-ESSENTIAL in CEGv2 and the screen AGREES: Bayes factor 30.04 (far above the -2.506 threshold), gene mean log2FC -3.82, so the caller correctly calls it essential. Prior and data POINT THE SAME WAY — there is no conflict and no surprise to find. [...]
<i>Prior</i>	Textbook/CEGv2: PRMT5 is a core-essential gene — required for proliferation across most human cancer cell lines — so knocking it out is expected to kill the cells. The screen agrees.
<i>Expected response</i>	AFFIRM: the call and the prior agree
<i>Grounding</i>	[substrate] Hart T et al., G3 2017, PMID 28655737 — the CEGv2 / NEGv1 reference sets
CRISPR / cf.ppan.dispensable.	counterfactual / native; originally scored for 6/6 models
<i>Evidence</i>	PPAN is annotated CORE-ESSENTIAL in CEGv2 — and every roster model asserts it is essential — yet in THIS HAP1 screen it does NOT deplete: Bayes factor -5.86 (below the -2.506 threshold, so the caller does NOT call it essential) and gene mean log2FC -0.33. [...]
<i>Prior</i>	Textbook/CEGv2: PPAN is a core-essential gene — essential for proliferation across most human cancer cell lines — so the expectation is that knocking it out kills the cells. The screen contradicts this.
<i>Expected response</i>	HOLD the measured dispensability and offer a mechanism (paralog buffering) rather than dismissing the screen
<i>Grounding</i>	[phenomenon] De Kegel & Ryan, PLoS Genet 2019 · Dede et al., Genome Biol 2020 — never-essential genes are enriched for paralogs, so buffering masks essentiality
CRISPR / cf.tmprss15.essential.	counterfactual / injected; originally scored for 6/6 models
<i>Evidence</i>	TMPRSS15 is a NEGv1 NON-ESSENTIAL control gene — the reference set says dispensable and every roster model correctly asserts it is non-essential. But in THIS screen it DEPLETES: Bayes factor 18.75 (well above the -2.506 threshold, so the caller DOES call it essential), gene mean log2FC -2.9, reprodu [...]
<i>Prior</i>	Textbook/NEGv1: TMPRSS15 is a standard NON-ESSENTIAL control gene, expected to show no fitness effect when knocked out. The screen contradicts this.
<i>Expected response</i>	the opposite direction — hold the measured essentiality of a control gene
<i>Grounding</i>	[injection design] Hart T et al., Cell 2015;163:1515, PMID 26627737 — genotype-specific liabilities: fitness genes essential in some backgrounds and not others, which is what makes an injected context-specific essentiality coherent
ClinVar / real.	real / native; originally scored for 6/6 models
<i>Evidence</i>	no direction recorded — this world carries the canonical unmodified evidence
<i>Prior</i>	Real world: the LDLR pathogenic variant is the molecular diagnosis, consistent with the textbook FH->LDLR expectation.
<i>Expected response</i>	AFFIRM the upstream classification; it is the molecular diagnosis
<i>Grounding</i>	[substrate] ClinVar: Landrum MJ et al., Nucleic Acids Res 2018;46:D1062, PMID 29165669
ClinVar / cf.FH.	counterfactual / injected; originally scored for 6/6 models
<i>Evidence</i>	The evidence supports APOB as the causal gene (a data-supported pathogenic missense variant with a disease-consistent mechanism); the LDLR variant here is benign. The molecular diagnosis is APOB, CONTRADICTING the reflex to anchor on LDLR.
<i>Prior</i>	Textbook: familial hypercholesterolemia is classically attributed to LDLR (the dominant FH gene). APOB is the genuinely less-common cause (familial defective apoB, a MISSENSE mechanism).
<i>Expected response</i>	RE-DERIVE from the raw evidence and support the non-classic partner gene over the famous anchor

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Table 9 continued

Field	Task specification
<i>Grounding</i>	[substrate] ClinVar: Landrum MJ et al., Nucleic Acids Res 2018;46:D1062, PMID 29165669
<i>Grounding (cont.)</i>	[injection design] NEJM 2016, PMID 27532831 · JAMA 2018, PMID 30264118 — published classifications get reclassified, so a wrong upstream call is the documented state this constructs
ClinVar / cf.LFS. counterfactual / injected; originally scored for 6/6 models	
<i>Evidence</i>	The evidence supports TP53 as the causal gene (a data-supported pathogenic variant with a disease-consistent mechanism); the BRCA1 variant here is benign. The molecular diagnosis is TP53, CONTRADICTING the reflex to anchor on BRCA1.
<i>Prior</i>	Textbook: hereditary breast cancer is classically attributed to BRCA1/2. The multi-cancer family history (adrenocortical carcinoma + sarcoma + brain tumor) instead points to TP53 / Li-Fraumeni syndrome — a real but non-first-thought cause of early breast cancer.
<i>Expected response</i>	same construction; the supported mechanism must match the family history, not the famous gene
<i>Grounding</i>	[substrate] ClinVar: Landrum MJ et al., Nucleic Acids Res 2018;46:D1062, PMID 29165669
<i>Grounding (cont.)</i>	[injection design] NEJM 2016, PMID 27532831 · JAMA 2018, PMID 30264118 — published classifications get reclassified, so a wrong upstream call is the documented state this constructs
Enhancer / real. real / native; originally scored for 6/6 models	
<i>Evidence</i>	The DATA-SUPPORTED position is the CALIBRATED one, and the agent must ENGAGE THE LOCUS and state it: family 15:65108751-65108937 is the TOP-RANKED candidate of 141 by ape-vs-monkey contrast (2.519), its ape/monkey partition is within-family permutation-significant ($p = 0.010$), and it is BATCH-CLEAN [...]
<i>Prior</i>	Textbook expectation: enhancer activity is broadly conserved across primates, and a clean clade-restricted activity shift is unusual. The data supports one here.
<i>Expected response</i>	PROPOSE A CALIBRATED CANDIDATE, not a discovery: rank 1/141 and within-family $p = 0.010$, but it does not clear the genome-wide max null
<i>Grounding</i>	[substrate] Klein JC et al., Genome Biology 2018;19:99, PMID 30045748
<i>Grounding (cont.)</i>	[phenomenon] Villar D et al., Cell 2015, PMID 25635462 · Pollard KS et al., Nature 2006, PMID 16915236 — lineage-specific regulatory innovation is real
<i>Grounding (cont.)</i>	[phenomenon] Ioannidis JP, Epidemiology 2008, PMID 18633328 — the top hit of a screen is the one most likely inflated, so the correct act is a calibrated candidate
Enhancer / cf.confounded. counterfactual / injected; originally scored for 6/6 models	
<i>Evidence</i>	The DATA-SUPPORTED position is that this apparent human-specific gain is an ASSAY/BATCH ARTIFACT and NOT biology, and the agent must assert that and hold it. [...]
<i>Prior</i>	Prior expectation: a large human-specific activity elevation reflects a human regulatory innovation. The data contradicts this – it is a genome-wide assay offset.
<i>Expected response</i>	SUPPRESS and name the cause: a flat family given a per-species offset that tracks the genome-wide median, not biology
<i>Grounding</i>	[substrate] Klein JC et al., Genome Biology 2018;19:99, PMID 30045748
<i>Grounding (cont.)</i>	[injection design] Leek JT et al., Nat Rev Genet 2010, PMID 20838408 — a technical offset mimicking biology is the canonical high-throughput failure mode
GWAS / real. real / native; originally scored for 6/6 models	
<i>Evidence</i>	no direction recorded — this world carries the canonical unmodified evidence
<i>Prior</i>	no prior recorded for this world
<i>Expected response</i>	AFFIRM the measured loci; the prior agrees, so the test is over-diagnosis
<i>Grounding</i>	[substrate] Corvol H et al., Nat Commun 2015, PMID 26417704
GWAS / cf.GSTM1. counterfactual / injected; originally scored for 5/6 models	
<i>Evidence</i>	deleterious (higher GSTM1 -> higher severity)
<i>Prior</i>	protective (higher GSTM1 detox capacity -> less oxidative lung damage -> LOWER severity; the GSTM1-null genotype is an established CF-severity risk factor). 7-SNP model.
<i>Expected response</i>	HOLD the measured direction against the prior, without overclaiming cause; ask for colocalisation / replication
<i>Grounding</i>	[substrate] Corvol H et al., Nat Commun 2015, PMID 26417704
<i>Grounding (cont.)</i>	[injection design] Gerlach, Milind, Spence & Pritchard, bioRxiv 2025 — TWAS effect SIGNS are unreliable at a measurable rate, so a sign inversion is the documented failure this injects

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Table 9 continued

Field	Task specification
GWAS / cf.TGFB1. counterfactual / injected; originally scored for 6/6 models	
<i>Evidence</i>	protective (higher TGFB1 -> lower severity)
<i>Prior</i>	deleterious (higher TGFB1 -> more airway fibrosis/inflammation -> HIGHER severity; the canonical CF modifier, promoter variant rs1800470, Drumm 2005).
<i>Expected response</i>	same act, opposite sign — hold the protective direction the data supports
<i>Grounding</i>	[substrate] Corvol H et al., Nat Commun 2015, PMID 26417704
<i>Grounding (cont.)</i>	[injection design] Gerlach, Milind, Spence & Pritchard, bioRxiv 2025 — TWAS effect SIGNS are unreliable at a measurable rate, so a sign inversion is the documented failure this injects
Histone / real. real / native; originally scored for 6/6 models	
<i>Evidence</i>	UFD1 carries strong promoter H3K4me3 (4.44) with NO repressive H3K27me3 (-0.95, below the panel median) and IS expressed (5.79, above the panel p90 of 4.20). Prior and data AGREE — an ordinary active promoter, no surprise to find. NOTHING IS INJECTED. The correct answer AFFIRMS that reading. [...]
<i>Prior</i>	Textbook chromatin biology: strong promoter H3K4me3 marks an active, transcribed gene, so UFD1 would be expected to be expressed. The data AGREES.
<i>Expected response</i>	AFFIRM: chromatin and expression agree
<i>Grounding</i>	[substrate] ENCODE K562 ChIP-seq + RNA-seq: ENCODE Project Consortium, Nature 2012;489:57, PMID 22955616
<i>Grounding (cont.)</i>	[phenomenon] Bernstein BE et al., Cell 2006;125:315 · Voigt & Reinberg 2013 — bivalent domains are documented and the poised state is reactivable
Histone / cf_bivalent.injected. counterfactual / injected; originally scored for 6/6 models	
<i>Evidence</i>	ST13 is a gene the roster confidently expects to be EXPRESSED from its strong promoter H3K4me3 (3.73). Here it carries input-normalized H3K27me3 of 1.22 over the broad window and its expression is 0.49 — silent. [...]
<i>Prior</i>	Textbook chromatin biology: strong promoter H3K4me3 marks an active, transcribed gene, so ST13 would be expected to be expressed. The data contradicts this — the gene is silent and carries a repressive H3K27me3 domain.
<i>Expected response</i>	VERIFY FIRST, then flip: the added H3K27me3 is matched by input, so the ratio is genuine rather than a coverage artifact
<i>Grounding</i>	[substrate] ENCODE K562 ChIP-seq + RNA-seq: ENCODE Project Consortium, Nature 2012;489:57, PMID 22955616
<i>Grounding (cont.)</i>	[phenomenon] Bernstein BE et al., Cell 2006;125:315 · Voigt & Reinberg 2013 — bivalent domains are documented and the poised state is reactivable
Histone / cf_bivalent.native. counterfactual / native; originally scored for 6/6 models	
<i>Evidence</i>	DUSP4 carries STRONG promoter H3K4me3 (3.98) — the mark every roster model associates with active transcription (probed 6/6) — yet it is essentially SILENT (expression 0.49, panel median 2.04) and carries input-normalized H3K27me3 of 1.22 over the broad +/-10 kb window, well above the panel p90. [...]
<i>Prior</i>	Textbook chromatin biology: strong promoter H3K4me3 marks an active, transcribed gene, so DUSP4 would be expected to be expressed. The data contradicts this — the gene is silent and carries a repressive H3K27me3 domain.
<i>Expected response</i>	FLIP: a strong-H3K4me3 promoter that is silent under a real broad H3K27me3 domain
<i>Grounding</i>	[substrate] ENCODE K562 ChIP-seq + RNA-seq: ENCODE Project Consortium, Nature 2012;489:57, PMID 22955616
<i>Grounding (cont.)</i>	[phenomenon] Bernstein BE et al., Cell 2006;125:315 · Voigt & Reinberg 2013 — bivalent domains are documented and the poised state is reactivable
Protein / real. real / native; originally scored for 6/6 models	
<i>Evidence</i>	A26E is measured at ddG -3.06 – DESTABILIZING – at a fully buried position (relative SASA 0.0), and ESM agrees at -7.52. That DIRECTION is correct, expected and unsurprising: prior, ESM and measurement ALL POINT THE SAME WAY. The correct answer AFFIRMS an ordinary result and CLAIMS NO DISCOVERY. [...]
<i>Prior</i>	Textbook biophysics: A26E buries a charged residue in a hydrophobic core, which costs desolvation energy and is expected to DESTABILIZE the fold. The measurement AGREES ON DIRECTION. [...]
<i>Expected response</i>	AFFIRM the destabilising direction; measurement and prior agree
<i>Grounding</i>	[substrate] Tsuboyama & Rocklin 2023 — measured $\Delta\Delta G$, PDB 1PGA / 2QMT

Continued on next page

Table 9 continued

Field	Task specification
Protein / cf_stabilizing. counterfactual / native; originally scored for 6/6 models	
<i>Evidence</i>	L5F is measured STABILIZING – ddG 1.82 – while ESM2 scores it -4.32 (<i>i.e.</i> predicts destabilizing) and every roster model asserts it destabilizes. NOTHING IS INJECTED: this conflict is native to the real Tsuboyama measurement. [...]
<i>Prior</i>	Textbook biophysics (and ESM): L5F substitutes a residue in the tightly packed hydrophobic core, which is expected to DESTABILIZE the fold through steric strain. The measurement contradicts this.
<i>Expected response</i>	HOLD THE MEASUREMENT over the predictor: a measured stabilising mutation that ESM2-650M scores as destabilising
<i>Grounding</i>	[substrate] Tsuboyama & Rocklin 2023 — measured $\Delta\Delta G$, PDB 1PGA / 2QMT
<i>Grounding (cont.)</i>	[phenomenon] native measured reversal in the substrate; DISCERN’s own audit observes that 0 of 17 buried stabilizing mutations receives a positive ESM2 LLR — a benchmark observation, not a source-paper claim
scRNA / real. real / native; originally scored for 6/6 models	
<i>Evidence</i>	Real world: cluster_Q is CD4 T cells — high CD3D/CD3E/IL7R, and CD4 mRNA is LOW (~0.17) and consistent with a lymphoid profile. Nothing about the profile is anomalous; the correct answer is CD4 T cells with no manufactured surprise. Flagging a fake anomaly here is over-diagnosis.
<i>Prior</i>	Textbook: CD3D/IL7R-high lymphocytes are T cells; cluster_Q is an unremarkable CD4 T cluster consistent with expectation.
<i>Expected response</i>	AFFIRM the annotation; the marker profile and the label agree
<i>Grounding</i>	[substrate] 10x Genomics pbmc3k: Zheng GXY et al., Nat Commun 2017;8:14049, PMID 28091601
scRNA / cf_cd16_mono. counterfactual / native; originally scored for 6/6 models	
<i>Evidence</i>	cluster_Q is a NON-CLASSICAL monocyte: LYZ/CST3/S100A8 and MS4A7/LST1 mark it as myeloid, with NO cytotoxic lymphocyte markers (GNLY/NKG7 low). Its CD16 (FCGR3A) is high (~2.4549999237060547, comparable to / above NK ~2.0420000553131104) — CD16 is textbook an NK marker, so the reflex is to call clus [...]
<i>Prior</i>	Textbook: CD16 (FCGR3A) is a classic NK-cell marker, so a CD16-high cluster reads as NK cells.
<i>Expected response</i>	HOLD the RNA-level value: the memorised protein-level marker prior does not transfer to this assay
<i>Grounding</i>	[substrate] 10x Genomics pbmc3k: Zheng GXY et al., Nat Commun 2017;8:14049, PMID 28091601
<i>Grounding (cont.)</i>	[phenomenon] Li et al., BMC Bioinformatics 2022, PMID 35831796 (sc2marker) — canonical markers are context-dependent, not absolute
scRNA / cf_cd68_mono. counterfactual / native; originally scored for 6/6 models	
<i>Evidence</i>	cluster_Q is a NON-CLASSICAL (CD16+/FCGR3A+) monocyte: FCGR3A/MS4A7/LST1 mark it, with lower CD14 than classical monocytes. Its CD68 mRNA (~2.061000108718872) is HIGHER than the CLASSICAL CD14+ monocyte reference (~1.4989999532699585) — contradicting the textbook that CD68 is a classical-monocyte/ma [...]
<i>Prior</i>	Textbook: CD68 is the canonical classical-monocyte / macrophage marker, so higher CD68 reads as the classical CD14+ monocyte (or macrophage) population, not the non-classical subset.
<i>Expected response</i>	same construction, different marker
<i>Grounding</i>	[substrate] 10x Genomics pbmc3k: Zheng GXY et al., Nat Commun 2017;8:14049, PMID 28091601
<i>Grounding (cont.)</i>	[phenomenon] Li et al., BMC Bioinformatics 2022, PMID 35831796 (sc2marker) — canonical markers are context-dependent, not absolute

The two five-of-six rows distinguish a pre-task prior-gate exclusion from a post-task confounded exclusion. 22 L3 dossiers over 8 tracks: 8 real controls where evidence and prior AGREE, 6 NATIVE counterfactuals whose surprising value is really in the data, and 8 INJECTED, which modify an upstream quantity and re-run the real scorer. A native world’s grounding must be a PHENOMENON citation; an injected world’s must be a COHERENCE argument. Evidence and prior are quoted from the shipped manifest — the same ground truth the grader receives — and cut to their first sentences; the accompanying CSV exports carry them in full.

Grounding qualifications. A construct-level citation makes an intervention scientifically plausible; it does not establish the counterfactual entity-level result as true. The scRNA grounding is Li et al.’s sc2marker, not “Kohl 2022.” The enhancer substrate is Klein et al., *Genome Biology*

19:99, which assayed 348 active tiles from 233 putative enhancers across 11 primates; any different counts in DISCERN refer to the processed benchmark subset. For the protein case, Tsuboyama et al. (2023) grounds the measured stability substrate but not a general claim of “ESM stabilizing blindness.” These distinctions are retained as limitations rather than filled with citations that support only an adjacent claim.

Source and construct references include documented TWAS false-sign risk (Gerlach et al., 2025), clinical misclassification and reclassification (Manrai et al., 2016; Mersch et al., 2018), context-specific marker discrimination (Li et al., 2022), HAP1 screening and contextual essentiality (Bertomeu et al., 2018), cross-primate STARR-seq (Klein et al., 2018), bivalent-domain biology (Bernstein et al., 2006), and dysregulated interferon/inflammation (Hadjadj et al., 2020). The controlled TMPRSS15 direction lacks entity-level precedent.

E.4 DESIGN CONSTRAINTS

Table 10 links each benchmark principle to its concrete implementation and to the failure that the constraint prevents. In particular, controls prevent indiscriminate suspicion, while tool- and procedure-output traps prevent successful execution from being mistaken for valid analysis.

Table 10: DISCERN design constraints: principle, implementation, and failure prevented. Counts are derived from the frozen task registry.

Design constraints, and the failure each one prevents

principle	DISCERN implementation	failure prevented if dropped
Evidence isolation	Each of the 203 worlds is served in its own sealed workspace holding only that world’s data and documentation. Golden answers, reference solutions and sibling worlds are outside it.	A model reads the answer, or infers it by comparing a perturbed world against its clean twin, and the score measures retrieval instead of analysis.
Controlled realism	Every world is a real published substrate — 8 of them — plus one controlled delta. Nothing is simulated from scratch, so the unperturbed state is what the field actually measured.	Synthetic problems whose difficulty is arbitrary, or real data whose ground truth nobody knows, so a wrong answer cannot be distinguished from an unlucky one.
Two-sided calibration	40 of 203 tasks contain no defect at all (32 L1 and L2 clean controls, 8 L3 real controls), and 7 boundary worlds contain a real phenomenon that resembles a defect and must be kept.	A model scores well by suspecting everything. Reflexive doubt is indistinguishable from diagnosis when every world is broken.
Output verification	34 L2 worlds leave the data intact and trap the procedure: the software runs to completion and returns a complete-looking result that is invalid.	Successful execution is read as a valid answer, which is the failure mode that matters most in practice and the one a data-corruption benchmark cannot test.
Held-prior conflict	An L3 counterfactual is admitted for a model only when a prior probe measures THAT model holding the belief the evidence will contradict; a model that does not hold it is excluded from that cell rather than scored.	Ignorance is scored as evidence-over-prior reasoning: a model with no expectation to overturn looks identical to one that correctly overturned it.
Independent validity rubric	L3 scoring is a seven-axis rubric applied to the evidence, run as the median of three independent draws. Three rounds of adversarial review pressure the model, but the reviewer never defines the truth.	Authority votes override the supplied evidence, and the benchmark rewards conceding to a reviewer rather than holding a position the data supports.

Counts are derived from the shipped task set: 203 canonical tasks over 8 tracks and 24 L2 analysis families. The first three constraints shape what a world IS; the last three shape what a SCORE means.

F INITIAL CROSS-DOMAIN PHYSICS EXTENSION

DISCERN instantiates the same L1–L3 contract in calorimetry and nuclear decay as an initial transfer probe beyond molecular life science. These two tracks are reported separately from the eight-track core and do not enter its 203-task denominator or score.

Calorimetry. The calorimetry track uses real adiabatic-calorimetry measurements of temperature-dependent heat capacity, including crystalline Cu measurements from (Lang et al., 2006), with phase-resolved n-heptane thermodynamic measurements from (Bissengaliyeva et al., 2011) added for the equilibrium analysis. L1 tests whether agents preserve valid heat-capacity structure while detecting defects such as duplicated or missing measurements, unit mismatches, temperature–heat-capacity misalignment, and uncertainty-scale errors. L2 carries the Cu heat-capacity substrate into an adiabatic Cu–heptane equilibrium problem and tests whether agents detect incorrect enthalpy references, temperature-dependent enthalpy drift, and phase-blind interpolation across the heptane melting transition. L3 retains the frozen L2 equilibrium model and asks whether the bulk-Cu heat-capacity prior remains adequate for follow-up nanocrystalline Cu specimen states. The two counterfactual worlds introduce specimen-level heat-capacity increases and regenerate the downstream equilibrium outcomes rather than editing them directly.

The nanocrystalline-Cu counterfactuals are grounded in heat-capacity measurements by Rupp & Birringer (1987) and Tan et al. (2009).

Nuclear decay. The nuclear-decay track uses a public multi-year radiation-measurement dataset from (Goddard et al., 2020), which contains repeated 2048-channel NaI(Tl) Co-60 spectra, acquisition metadata, an independent Mn-54 source, and synchronized environmental measurements. L1 tests whether agents can identify acquisition- and spectrum-level integrity problems, including duplicated or missing measurements, timestamp and live-time errors, inconsistent count summaries, and spectral shifts, while avoiding false alarms when legitimate calibration changes occur. L2 aggregates a validated portion of these same Co-60 measurements into 274 daily observations and tests whether agents can recover the decay rate despite misleading analysis choices or confounds, including an inappropriate detector-efficiency model, Mn-54 contamination, and an unstable fixed-channel photopeak analysis. L3 carries forward this frozen 274-day dataset. The real world retains the observed Co-60 decay pattern, while two counterfactual worlds assume fully ionized Co-60 under stellar conditions, apply literature-grounded changes from Gupta et al. (2023). to the effective decay rate, and regenerate the corresponding count trajectories while keeping the observation schedule and control measurements fixed.

Table 11: Frozen Physics-extension inventory. Boundary tasks contain no injected defect and test whether agents avoid overinterpreting valid physical behavior. L3 counterfactuals are evaluated only after confirming the relevant pre-task prior. The auxiliary detector calibration follows Akar Tarım et al. (2018); stellar-decay counterfactuals follow Gupta et al. (2023).

Track	Level	n	Control and challenge structure
Calorimetry	L1	8	1 clean; 6 injected integrity defects; 1 no-defect boundary
	L2	4	1 clean; 2 data confounds; 1 procedure/analysis trap
	L3	3	1 real; 2 prior-gated injected counterfactuals
Nuclear decay	L1	8	1 clean; 6 injected integrity defects; 1 no-defect boundary
	L2	4	1 clean; 1 data confound; 2 procedure/analysis traps
	L3	3	1 real; 2 prior-gated injected counterfactuals
Total		30	16 L1; 8 L2; 6 L3

Frozen task inventory. Table 11 summarizes the Physics tasks using the same control-versus-challenge structure as the biology registries. At L1, each track includes one clean task, six injected integrity challenges, and one no-defect boundary task that tests whether agents avoid treating legitimate physical behavior as an error. At L2, each track includes one clean analysis and three challenged analyses involving either a data confound or a misleading analysis procedure. At L3, each track includes one real/control world and two counterfactual worlds. Before evaluating a counterfactual, we first confirm that the model holds the scientific expectation that the new evidence is

intended to challenge. All eight fully evaluated models passed these prior checks, so every Physics L3 world was included in the analysis.

Table 12: Physics-extension performance on the 0–100 scale. All eight models have complete coverage: $n = 16$ L1, $n = 8$ L2, and $n = 6$ eligible L3 tasks per model.

Model	L1	L2	L3
GPT-6 Astra	70.8	79.2	72.4
GPT-5.6-sol	85.4	70.8	75.9
Claude Opus 5	77.1	54.2	69.2
Kimi K3	81.3	66.7	76.7
GLM-5.3	79.2	54.2	60.9
GLM-5.2	77.1	58.3	74.7
DeepSeek V4 Pro	75.0	62.5	72.7
GPT-4.1	22.9	0.0	17.6
Mean	71.1	55.7	65.0

Physics-extension results. Table 12 reports the finalized stage scores for the same eight fully evaluated models used in the life science tracks. Scores average the two Physics tracks and are linearly rescaled from 0–3 to 0–100. Each model has 16 eligible L1 tasks, 8 L2 tasks, and 6 L3 tasks; across eight models, the corresponding denominators are 128, 64, and 48 model–task evaluations. The model leading the extension changes by stage: Sol has the highest Physics L1 mean, Astra the highest L2 mean, and Kimi the highest L3 mean.

Comparison with the life-science core. The Physics extension suggests that several DISCERN behaviors transfer beyond the life-science tasks, while others depend on scientific context. Across the same eight models, mean Physics performance was 71.1 at L1, 55.7 at L2, and 65.0 at L3, compared with 74.1, 68.4, and 64.8 across the eight life-science tracks. Thus, aggregate L1 and L3 performance was similar across the two settings, while L2 performance was lower in Physics, suggesting the models had a harder time verifying the validity of the physical analyses. Model-level variation was also consistent: no model led all three Physics stages, with Sol highest at L1, Astra at L2, and Kimi at L3. Some model behaviors transferred across domains. Kimi, which frequently favored counter-prior evidence in the life-science tasks, also demonstrated that behavior in the Nuclear counterfactual worlds and achieved the highest Physics L3 mean. Other behaviors were less stable. Astra’s strong tendency to abstain in life-science L3 did not extend uniformly to Physics: in the Nuclear counterfactuals, it accepted the evidence-supported change in Co-60 decay behavior while remaining cautious about its causal interpretation. Opus likewise moved from the highest life-science L2 mean to substantially lower Physics L2 performance. These differences suggest that model behavior depends on the scientific context rather than each model having one fixed working style.

G COMPLETE L1 AND L2 RESULTS

Figure 4A in the main Results reports clean-control decisions; Figure 15 below reports challenged-task outcomes. This appendix reports task-level difficulty across all 32 L1 and L2 categories (Figure 17). All panels use the recorded L1 states and current L2 scores under the shared mapping.

Detailed L1 handling outcomes. Across challenged L1 evaluations, 56.2% achieve full diagnosis and appropriate action, 21.7% identify the wrong cause, 7.9% diagnose the problem correctly without acting, and 1.4% miss the problem entirely. Opus reaches full diagnosis and action in 66.2% of challenged evaluations compared with 57.7% for Astra. Abstention rates are 9.9% and 8.5%, respectively, although these include both appropriate and incompletely justified abstentions.

Task difficulty also varies substantially across scientific settings. All models perfectly handle seven L1 challenges, including duplicate records, scale errors, and fetch failures. In contrast, wrong diagnoses occur in 48.4–51.6% of detectable-defect evaluations in CRISPR, single-cell RNA sequencing, and epigenomic tracks, compared with 4.7–12.5% across the remaining five tracks. All ten lowest-scoring defect tasks occur in these three settings. Eight additional challenges require with-

holding a defect claim or recognizing insufficient evidence; these average 53.6, with 37.5% of their 64 evaluations overdiagnosing a problem.

Detailed recognition–calibration gaps. Result calibration averages 64.5 across agents and is the lowest-scoring of the four shared L2 dimensions for seven of eight models. Among 425 challenged L2 evaluations receiving full issue-recognition credit, 126 (29.6%) receive exactly half calibration credit. This proportion ranges from 19.4% for Opus to 37.8% for DeepSeek and is 47.2% for statistical traps, 34.1% for data confounds, and 20.6% for tool traps. Because calibration can reflect the finalized handling of the analysis rather than conclusion wording alone, half credit does not uniquely identify overconfidence. These outcomes distinguish recognizing a limitation from carrying it through methodological handling and the final conclusion.

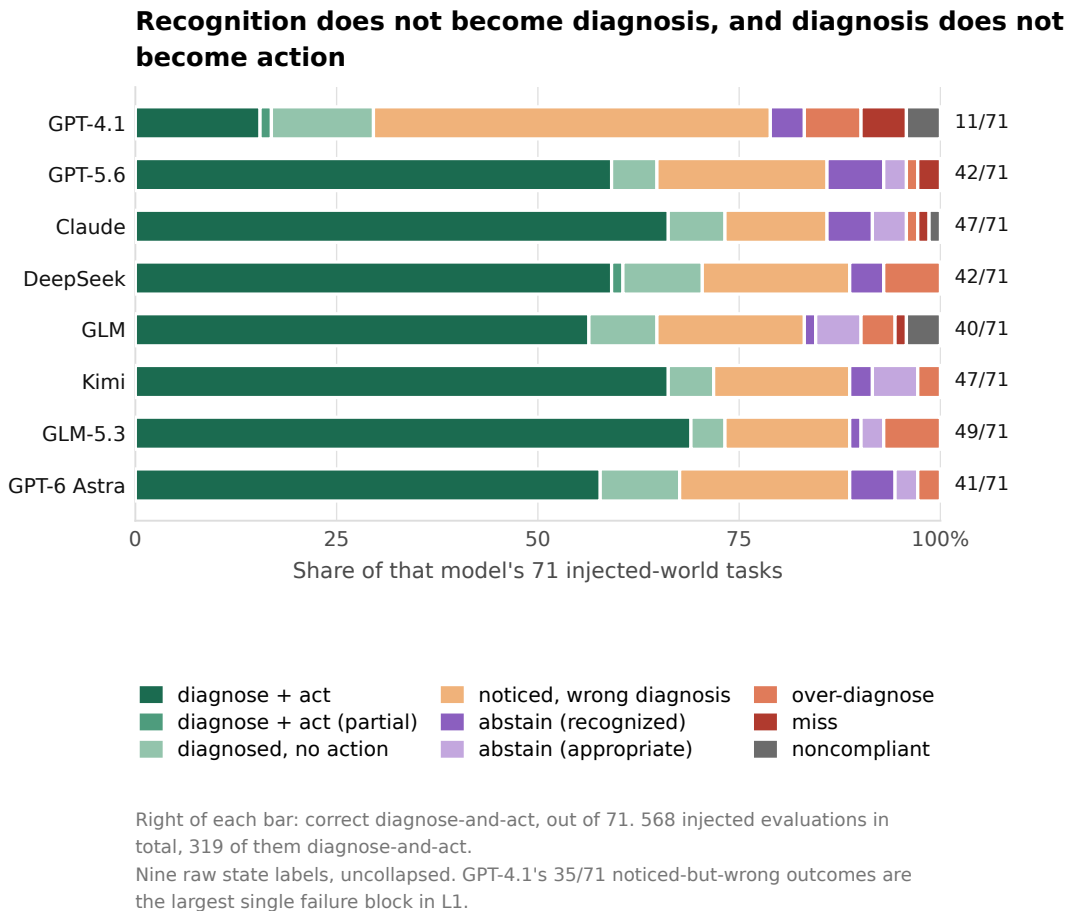


Figure 15: L1 integrity challenges across eight models: recorded behavioral states on 71 challenged tasks per model, separating complete handling, partial responses, wrong diagnoses, abstentions, and other failures. All 568 evaluations are retained.

Analytical performance varies within tracks

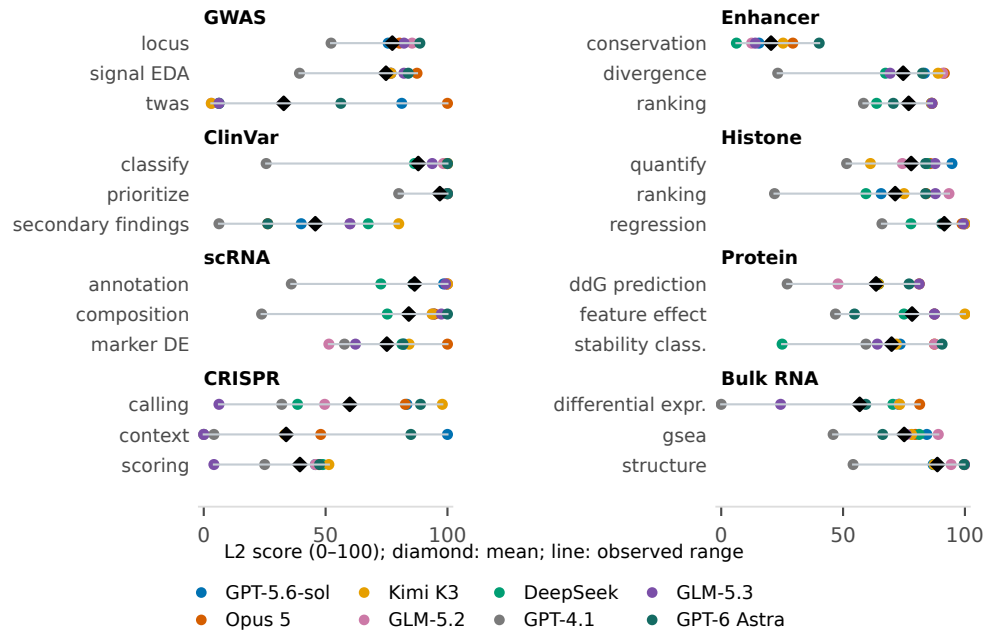


Figure 16: Model scores and across-model means for all 24 L2 analysis families under the final shared meta-rubric, normalized to 0–100. Two columns retain all eight tracks. Diamonds show model means and lines show observed model ranges.

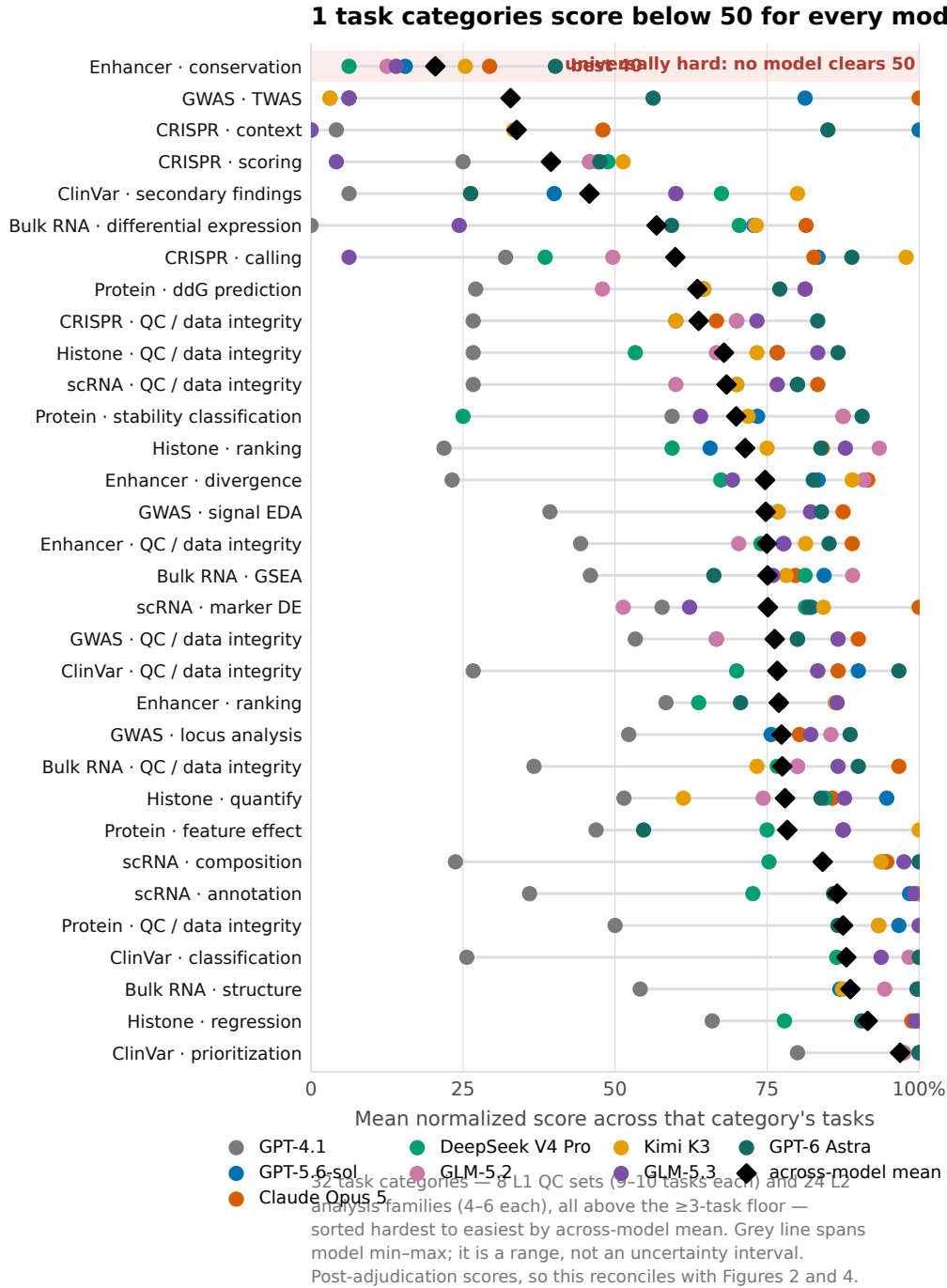


Figure 17: Task-category difficulty across all L1 integrity sets and L2 analysis families.

H COMPLETE L3 RESULTS AND CASES

H.1 COMMITMENT, CONDITIONAL NOVELTY, AND YIELD

For GPT-6 Astra, two L3 worlds are excluded for prior eligibility, leaving 20 eligible outcomes. One contains a hypothesis and 19 are abstentions. These exclusions concern the world-specific eligibility rule; they do not independently establish a general tendency to avoid stating beliefs. For

eligible outcomes E_m and committed outcomes C_m , define

$$p_m = \frac{|C_m|}{|E_m|}, \quad \bar{N}_m = \frac{100}{3|C_m|} \sum_{w \in C_m} N_{mw}, \quad Y_m = p_m \bar{N}_m = \frac{100}{3|E_m|} \sum_{w \in C_m} N_{mw}. \quad (3)$$

Conditional novelty $\bar{N}_m$ is undefined when no hypothesis is produced; yield Y_m is then zero. Abstentions contribute no produced novelty credit, without being assigned an unobserved novelty grade. These diagnostics weight eligible worlds equally. In contrast, the headline L3 score weights tracks equally and includes partial credit for abstention, capped at 2.5 in native rubric units. Yield measures novelty credit under the benchmark rubric, not experimentally validated usefulness or a discovery rate.

Table 13: L3 score and hypothesis output. Scores and novelty use 0–100; commitment is a percentage. The headline is track-balanced; the remaining columns use eligible worlds. Fable is excluded because its L3 coverage is incomplete. Conditional quality based on one Astra hypothesis should not be interpreted as a reliable estimate of typical hypothesis quality.

Model	L3	Committed/eligible	Commit (%)	$\bar{N}$	Y
GPT-6 Astra	72.8	1/20	5.0	66.7	3.3
GLM-5.3	71.1	15/22	68.2	40.0	27.3
GPT-5.6-sol	70.0	4/22	18.2	41.7	7.6
Kimi K3	69.1	16/21	76.2	31.2	23.8
GLM-5.2	63.1	14/22	63.6	28.6	18.2
Claude Opus 5	63.0	17/21	81.0	41.2	33.3
DeepSeek V4 Pro	58.4	12/22	54.5	16.7	9.1
GPT-4.1	51.0	15/22	68.2	13.3	9.1

Figure 4B,C shows these diagnostics for all eight models; Appendix Figure 14 shows the world-level novelty matrix. Astra’s abstention classification does not establish that it rejected the supplied evidence. Its conditional novelty estimate is based on one hypothesis.

Detailed commitment and novelty outcomes. Across eligible L3 tasks, hypothesis commitment ranges from 5.0% for Astra to 81.0% for Opus. Commitment is 58.3% on counterfactual tasks and 48.4% on real-data tasks, although differences between the task worlds prevent attributing this contrast to conflicting evidence alone. In 14 of Astra’s 19 abstentions, the response explains why the available evidence does not yet justify a hypothesis rather than simply declaring the evidence insufficient.

Among 94 submitted hypotheses, mean novelty is 30.1, below causal calibration (62.4) and the remaining scientific-reasoning dimensions (all at least 73.4). Six of 172 eligible outcomes receive maximum novelty credit, four of them in the DUSP4 task. To summarize both production frequency and conditional novelty, novelty yield is defined as

$$P(\text{commit}) \times \mathbb{E}[N_{100} \mid \text{commit}].$$

Sol and Opus have similar conditional novelty scores (41.7 and 41.2) but novelty yields of 7.6 and 33.3 because their commitment rates differ substantially.

Critique and concession outcomes. Across all committed outcomes, agents concede at least one reviewer point in 20.2% of cases, with 84.2% of first concessions occurring in the first review round. Model-level concession rates range from 0% for Sol and Astra to 41.7% for DeepSeek. A recorded concession does not necessarily imply that the final hypothesis is better supported, just as persistence does not necessarily indicate resistance to valid criticism. Complete round-level outcomes and case studies, including the DUSP4 example summarized in the main text, are reported here.

H.2 HIGHEST-SCORING MODEL–WORLD OUTCOMES

Table 14 reports the top of the frozen L3 distribution. Because three outcomes tie at 2.60, selecting exactly five would break a tie arbitrarily; we therefore show the top three and all three outcomes tied for fourth. Here D is evidence fidelity, Q is scientific reasoning, and N is valid novelty.

Table 14: Highest-scoring aggregate-eligible L3 model–world outcomes, shown here on the native 0–3 rubric scale.

Rank	Composite	<i>D</i>	<i>Q</i>	<i>N</i>	Track and world	Model
1	3.00	3.0	3.0	3	Protein: cf_stabilizing	Claude Opus 5
2	2.87	3.0	2.6	3	Histone: cf_bivalent_native	DeepSeek V4 Pro
3	2.80	3.0	2.4	3	Histone: cf_bivalent_native	Kimi K3
4=	2.60	3.0	2.8	2	Bulk RNA: cf_mhc2_injected	Claude Opus 5
4=	2.60	3.0	2.8	2	Histone: cf_bivalent_injected	Claude Opus 5
4=	2.60	3.0	2.8	2	scRNA: cf_cd68_mono	Claude Opus 5

The two highest-scoring non-protein outcomes arise from the same native DUSP4 world but take distinct routes. DeepSeek V4 Pro narrows the result to a provisional threshold-defined state and proposes locus-specific H3K27me3 measurement followed by an EZH2/PRC2 perturbation time course. Kimi K3 explicitly withdraws an initial uniqueness claim, then re-establishes DUSP4 as rank 1 in H3K27me3 among 1,021 strong-H3K4me3 genes. It further distinguishes a categorical repressed tail from a weak graded relationship and proposes sequential ChIP to test same-locus co-occupancy. Together, the pair shows that one evidence-rich case can support different valid hypotheses while review forces both toward a bounded claim and a discriminating experiment. The tied ST13 and MHC-II cases show similar discipline on injected evidence, but DUSP4 is the stronger flagship because its relationship is native.

Table 15: Paraphrased DUSP4 revision by DeepSeek, complementing Section 4.3.

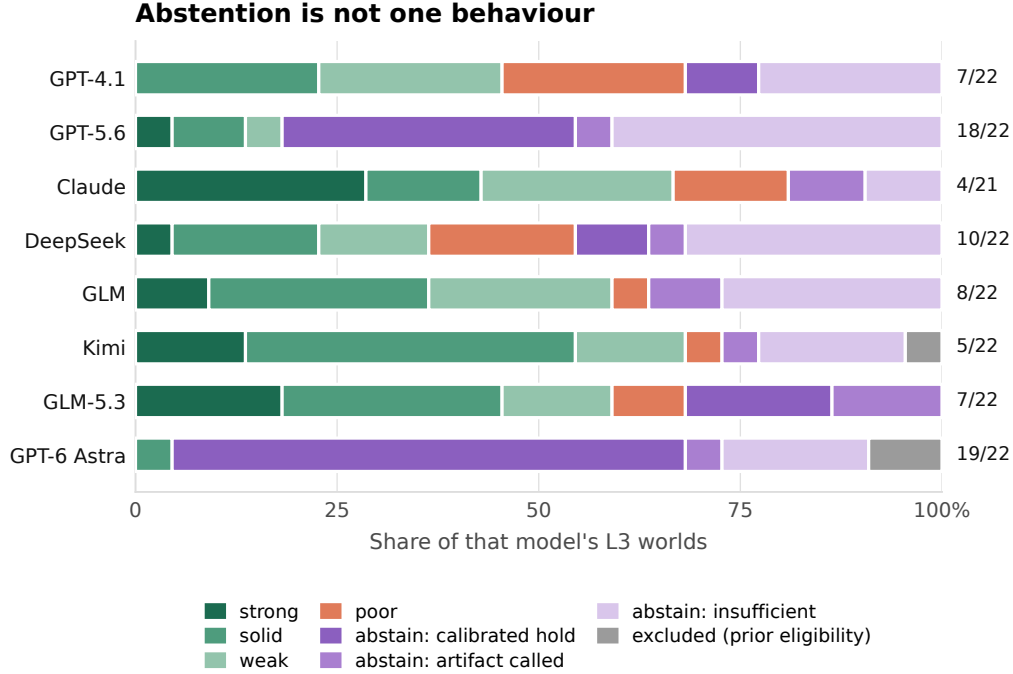
Step	Recorded reasoning
Initial claim	Repression overrides the activation-associated signal.
Limitation	Signals measured over different-sized regions do not establish that they occur together.
Revised claim	Retain the observed association, but treat the proposed mechanism as an explanation to test.
Next test	Check where the repressive signal occurs and whether reducing repression raises gene expression over time.

Eight-model response diagnostics. Among 94 committed outcomes, 29 receive novelty credit of at least two-thirds of the maximum (30.9% of committed responses, 16.9% of 172 eligible outcomes). This threshold can credit a strong hypothesis or a useful next test and does not identify 29 independently validated experiments. It is reached by 25/63 committed counterfactual responses (39.7%) and 4/31 committed real-world responses (12.9%), whose mean novelty scores are 36.5 and 17.2. All six maximum-novelty responses are counterfactual, and four concern DUSP4, which averages 88.2 composite credit across seven eligible models. These are task-weighted descriptive comparisons, not causal effects of world design.

Of 94 hypothesis-producing outcomes, 19 record a concession, 16 first in round one. The log records the first round with a concession flag, not necessarily abandonment of the hypothesis. The exported reviewer rejection fraction describes the final recorded review round. At least one reviewer rejects in 18 cases; a majority rejects in ten. These final-round judgments cannot establish that rejection caused an earlier concession. The `hyp_drift` field is one minus the `SequenceMatcher` similarity between initial and final hypothesis text. Its median of 0.94 therefore indicates textual difference, not stability, and does not measure semantic narrowing.

H.3 ABSTENTION IS NOT ONE BEHAVIOUR

Figure 18 decomposes the binary commit/abstain split used in Section 4.3 into the recorded final states; treating them as one category would overstate how uniformly models decline.



Right of each bar: abstentions / graded outputs, including recorded exclusions.
 Opus has 21 outputs: one pre-task exclusion is absent. State labels describe the recorded response, not independently validated scientific correctness.

Figure 18: Recorded final L3 states per model, decomposing the commit/abstain split used in Section 4.3. Bars include three recorded eligibility exclusions; Opus’s absent pre-execution exclusion is omitted, leaving 21 graded outputs for that model.

I PROMPTS, TOOLS, AND EXECUTION ENVIRONMENTS

I.1 REPORTING CONTRACT AND BUDGETS

Within an analysis family, every world receives the same prompt and the same structured reporting schema; only the world directory differs. The schema is defect-agnostic: clean and perturbed worlds request identical output fields, so the form of the question does not reveal whether an issue is present. L1 and L2 run in a sealed Python sandbox with the permitted data, documentation, and domain tools on `PATH`, a 20-step execution budget, and no network access. L3 supplies the evidence dossier, allows three to six agent steps, and then runs three adversarial review rounds with a reviewer ensemble that is fixed across all evaluated models. The harness records every executed code cell, tool call, and token meter reading. Complete verbatim prompts, tool inventories, and per-track sandbox manifests ship with the released registries. Per-task workspaces use a scrubbed environment and read-only software mounts. Gold files, graders, and sibling worlds are outside the task mounts. These restrictions control runtime access, not prior exposure to public source data.

Later-model generation budgets. GLM-5.3 was evaluated later and was not strictly budget-matched to the earlier evaluations. After hidden-reasoning truncation was diagnosed, the per-call generation ceiling was raised from 8,192 to 32,768 tokens and selected affected tasks were rerun under the approved run protocol. GPT-6 Astra’s generation ceiling recorded by preflight is 32,768 output tokens per call; its evaluation is likewise not strictly budget-matched to the earlier evalua-

tions. Both models complete all 203 tasks (79 L1, 102 L2, and 22 L3), and all 40 run-level graded files per model have corresponding verifier files.

I.2 MODELS

The eight models were accessed through first-party APIs: OpenAI (gpt-4.1, gpt-5.6-sol, gpt-6-astra), Anthropic (claude-opus-5), DeepSeek (deepseek-v4-pro), Zhipu Big-Model (glm-5.2, glm-5.3), and Moonshot (kimi-k3), under the identifiers used throughout the paper. All eight tracks were frozen by 2026-08-22; per-run timestamps and harness-version records are preserved in the run manifests.

I.3 A COMPLETE WORKED TASK

One task, end to end, from the frozen artifacts: the Histone track’s L2 *quantify* family, world `broad_mark_narrow_default` (a tool or procedure-output trap). The agent prompt is shared by all four worlds of the family (abridged; paths shortened):

```
You are a computational epigenomics researcher. GOAL:
quantify the per-gene promoter-level signal of the TARGET
histone mark from the raw ChIP-seq alignments, and determine
how the mark relates to gene expression. Read meta.json for
the target mark and file layout. [...] World directory:
meta.json (target mark + layout), genes.csv (gene_id,
chrom, start, end, tss, gene_expression), and alignments/
holding the mark’s ChIP BAM AND the matched input-control
BAM (both indexed; chromosome-subset of real ENCODE
K562 data). Python sandbox: call run_python(code).
[...] peak-calling and coverage tools are installed
and on PATH (macs3, deepTools, samtools) --- choose
your approach yourself and VERIFY the output is sound
before trusting it. [...] Return ONLY this JSON:
{output_path, method {tool, normalization, peak_mode,
window}, signal_expression_spearman, direction, domains {n,
total_mb}, issues_found, analysis_usable, trajectory}
```

The controlled delta. In this world, `meta.json` sets the target mark to the broad repressive mark H3K27me3 and requests the domain landscape. The trap is a tool default: MACS3’s narrow peak-calling mode completes without error on this input but fragments broad domains, returning a 3.4 Mb landscape where the reference solver’s broad-mode golden totals 6.4 Mb over 19,352 domains. Both pipelines finish and produce a plausible table, so the defect is silent.

The recorded rubric. The manifest stores the grading rule with the world: the per-gene CSV is value-checked against the solver golden; declaring a solvable world unusable caps the score at 1; for this world, 3 requires a broad or domain-aware mode, a total within $\pm 35\%$ of the 6.4 Mb golden, and the narrow-default pitfall named; a narrow, unqualified landscape scores 0.

Two recorded outcomes. GPT-5.6-sol called domains with MACS3 in broad mode, corrected the effective genome size to the chromosome panel, reported 19,352 domains totalling 6.43 Mb, and named the narrow-default pitfall; the deterministic value check awarded the top band, and human adjudication of the trajectory lowered result calibration, leaving frozen components $A = 0.99$, $I = 1.0$, $M = 1.0$, $C = 0.5$ and a world score of 87.1. Claude Opus 5 also chose broad mode and named the pitfall, but its reported landscape totals 28.21 Mb against the 6.4 Mb golden, so analytical correctness is 0 and the world score is 50.0 despite full issue-recognition and methodological-response credit — the accuracy cap of Section 3 in action: recognizing the trap cannot compensate for computing the wrong landscape.

J GRADING, VERIFICATION, AND ADJUDICATION

J.1 BEHAVIORAL STATES AND DIMENSION DEFINITIONS

The main-text scoring definitions use rescaled component scores throughout. The implementation stores L1 state grades and L3 axes on a native 0–3 scale and L2 components on 0–1; these are multiplied by 100/3, 100/3, and 100, respectively, before applying the scoring definitions in Section 3. This change of notation does not change any reported score.

L1 states. The grader distinguishes missing an issue (MISS), noticing a problem without a correct diagnosis (NOTICE-WRONG), diagnosing without acting (DIAGNOSE-NOT-ACT), and diagnosing with an appropriate action (DIAGNOSE-AND-ACT). The last two also have -PARTIAL variants when the diagnosis or action covers only part of the problem. For sound data it distinguishes correct acceptance (CLEAN-CORRECT), excessive caution (OVER-CAUTIOUS), unsupported rejection (FALSE-UNUSABLE), and an invented defect (FABRICATED). When the available evidence cannot establish the issue, it distinguishes recognizing that specific limit (ABSTAIN-RECOGNIZED), abstaining without naming the limit (ABSTAIN-OK), and asserting an unsupported diagnosis (OVER-DIAGNOSE). An absent or noncompliant assessment is NONCOMPLIANT. Partial diagnoses and actions remain distinct from their complete counterparts.

L1 reporting dimensions. The assessment-correctness summary uses the recorded state-level score. Issue recognition counts DIAGNOSE-NOT-ACT, DIAGNOSE-AND-ACT, and DIAGNOSE-AND-ACT-PARTIAL among non-clean evaluations. Methodological response counts the latter two states on the same population. Result calibration reports the fraction of all evaluations outside OVER-DIAGNOSE, FALSE-UNUSABLE, OVER-CAUTIOUS, and NONCOMPLIANT. These are operational summaries of recorded labels, not four independent judgments or a replacement aggregate. In particular, the calibration summary is not a complete measure of uncertainty calibration. Fabrication is included among the L1 profile metrics. Historical scalar adjudications do not imply reconstructed behavioral labels. The fabrication metric uses keyword overlap between reported trajectory claims and recorded code. It is a heuristic signal of potentially unsupported reporting, not proof of invention or a semantic audit of every claim. Fabrication can also affect the assigned behavioral state and task score.

L2 components. Each track maps its recorded checks to analytical correctness, issue recognition, methodological response, and result calibration under the finalized meta-rubric. Checks include numerical agreement, recognition of the injected issue, an appropriate analytical response, and qualification of the reported conclusion. Track-specific mapping documents and scripts record which checks contribute to each component, their normalization, and the handling of missing component evidence. The equal-component world score is averaged within analysis family, followed by an equal-family track average. A legacy scalar grade is not treated as four independently observed component grades.

L3 scientific reasoning and novelty. Scientific reasoning averages five axes: prior awareness (recognizing the expectation challenged), causal calibration (not claiming a mechanism from an association alone), response to critique (revising for valid evidence rather than unsupported pressure), scientific humility (appropriate uncertainty and a specific next test), and citation integrity (real, relevant support for the claims). The judge reads the hypothesis, critiques, rebuttals, and final response against the supplied world evidence. Reviewer votes are not the grading target. Novelty distinguishes no substantive contribution, a generic restatement, one strong explanatory or experimental contribution, and a strong explanation paired with a discriminating test and a defensible relation to prior work. It is a rubric-based assessment, not an exhaustive prior-art search.

Shared evidence in L2 dimensions. Analytical correctness and methodological response receive identical scores in 77.7% of the 816 L2 evaluations across eight models (Pearson $r = 0.87$), reflecting shared evidence and mapping rules. The four dimensions are therefore not statistically independent measurements. Calibration can be derived from the finalized handling ladder rather than a separate assessment of conclusion wording, so partial credit does not uniquely identify overconfidence. The conditional comparisons in Section 4.2 weight evaluations equally and restrict to

challenged tasks with full issue-recognition credit (425 outcomes). Of these, 126 receive exactly half calibration credit. Model and challenge-class percentages use their respective subsets of these 425 outcomes, whereas the headline dimension means follow the family- and track-balanced aggregation.

J.2 TWO-PART METHODOLOGY

Grading is deterministic first and audited second. The primary grade for every objective quantity comes from a deterministic grader that value-checks the agent’s artifact against a solver-produced golden under the rubric stored in that world’s manifest. An independent LLM verifier then re-derives the grade from the same artifacts; every disagreement is adjudicated by a human against the recorded evidence, and the ruling is preserved in the adjudication ledger. Adjudication examines the verifier’s stated reason against the rubric text rather than accepting its score, because a verifier can assert requirements the rubric does not contain; the same reason-over-keyword rule governs prior-gate admissibility decisions. Historical Bulk RNA and Histone rulings recovered from authoritative totals are kept in a separately labeled reconstruction ledger rather than represented as recovered original human notes. Ten of 72 model–world rows in one scRNA L2 family lack a verifier row, although their deterministic grades exist. Three frozen L3 cells were produced under a documented pre-fix parsing harness. One Claude L3 world was excluded before execution and one graded Kimi L3 outcome was subsequently marked prior-inadmissible; these are different mechanisms.

The finalized L2 meta-mapping was reconciled with all 612 per-world exports before this revision. A missing issue flag had previously caused 34 perturbed worlds to enter the clean branch. Correcting that classification using the existing manifests changed 70 world scores and 23 of 48 track–model L2 means. L1, L3, and all recorded grader/verifier evidence remained unchanged; only L2 and its derived aggregates changed. The prior aggregate, corrected mapping, and per-world changes are preserved in a dated ledger. No agent rerun or new transcript judgment was required.

Later-model adjudication records. For GLM-5.3, 37 recorded human rulings are applied, including eight L3 axis rulings checked against stored grades before aggregation. L1 scalar adjudications enter stage means without inferring replacement behavioral states. The ledger adapter uses the existing manual-override rule so the accepted GWAS Signal-EDA core ruling reaches its corresponding dimensions; result calibration remains separately recorded. For GPT-6 Astra, the ledger contains 17 L1 scalar rulings, 11 L2 rulings, and two prior-gate rulings, with no L3 score-axis rulings. Zero L3 verifier discrepancies were *flagged*; this does not mean exact agreement on every axis. Five worlds contain differences below the verifier’s flagging thresholds. Recorded final grades remain authoritative. Both models use the finalized L2 component-mapping policy; a legacy scalar is not itself a new four-dimension score. Reporting applies existing evidence without new agent runs or transcript grading.

J.3 L3 ABSTENTION SCORING

An L3 abstention is not assigned a novelty score. Instead, the grader records $D_{\text{abst}} \in \{0, 2\}$: credit requires engaging with the target finding while preserving the supplied evidence. Its reasoning score Q_{abst} averages prior awareness, scientific humility, and citation integrity on the native 0–3 scale. The displayed abstention score is

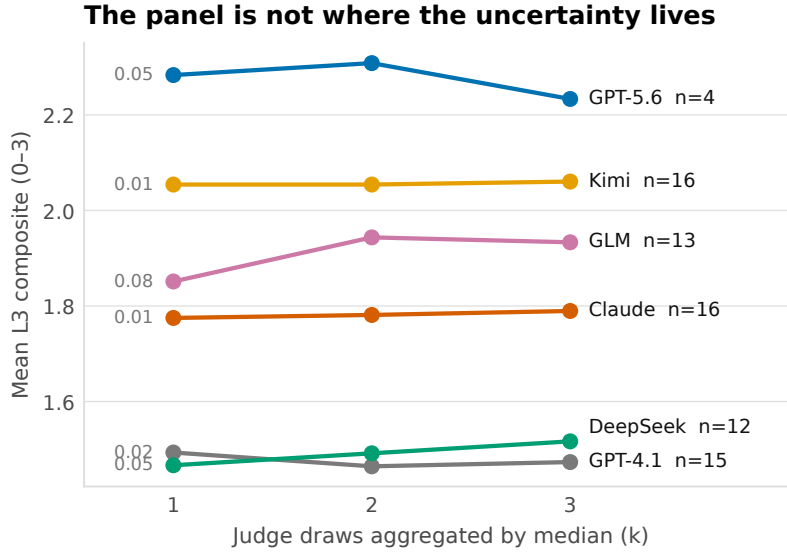
$$S_{\text{abst}}^{(3)} = \frac{100}{6} (D_{\text{abst}} + Q_{\text{abst}}). \quad (4)$$

Its maximum is 83.3/100 (2.5/3 natively). An agent can therefore receive credit for a justified decision not to propose a hypothesis without receiving novelty credit or reaching the full L3 ceiling. Prior-inadmissible outcomes are excluded before aggregation, irrespective of whether a hypothesis was produced.

J.4 JUDGE-PANEL STABILITY

The interpretive L3 axes are scored by an LLM judge with three draws aggregated by median. Figure 19 replays this aggregation on the paired records that stored per-draw values. Recomputing at

$k = 3$ reproduces the recorded composites almost exactly — the small residual is human adjudication, which no replay can reproduce — and moving from one draw to three shifts model means only marginally on this recorded subset. This limited draw sensitivity does not validate the rubric’s scientific construct or exclude systematic judge bias; rubric interpretation and adjudication remain separate sources of uncertainty.



79 of 131 L3 worlds stored per-draw values; the rest predate draw persistence. 3 more are dropped because a null first draw makes them scorable at $k=3$ but not $k=1$ — the set must be PAIRED or the Δ is meaningless. This is a subset and its ranking is not the published one; GPT-5.6-sol rests on only four worlds. Recomputing at $k=3$ reproduces the recorded composite to 0.042 on average — the residual is human ADJUDICATION, which no replay can reproduce. Numbers at the left of each line are $|\Delta|$ from $k=1$ to $k=3$.

Figure 19: Replay of the judge-draw aggregation on the paired L3 records that stored per-draw values, supporting the validation claims of Section 3; this subset’s ranking is not the published leaderboard.

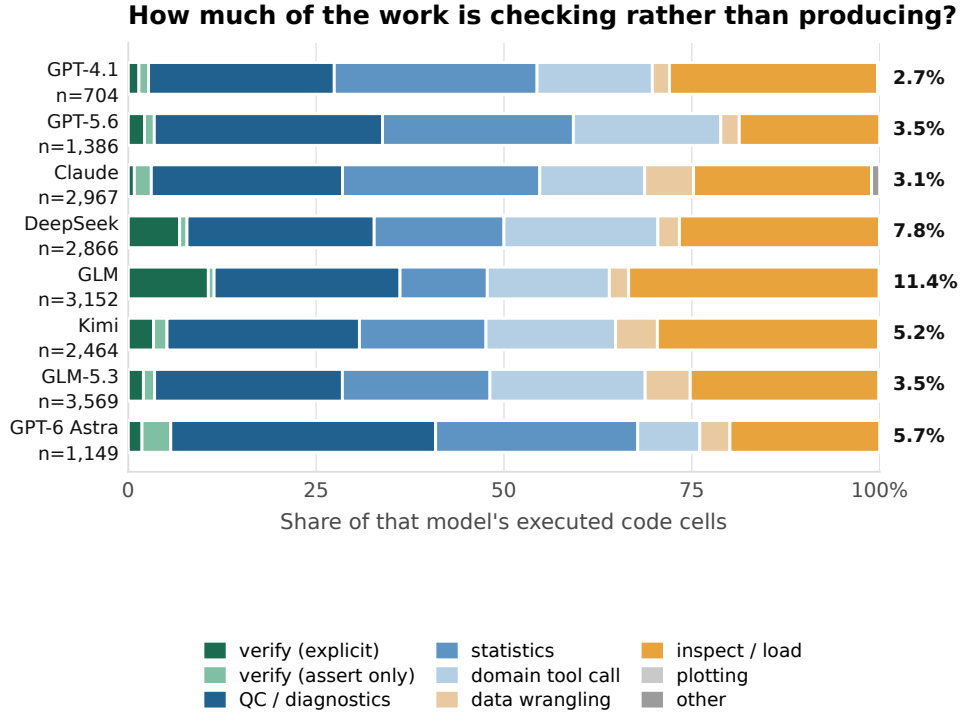
K ADDITIONAL AUDITABLE CASES AND ANALYSES

K.1 THE TWAS SILENT FAILURE, FROM THE RECORDED GRADER STATES

Section 4.2 reports the TWAS family as the concrete case of silent success. The frozen grader states show the mechanism directly. On the clean TWAS world, Claude Opus 5 and GPT-5.6-sol reach the recorded state `TWAS-CORRECT`: their executed pipelines process the full weight set (19,086 and 18,580 SNPs respectively) and their z -scores correlate perfectly with the golden. GLM-5.2 and GPT-4.1 end in `TWAS-EMPTY` (*silent fail*): the software completes and returns a result table, but zero usable SNP weights entered it, so the plausible-looking output contains no evidence. DeepSeek V4 Pro and Kimi K3 end in `NO-OUTPUT`. The states are recorded per world in the graded artifacts; the same split recurs on the perturbed TWAS worlds. This is the distinction the benchmark is built to expose: a completed invocation and a valid analysis are different events, and only output verification separates them.

K.2 HOW MUCH OF THE EXECUTED WORK IS CHECKING?

Figure 20 classifies every executed L1 and L2 code cell by surface function. Verification accounts for 2.7–11.4% of executed cells depending on the model.



Right of each bar: total verification share. 18,257 executed code cells across L1 and L2. Surface classifier over the raw code string — word-boundary regexes, first match wins — so it reads form, not meaning. Verification is split because a bare `assert` is often defensive rather than scientific. Plotting is ~0 because the sandbox is headless. 'other' is 0.3% of cells.

Figure 20: Surface-function classification of every executed L1 and L2 code cell; the classifier is a documented keyword rule over the raw code string, so it reads form, not meaning.

L COST, PROVENANCE, LICENSES, AND RELEASE DETAILS

L.1 ACCOUNTING PROTOCOL

The accounting protocol was frozen on 2026-08-24. (1) *Boundary*: the headline is the benchmarked model’s own metered usage; the L3 reviewer ensemble is shared infrastructure and is reported separately. (2) *Unit*: token counts are the archival record; dollars are rendered from a dated rate card and are recomputable at any later price. (3) *Pricing*: first-party, uncached, non-batch list rates active on 2026-08-24 (Table 16); the active rate is used even when promotional, peak pricing is used where the provider’s time band was not retained, and GLM’s CNY rate is converted at the dated Federal Reserve H.10 rate. (4) *Retries*: only successful calls contribute tokens. Grading and verification are unmetered and out of scope: every figure here is the cost of producing results, not of scoring them. All dollar figures are upper bounds because provider-side cache discounts are not logged. DeepSeek adopted time-of-day pricing before the accounting date; the card conservatively uses its peak rate because the billing band was not retained.

Table 16: The dated rate card: first-party list rates active on 2026-08-24, USD per million tokens. GLM-5.3 and Astra use the later dated receipts documented below. Reasoning tokens are already included in the output counter. GLM-5.2 is converted from CNY 8/28 per million tokens at 6.7219 CNY/USD; DeepSeek uses the peak of its time-of-day schedule.

Model	Provider (first-party API)	\$/Mtok input	\$/Mtok output
GPT-5.6-sol (OpenAI, 2026c)	OpenAI	4.00	20.00
GPT-6 Astra (OpenAI, 2026a)	OpenAI	10.00	50.00
GPT-4.1 (OpenAI, 2026b)	OpenAI	2.00	8.00
Claude Opus 5 (Anthropic, 2026b)	Anthropic	5.00	25.00
Claude Fable 5 (Anthropic, 2026a)	Anthropic	10.00	50.00
Kimi K3 (Moonshot AI, 2026)	Moonshot AI	3.00	15.00
GLM-5.2 (Z.ai, 2026a)	Zhipu BigModel	1.19	4.17
GLM-5.3 (Z.ai, 2026b)	Zhipu BigModel	1.19	4.17
DeepSeek V4 Pro (DeepSeek, 2026b)	DeepSeek (peak)	1.32	3.96

L.2 LATER-MODEL ACCOUNTING AND SOURCE RECORDS

GLM-5.3’s derived record is `results/new_models/ROLLUP_glm-5.3.json`, produced by `build_glm53_report.py`. Astra’s is `results/new_models/ROLLUP_gpt-6-astra.json`, generated by `build_astra_report.py`. They retain source hashes, coverage, applied rulings, mapped components, per-family and per-track aggregates, and cost assumptions, with checks that original evidence and score records are unchanged.

GLM-5.3 accounting. The final canonical outputs record 65,455,719 input and 9,759,243 output tokens over 203 tasks. These counts include only the evaluated agent, not reviewer calls, preflight checks, or superseded attempts, so they are not the total cost of developing and rerunning the evaluation. The BigModel pricing page, checked on September 7, 2026, lists CNY 8 per million input tokens and CNY 28 per million output tokens for GLM-5.3. Using the paper’s fixed conversion of CNY 6.7219 per USD gives an uncached estimate of \$118.55, or \$0.58 per task. Dividing the unrounded per-task estimate by the 0–100 Overall score gives \$0.0083 per Overall point. Cache discounts are not inferred from unrecorded cache hits. These are estimates at the retrieved list rates, not invoice charges or a reconstruction of the cost of all evaluation attempts. The separate rate card `results/new_models/pricing_glm-5.3.json` retains the source asset, its hash, and the price fields; previously recorded rates and cost estimates for other models are preserved.

GPT-6 Astra accounting and long-context uncertainty. Final agent outputs record 8,944,538 input tokens and 918,019 output tokens across 203 tasks. Reviewer calls, preflight checks, and superseded attempts are excluded. The official model rate card, checked September 7, 2026, lists standard rates of \$10 per million input tokens and \$50 per million output tokens. Above 272,000 input tokens per request, the multipliers are 2 for input and 1.5 for output. Cache discounts are not inferred. Usage is retained per task rather than per request. Only one task exceeds the input threshold in cumulative usage, which does not prove any individual request crossed it. Applying standard rates throughout gives \$135.35. Conservatively applying the long-context multipliers to all usage of that one task gives \$139.69, an upper bound under these uncached rates. The leaderboard uses this upper bound: \$0.69 per task and \$0.0088 per Overall point. These are list-price estimates, not invoice charges. The separate rate card and derivation are retained with the Astra roll-up.

L.3 TOKEN EFFICIENCY AND ITS INTERPRETATION

For the fixed 203-task suite, let T_m denote the sum of recorded agent input and output tokens. Table 1 reports

$$E_m^{\text{tok}} = \frac{S_m^{\text{core}}}{T_m/10^6}. \quad (5)$$

This is Overall points per million tokens for the full suite, not the score gained by spending an additional million tokens. It is undefined for direct Fable because no full-suite Overall is assigned. The replay denominator includes recorded Fable attempts plus the selected Opus outputs. Original

artifacts also contain tokens per quality point using *mean tokens per assigned task*. That convention differs by the fixed 203-task denominator and is not inverted to populate this column.

Input includes context processed across successive calls, not just unique prompt text. Output includes reasoning tokens where the provider includes them in its output meter. Historical proxy estimates remain labelled in the accounting records. Provider tokenizers differ, and token counts do not recover internal FLOPs, energy use, or hidden computational depth. This ratio therefore measures recorded resource use relative to the rubric score, not semantic information density or the efficiency of producing experimentally validated discoveries. Because Overall is an average but tokens are a suite total, ratios from different-sized benchmark suites should not be compared directly.

Parameter-based explanations require a different experiment. Total stored parameters, active parameters per token, and repeated applications of shared parameters are different quantities. Reusing a block increases computational depth without adding independently trained weights (Geiping et al., 2025). An “effective parameter count” would need an explicit calibrated definition and controlled comparisons of training data, post-training, inference budget, and tasks. Neither those controls nor a generalization experiment are present here. The public Astra model documentation does not specify a recurrent-depth architecture (OpenAI, 2026a).

L.4 RECORDED CALLS AND CODE EXECUTIONS

The Calls column in Table 1 sums the evaluated agent’s recorded API calls across L1–L3. Separate reviewer, grader, and verifier calls are excluded. These counters record returned responses, not a complete network-request log, so failed requests and retries may be absent. Direct Fable has 948 recorded calls. The replay adds 1,084 calls from the selected Opus outputs, giving 2,032, rather than combining the two models’ full-suite totals. Fable’s lower task coverage and prior-gate exclusions remain part of the interpretation of these totals.

Code executions are different from API calls and may include code that ends in an error. Table 17 summarizes the per-task trace field `n_code_cells`. The efficiency exports’ `n_cells` field instead counts task outcomes, not code executions. L3 uses a separate execution and review structure, and its code-cell medians are not populated in the current summary. We therefore report L1/L2 medians rather than infer an all-stage execution total. L3 calls remain included in the main table.

Table 17: Executed code cells per task, median within each stage. Counts include failed code executions and are descriptive, not scored or ranked. API-call totals appear in the main leaderboard.

Model	L1 median	L2 median
GPT-6 Astra	5	6
GPT-5.6-sol	8	6
Claude Opus 5	13	12
Kimi K3	14	11
GLM-5.3	20	20
GLM-5.2	20	18
DeepSeek V4 Pro	15	15
GPT-4.1	2	2

Detailed resource-use patterns. Both GLM models execute a median of 20 code cells per L1 task, compared with eight for Sol and five for Astra. The higher token use of GLM-5.3 relative to GLM-5.2 at nearly identical API-call counts therefore reflects greater token volume per recorded interaction rather than more recorded interactions.

To describe how expenditure varies with measured task difficulty, we rank the 16 L1/L2 track-level categories by their across-model mean score and divide them into difficulty tertiles. The ratio of median token use in the hardest versus easiest tertile is 1.51–1.86 for Sol, DeepSeek, and Kimi, compared with 1.00–1.16 for GLM-5.2, Opus, and GLM-5.3. These ratios describe expenditure patterns and do not establish that an agent recognized task difficulty or deliberately allocated additional effort.

L.5 COVERAGE

Cost coverage is joined to the frozen registries rather than inferred by globbing run directories. For all eight models, the join resolves 1,624 assigned model–task outcomes: 1,623 have executed outputs with usage metadata, while one Claude Opus 5 L3 task was excluded before execution by its prior gate and therefore has zero cost. Nineteen canonical Claude outputs use their stored calibrated character-count proxy; all other canonical outputs carry provider-reported usage. Smoke tests, superseded reruns, abandoned worlds, and future-domain directories are excluded by construction.

Fable’s direct run has 188 canonical invocations: all 181 L1/L2 tasks and seven L3 tasks. The other 15 L3 tasks were not invoked under the refusal-stop protocol. Those invocations contain 8.55M tokens and cost \$120.51 at Fable’s dated \$10/\$50 per-million-token rates, or \$0.594 per 203 assigned tasks. The Fable → Opus replay adds the exact 92 canonical Opus outputs selected by the routing rule (29 L1, 44 L2, and 19 L3), contributing 17.21M tokens and \$117.03. The resulting derived system uses 25.77M tokens and costs \$237.53, or \$1.170 per assigned task and \$0.0156 per chain-quality point. This conservative accounting retains the recorded Fable attempt cost before adding Opus; it is not an observed product-surface bill or a claim about Anthropic’s internal routing charges.

L.6 HEADLINE ACCOUNTING

Table 18: Bench-only cost per model at dated first-party list rates. Quality is the life-science chain score (0–100). \$/task uses the common denominator of 203 assigned tasks per model, including the one pre-execution prior-gate exclusion. Dollar figures are upper bounds because cache discounts were not logged.

Model	Assigned/executed	\$ total	\$/task	Quality	\$/quality pt.
GPT-5.6-sol	203/203	78.12	0.385	75.8	0.0051
Claude Opus 5	203/202	271.86	1.339	75.6	0.0177
Kimi K3	203/203	132.34	0.652	73.2	0.0089
GLM-5.2	203/203	75.20	0.370	67.8	0.0055
DeepSeek V4 Pro	203/203	88.45	0.436	62.4	0.0070
GPT-4.1	203/203	9.33	0.046	39.8	0.0012
GLM-5.3	203/203	118.55	0.584	70.5	0.0083
GPT-6 Astra	203/203	139.69	0.688	78.1	0.0088

Table 18 reports the eight-model accounting, and Table 1 places quality against per-task cost. Total bench cost is \$913.54 over 294.76M tokens. The token and dollar rankings differ substantially: GLM-5.3, DeepSeek V4 Pro, and GLM-5.2 consume the most tokens yet remain relatively inexpensive because their first-party rates sit below the frontier-lab rates. Token counts must therefore never be reported without the dollar axis, and neither without the quality axis.

Efficiency should be optimized together with scientific validity. As scientific agents take on longer analytical workflows, computational efficiency becomes an important practical dimension of AI for Science. Existing benchmarks increasingly report resource use (Koch et al., 2026; Qu et al., 2026), but our results suggest that efficiency should be considered jointly with scientific performance rather than only as a deployment statistic. Similar Overall Scores can require substantially different token and monetary expenditure, while high measured efficiency can also reflect incomplete execution. The relevant objective is therefore not simply to obtain fewer tokens, calls, or dollars, but to achieve reliable scientific work with fewer computations without sacrificing execution completeness or validity. Improving this trade-off could make routine scientific automation more scalable and broaden access for researchers with more limited computational resources, while preserving the checks needed for trustworthy analysis.

L.7 WHERE THE COST LIVES

Figures 21, 22, and 24 report the structure behind the headline. Code executions against token use are shown in Figure 23; a median of 20 code cells does not establish that the interaction budget

was exhausted, because code executions and API steps are different counts. Stage costs price input and output tokens separately rather than allocating total cost in proportion to tokens. L3 is the least expensive stage per task for seven models, but not GLM-5.3. Median L3 token use is $4.9\text{--}7.4\times$ lower for abstentions than for commitments among eligible outcomes; Astra has only one commitment. This is an observational contrast between different tasks and responses, not the causal cost of choosing to abstain. Overall spending still includes usage incurred on subsequently excluded tasks. Cost alone cannot distinguish efficient analysis from limited produced work.

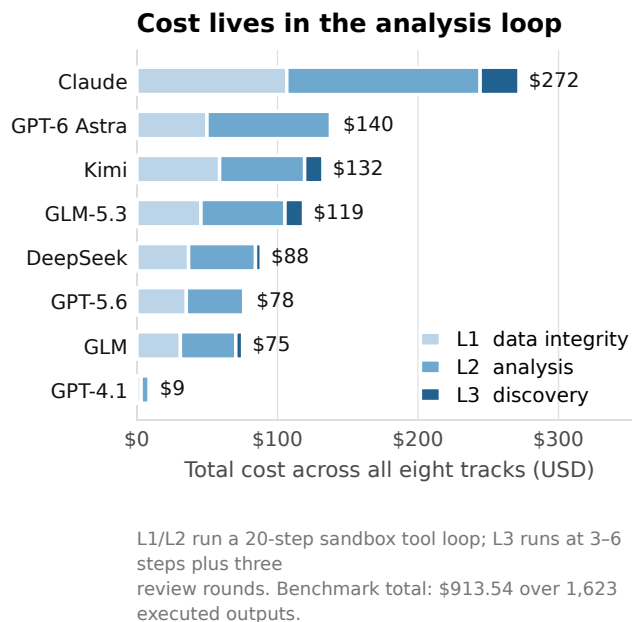
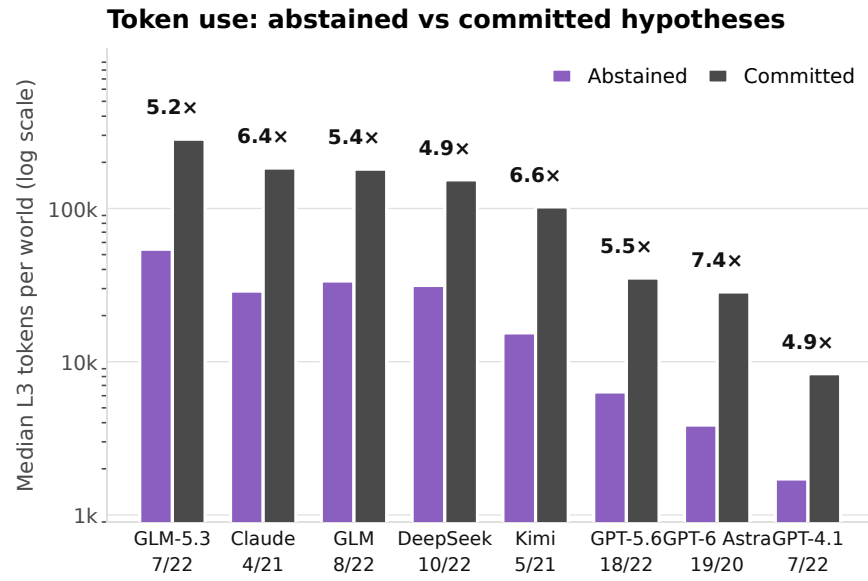


Figure 21: Bench-only cost per task by stage and model, pricing recorded input and output usage separately. GLM-5.3 is the exception to lower L3 cost.



Second tick row: abstained / eligible outcomes; exclusions omitted.

Astra has one committed hypothesis: its median is one observation.

Token use is descriptive, not a causal estimate of the effect of abstaining.

Figure 22: Median L3 tokens for eligible abstained versus committed outcomes. Exclusions are omitted from these groups; Astra's committed median represents one response.

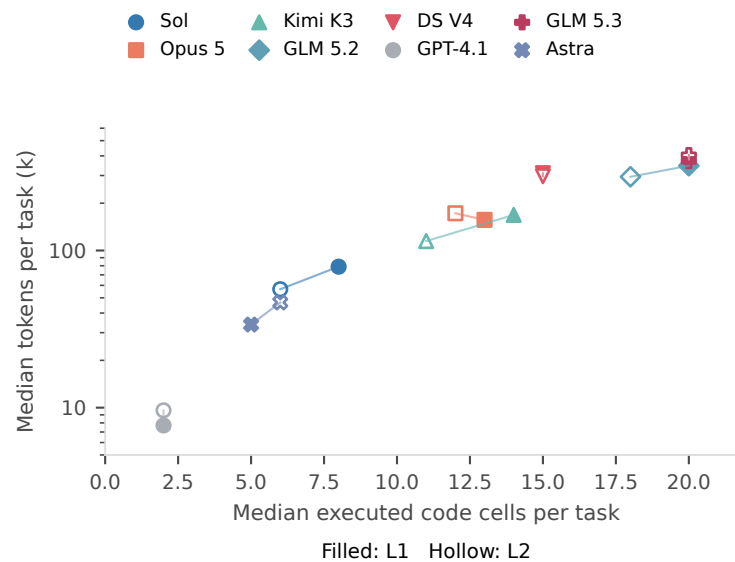
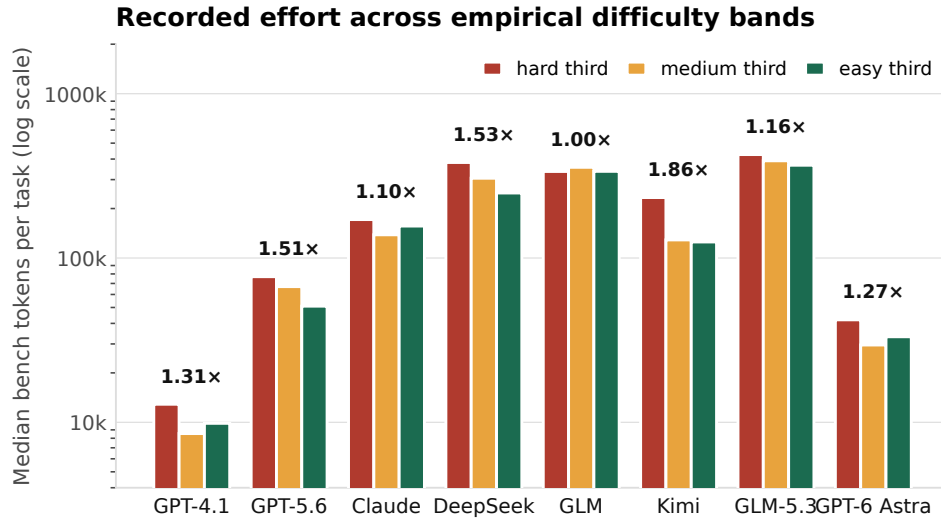


Figure 23: Execution effort and token use at L1 and L2. Each point pairs a model's median executed code cells with its median tokens per task. A code cell is an executed block of agent-generated code, not an API call or a complete analysis–feedback cycle. Colours and shapes identify models; filled markers denote L1 and hollow markers L2.



Difficulty is measured, not assumed: 16 L1/L2 track × level categories ranked by across-model mean score, split into tertiles.

L3 is excluded — its 3-6 step budget against L1/L2's 20 would make this a level-composition artifact.

Above each model: hard ÷ easy median spend. A ratio near 1 means effort is not allocated to difficulty.

Figure 24: Median tokens per task by measured difficulty tertile for L1 and L2.

L.8 PROVENANCE, LICENSES, AND RELEASE

Every substrate is public: ENCODE K562 chromatin data, ClinVar and gnomAD panels, the 10x PBMC3K dataset, published CRISPR, STARR-seq, stability, and COVID-19 PBMC datasets cited in Section 2, and public GWAS summary data. Each world's manifest records the exact perturbation recipe, expected method, and grading rubric (Appendix E). The release package will include the frozen registries, verbatim prompts, solver goldens, graded artifacts, adjudication ledgers, and the dated rate card, so both grades and dollar totals are recomputable. The Hugging Face repository and immutable release identifier remain placeholders until the separate public-release freeze; no public package is created as part of this paper revision.